PHYSICS GRADE 10: CURRENT ELECTRICITY

CBE Phenomenon-Driven Lesson Sequence — Sub-Strand 3.2 (12 Lessons)

SUB-STRAND OVERVIEW	
Grade Level	10
Subject	Physics
Strand	Strand 3.0: Electricity and Magnetism
Sub-Strand	Sub-Strand 3.2: Current Electricity
Total Duration	12 lessons × 40 minutes = 480 minutes total
Sub-Strand Content	– Electric current: definition, measurement, and SI units (amperes) – Electromotive force (EMF) and potential difference (voltage): definitions, distinctions, and units (volts) – Ohm's Law: relationship between voltage, current, and resistance; graphical verification – Resistance: definition, SI units (ohms), and factors affecting resistance (length, cross-sectional area, material, temperature) – Resistivity: definition, formula $\rho = RA/L$, and application to Kenyan wiring/electrical installations – Series circuits: current, voltage, and equivalent resistance rules; practical applications – Parallel circuits: current, voltage, and equivalent resistance rules; practical applications – Series-parallel (combination) circuits: analysis and problem-solving – Internal resistance and EMF: terminal voltage, lost volts, and the equation $V = E - Ir$ – Electrical energy and power: formulae $P = IV = I^2R = V^2/R$; energy $E = Pt$; Kenyan household electricity billing (Kenya Power units) – Heating effect of electric current: applications (electric cookers, heaters, immersion heaters common in Nairobi households) – Safety in electrical installations: fuses, circuit breakers, earthing, and safe practices in the Kenyan home and school context
Learning Outcomes	By the end of the sub-strand, the learner should be able to: a) define electric current, electromotive force, potential difference, resistance, and resistivity and state their SI units, b) verify Ohm's Law experimentally and apply it to solve problems involving voltage, current, and resistance, c) determine the factors affecting the resistance of a conductor and apply the resistivity formula $\rho = RA/L$, d) analyse series, parallel, and series-parallel circuits to determine equivalent resistance, current distribution, and voltage distribution, e) apply the concept of internal resistance and EMF to explain terminal voltage in practical cells and batteries, f) calculate electrical energy and power consumed in circuits and relate this to Kenya Power electricity billing, g) describe the heating effect of electric current and evaluate its applications in everyday Kenyan household devices, h) demonstrate safe practices in electrical installations and appreciate the importance of electrical safety in the home and community.
Core Competencies	– Communication and Collaboration: the learner works in groups to design and conduct circuit experiments, shares findings with peers, and collaboratively constructs explanations for circuit behaviour observed in Kenyan household electrical systems. – Critical Thinking and Problem Solving: the learner applies Ohm's Law, resistivity equations, and power formulae to analyse

	real-world circuit problems such as calculating electricity bills from Kenya Power meter readings and troubleshooting faulty circuits. – Learning to Learn: the learner reflects on prior knowledge of static electricity and systematically builds understanding of current electricity by revising models and updating the Driving Question Board across lessons. – Digital Literacy: the learner uses digital simulations (e.g., PhET Circuit Construction Kit) and online resources to explore circuit configurations and verify experimental results, developing competence with digital tools. – Creativity and Imagination: the learner designs safe and efficient circuits for model Kenyan household scenarios, proposing solutions to electrical safety challenges and energy-saving strategies.
Core Values	– Responsibility: the learner takes ownership of safe laboratory practices during circuit experiments, ensuring correct connections and proper handling of equipment to protect self and peers. – Integrity: the learner records experimental data honestly, even when results do not perfectly match theoretical predictions, and discusses sources of error transparently. – Respect: the learner listens to and values the contributions of all group members during circuit analysis activities, acknowledging diverse problem-solving approaches. – Unity: the learner collaborates equitably in groups during practical sessions, ensuring every member participates in setting up circuits, taking readings, and presenting conclusions.
Science & Engineering Practices	– SEP 1 – Asking Questions and Defining Problems: students generate and refine questions about why electrical devices in Kenyan homes behave differently under varying loads, anchoring inquiry throughout the unit. – SEP 3 – Planning and Carrying Out Investigations: students design controlled experiments to verify Ohm's Law and investigate factors affecting resistance, selecting appropriate variables and measuring instruments (ammeters, voltmeters, rheostats). – SEP 4 – Analysing and Interpreting Data: students plot V–I graphs, calculate gradients to determine resistance values, and interpret deviations from linearity as evidence of non-ohmic behaviour. – SEP 6 – Constructing Explanations and Designing Solutions: students construct explanations for why parallel wiring is used in Kenyan home installations and design circuits that meet specified power and safety requirements. – SEP 8 – Obtaining, Evaluating, and Communicating Information: students evaluate Kenya Power electricity bills, communicate findings on energy consumption, and present circuit diagrams with written justifications to the class.
Pertinent & Contemporary Issues (PCIs)	– Financial Literacy: the learner calculates electrical energy costs using Kenya Power billing rates (cost per kWh), enabling informed household energy budgeting and understanding of monthly electricity expenditure. – Safety and Security: the learner investigates causes of electrical fires and accidents in Kenyan homes and schools, and proposes mitigation measures such as correct fuse ratings, proper earthing, and avoiding overloading of sockets. – Environmental Education: the learner connects excessive electrical energy consumption to increased fossil-fuel-based generation and associated carbon emissions in Kenya, promoting energy conservation as an environmental responsibility. – Life Skills – Decision Making: the learner evaluates trade-offs between series and parallel circuit designs in practical household wiring scenarios, applying decision-making skills to real-life electrical installation choices.
Career Connections	– Electrical Engineer: sub-strand builds foundational circuit analysis skills (Ohm's Law, series/parallel networks, power calculations) directly applicable to electrical engineering design and infrastructure projects such as Kenya's national grid expansion. – Electrician / Electrical Technician: knowledge of internal resistance, safe wiring practices, fuse ratings, and circuit configurations prepares learners for vocational pathways in domestic and industrial electrical installation. – Electronics Technician: understanding of current, voltage, and resistance provides the essential groundwork for the subsequent Sub-Strand 3.3 Introduction to Electronics and careers in electronics repair and manufacturing.

	– Renewable Energy Technician: power and energy calculations connect to solar panel and battery system sizing, relevant to Kenya's expanding off-grid solar sector (e.g., M-KOPA solar installations in rural Kenya). – Physics Teacher / Lecturer: mastery of current electricity concepts equips learners who pursue education pathways to teach this foundational topic at secondary and tertiary levels in Kenyan schools.
Focus for Lessons	This unit develops learners' understanding of electric current, circuit behaviour, and electrical energy through a sequence of hands-on investigations and real-world problem-solving rooted in Kenyan household and community contexts. Starting from the puzzling anchoring phenomenon of variable brightness in household bulbs during Kenya Power load-shedding recovery, learners progressively build and revise conceptual models of how charge flows, how resistance governs that flow, and how energy is transferred and transformed in circuits. The unit balances quantitative problem-solving with qualitative sensemaking, ensuring learners can both calculate and explain electrical phenomena they encounter in daily Kenyan life.
Driving Question / Key Inquiry Questions	DRIVING QUESTION: Why do electrical devices in our homes sometimes receive less energy than expected, and how can we design circuits to deliver the right amount of electrical energy safely and efficiently? KICD KEY INQUIRY QUESTIONS: 1. What is the relationship between voltage, current, and resistance in an electric circuit? 2. How is electrical energy consumed and managed safely in our homes and communities?
Anchoring Phenomenon	ANCHORING PHENOMENON — "The Flickering Lights of Huruma Estate": During a power restoration after a Kenya Power load-shedding blackout in Huruma Estate, Nairobi, residents notice something puzzling: when electricity is first restored, the lights in the house glow very dimly and the electric kettle takes much longer than usual to boil water. A few minutes later, the lights brighten to their normal level and the kettle heats at its normal rate. At the same time, a neighbour using a bulb connected with a long, thin extension cord notices their bulb is always dimmer than one connected directly to the wall socket — even when power is fully restored. Students are shown a short video clip of this scenario and asked: why do the lights glow dimly at first? Why does the kettle take longer to heat? Why does the bulb on the long thin extension cord always appear dimmer? The phenomenon is puzzling because students expect electricity to either work or not work — they do not yet have a model to explain why the same power source can deliver different brightness and heating rates under different conditions. This drives the entire unit's inquiry into current, voltage, resistance, internal resistance, circuit configuration, and electrical power.
Storyline Thread	Lesson 1 — Anchoring Phenomenon & Predict Phase: Students observe the "Flickering Lights of Huruma Estate" video, record initial predictions about why brightness varies, and co-construct the initial Driving Question Board (DQB). Introduction to electric current as the flow of charge; definition of ampere and measurement using an ammeter. Lesson 2 — EMF and Potential Difference: Students investigate the difference between electromotive force (EMF) and potential difference (terminal voltage) using dry cells and voltmeters. Connect back to anchoring phenomenon: why is the voltage at the socket lower than expected during power restoration? Lesson 3 — Ohm's Law Investigation: Students design and carry out a controlled experiment using a nichrome wire, rheostat, ammeter, voltmeter, and battery to verify Ohm's Law. Plot V–I graphs and calculate resistance from gradient. Update DQB with the concept of resistance as the factor controlling current for a given voltage.

	Lesson 4 — Resistance and Resistivity: Students investigate factors affecting resistance (length, cross-sectional area, material) using wires of different gauges — connecting to the "hot extension cord" supporting phenomenon. Derive and apply the resistivity formula $\rho = RA/L$. Link thicker vs. thinner cables in Kenyan home wiring to resistivity. Lesson 5 — Series Circuits: Students build series circuits with multiple bulbs, measure current and voltage at multiple points, and derive rules for series circuits (same current, voltages add, resistances add). Revisit Christmas lights supporting phenomenon; explain why bulbs dim when more are added in series. Lesson 6 — Parallel Circuits: Students build parallel circuits, measure current and voltage at multiple points, and derive rules for parallel circuits (voltage same, currents add, equivalent resistance decreases). Connect to Kenyan home wiring: explain why all appliances operate at 240 V independently and why adding appliances increases total current drawn. Lesson 7 — Series-Parallel Combination Circuits: Students analyse combination circuits through worked examples and problem-solving tasks set in Kenyan household and school contexts. Apply equivalent resistance methods step-by-step. Update circuit model on DQB to include both circuit types. Lesson 8 — Internal Resistance and EMF: Students investigate internal resistance using a battery, rheostat, voltmeter, and ammeter; plot terminal voltage vs. current graphs; derive $V = E - Ir$. Connect to "The Weakening Torch" and "Flickering Lights" phenomena — explain why terminal voltage drops under high current demand during power restoration surge. Lesson 9 — Electrical Power: Students derive and apply power formulae $P = IV = I^2R = V^2/R$ using data from previous experiments. Calculate the power rating of common Kenyan household devices (electric jiko, immersion heater, phone charger). Relate power dissipation to heating in wires — revisit "hot extension cord" phenomenon. Lesson 10 — Electrical Energy and Kenya Power Billing: Students calculate electrical energy consumption ($E = Pt$) and convert to kilowatt-hours. Analyse a sample Kenya Power electricity bill, calculating the cost of running specific household appliances. Discuss energy conservation strategies relevant to low-income Kenyan households. Lesson 11 — Heating Effect of Electric Current & Electrical Safety: Students investigate the heating effect of current and evaluate applications (electric cooker, immersion heater, toaster). Study electrical safety: fuses, circuit breakers, earthing, and safe practices. Analyse causes of electrical fires in Kenyan homes; design safety recommendations. Connect fuse-blowing supporting phenomenon to calculated current limits. Lesson 12 — Model Building, DQB Resolution & Assessment: Students revise their cumulative circuit models incorporating all concepts (Ohm's Law, resistivity, series/parallel rules, internal resistance, power, energy, safety). Present final explanations for all anchoring and supporting phenomena. Complete a summative performance task: design a safe, efficient wiring plan for a model Kenyan household with a given appliance list and Kenya Power budget.
Supporting Phenomena	– Supporting Phenomenon 1 — "The Dimmer Bulb in the Series Christmas Lights": a string of decorative lights sold at Nairobi's Gikomba market is wired in series; when one bulb is removed or blows, all others go out, and when multiple bulbs are added the remaining ones all dim — puzzling students about how adding more devices reduces brightness. – Supporting Phenomenon 2 — "Why Does the Fuse Blow When We Plug In Too Many Devices?": in a Mombasa household, plugging a 2,000 W electric iron, a microwave, and a water pump into the same extension board trips the circuit breaker;

	students investigate why adding more devices in parallel eventually causes the fuse to blow. – Supporting Phenomenon 3 — "The Hot Extension Cord": after a long cooking session in a Kisumu home, the thin extension cord connecting the electric cooker feels very warm to the touch, while the thicker main cable from the wall socket remains cool — students investigate why the same current produces different heating in different conductors. – Supporting Phenomenon 4 — "The Weakening Torch": a torch (mkebe) used by a watchman in Rift Valley runs brightly at the start of the night but grows progressively dimmer by morning even though the bulb and wiring are undamaged — students connect this to internal resistance increasing as the battery discharges, linking EMF, terminal voltage, and internal resistance.

LESSON 1 (40 minutes): The Flickering Lights of Huruma Estate: Launching the Phenomenon	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To launch the unit by anchoring student curiosity in a real Nairobi power-restoration scenario, elicit prior knowledge about electricity, introduce electric current as the flow of charge, and co-construct the Driving Question Board that will guide the entire sub-strand inquiry.
Knowledge	Students will be able to define electric current as the rate of flow of electric charge, state the SI unit of current as the ampere (A), and identify an ammeter as the instrument used to measure current.
Skills	Students will practise making evidence-based predictions, recording observations systematically, formulating testable questions, and collaboratively constructing a class Driving Question Board.
Attitudes	Students will demonstrate curiosity about everyday electrical phenomena, openness to revising initial ideas, and collaborative responsibility for shared inquiry questions.
Key Inquiry Question	What is the relationship between voltage, current, and resistance in an electric circuit?
Purpose in Storyline	This is the opening lesson that launches the entire unit. It anchors all subsequent investigations in the puzzling Huruma Estate flickering-lights phenomenon, surfaces students existing mental models about electricity, and generates the questions that Lessons 2 through 12 will systematically answer.
Safety Notes	No live mains electricity is used in this lesson. If a physical torch or dry-cell battery is demonstrated, remind students never to short-circuit a battery. Ensure the video playback device is handled by the teacher only. Reinforce that students must never tamper with wall sockets or Kenya Power infrastructure.

B. LESSON OVERVIEW
Students begin by writing silent predictions about why lights in a Nairobi estate flicker and dim after a blackout. They then watch a short video reconstruction of the Huruma Estate scenario showing dim bulbs, a slow-boiling kettle, and a permanently dim bulb on a long thin extension cord. After observing, students share and compare their predictions with actual observations, surface the gaps in their understanding, receive a brief teacher introduction to electric current as charge flow and the ampere, and finally co-construct the class Driving Question Board with sticky notes organised into thematic clusters. The lesson closes with students sketching an initial particle model of what they think is happening inside a wire when current is low versus high.

C. LESSON IMPLEMENTATION FRAMEWORK				
Phase	**Learner Experience**	**Teacher Moves**	**Sensemaking Strategy**	**Formative Assessment Strategy**
Predict Phase 📹 **VIDEO:** [Introduction to circuits and Ohm's law] Source: Khan	Students receive a half-page Prediction Card. The teacher reads aloud the Huruma Estate scenario without showing the video yet: after a Kenya Power load-shedding blackout, electricity is	Distribute Prediction Cards face-down before reading the scenario. Say: Before I show you anything, I want to know what YOU already think. Flip your card over and write or draw your ideas. You have 3	Eliciting prior conceptions using a structured Prediction Card ensures all students commit to an initial model before observation, creating cognitive dissonance when the video reveals complexity.	Teacher scans Prediction Cards while circulating during silent writing to identify the range of existing models: students who think in binary on-off terms, students who mention voltage

Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Electric Charge and Electric Force (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	restored in a home. The lights glow dimly at first and the kettle is slow to boil. A neighbour using a long thin extension cord always has a dimmer bulb even after full power is restored. Students individually write or sketch answers to three prompts on the card: (1) Why do you think the lights are dim at first? (2) Why does the kettle take longer? (3) Why is the neighbour bulb always dimmer? Students work in silence for 3 minutes, then share predictions with a seat partner for 2 minutes. Selected pairs share with the class. Teacher records key prediction phrases verbatim on a section of the board labelled OUR PREDICTIONS.	minutes of thinking time — no talking yet. Start the timer visibly. After 3 minutes say: Now share with your partner for 2 minutes — listen carefully to where you agree and where you disagree. After partner talk, cold-call three pairs using equity sticks. For each response, say: Thank you. I am writing your exact words here — we will come back to check these at the end of the unit. Do NOT affirm or correct predictions at this stage. If a student says it is because there is less electricity, write that phrase exactly and say: Interesting — what do others think less electricity means? WAIT TIME: After each student response, pause for at least 5 seconds before calling on another student. This signals that thinking time is valued.	Recording verbatim predictions preserves authentic student language for later revisiting.	informally, and students who reference wires or cables. This informs pacing for the Explain Phase. No grades are given.
Observe Phase 📹 **VIDEO:** Introduction to circuits and Ohm's law Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Electric Charge and Electric Force (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	The teacher plays the 3-to-4-minute video clip reconstructing the Huruma Estate scenario. The video shows: a dark room during load-shedding, lights flickering on dimly at restoration, a kettle element glowing faintly and taking longer to boil, and a close-up comparison of two bulbs — one on a long thin extension cord (dim) and one plugged directly into the wall socket (bright). Students complete an Observation Record with two columns: What I see and What I notice is puzzling. After the video, students revisit their Prediction Cards and use a coloured pen to annotate: circle any prediction that matched, and draw a question mark next to anything the video made them unsure about.	Before playing the video say: Watch very carefully. You are scientists. Your job is to notice everything — not just the big things but the small details too. You will fill in your Observation Record as you watch. Play the video once through without pausing. Then say: I will play it one more time. This time, focus on the comparison between the two bulbs at the end. Play the second viewing. After viewing say: Look at your Prediction Card. With your coloured pen, circle anything you got right, and put a question mark next to anything that now seems incomplete or wrong. You have 2 minutes. Then ask: How many of you found something that surprised you? Hands up. Acknowledge the hands and say: Good — surprise is where science begins. What surprised you most?	The two-column Observation Record separates empirical observations from interpretive puzzlement, building the habit of distinguishing data from explanation. Annotating the Prediction Card with a contrasting colour makes the predict-observe-explain cycle visible and tangible for students.	Teacher collects Observation Records at the end of the lesson to check whether students can distinguish observation language (the bulb looks dimmer) from explanation language (the current is lower). This informs scaffolding needed in Lesson 3 when students design their own investigations.

		Take 3 responses. WAIT TIME: After asking what surprised you most, count silently to 7 before accepting any response, allowing processing time.		
Explain Phase 📹 **VIDEO:** Introduction to circuits and Ohm's law Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Electric Charge and Electric Force (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	The teacher facilitates a 7-minute whole-class discussion anchored to the observation. Students are guided to recognise that the scenario shows that electricity does not simply work or not work — something about the circuit is changing the amount of energy delivered. The teacher introduces the concept of electric current as the flow of electric charge through a conductor, using the analogy of water flowing through a pipe in Nairobi Water Supply: a wider pipe with good pressure carries more water per second, just as a good conducting path delivers more charge per second. Students are introduced to the formal definition: current I is the quantity of charge Q passing a point per unit time t, written as I equals Q divided by t. The unit is the ampere, named after French physicist Andre-Marie Ampere. The teacher shows a physical ammeter and explains its role. Students copy the definition and the formula into their notebooks with a labelled diagram of an ammeter connected in series in a simple circuit.	Begin by saying: Your observations show electricity is doing different amounts of work under different conditions. What word could we use for how much charge is flowing through the wire per second? Accept suggestions. Then say: Scientists use the word current. Electric current is the rate at which charge flows. Write on the board: I equals Q divided by t. Say: I stands for current, Q stands for charge measured in coulombs, and t stands for time. The unit of current is the ampere — we call it the amp for short. Display the ammeter and say: This instrument is called an ammeter. It measures current in amperes. It must always be connected in series — directly in the path of the current. Ask: Looking back at our Huruma scenario, if the lights are dim, what do you think is happening to the current — is more or less charge flowing per second? WAIT TIME: Pause for 6 seconds after this question. Then say: Discuss with your partner for 1 minute, then we will hear ideas. After partner talk, take 2 responses. Do not yet explain WHY the current differs — save that for Lessons 2 through 4.	The Nairobi water-pipe analogy connects an abstract concept to a locally familiar infrastructure experience. Deferring the explanation of WHY current varies maintains productive confusion that will be resolved progressively across the unit storyline.	Exit check: teacher poses an oral question — If a charge of 10 coulombs passes a point in 2 seconds, what is the current? Students write the answer on a mini whiteboard or scrap paper and hold it up. Teacher scans for correct application of I equals Q divided by t. Correct answer is 5 A.
Driving Question Board (DQB) Creation 📹 **VIDEO:** Introduction to circuits and Ohm's law	Each student receives three sticky notes in different colours: yellow for questions about what is happening, pink for questions about why it is happening, and blue for questions about what we could do about it. Students write one question per sticky note in	Distribute sticky notes and say: You now have 3 minutes to write your burning questions — one question per sticky note. Yellow notes are for WHAT questions, pink for WHY questions, and blue for WHAT CAN WE DO questions. After posting, say: Look at all	The three-colour sticky-note system scaffolds different epistemic question types, building metacognitive awareness of the difference between descriptive, causal, and design questions. Physical manipulation of the DQB gives ownership and makes the	Teacher photographs the DQB and reviews question quality after the lesson. Questions that are too vague (why does electricity do that?) are noted so the teacher can model more precise scientific question framing in Lesson 2. Questions that already mention

Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Electric Charge and Electric Force (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	response to the phenomenon, working individually for 3 minutes. Students post their sticky notes on a large sheet of paper fixed to the wall labelled THE HURUMA ESTATE MYSTERY. The teacher facilitates a 5-minute whole-class clustering activity, asking students to physically move sticky notes into thematic groups and suggest labels for each cluster. Expected clusters include: questions about voltage and power supply, questions about wires and cables, questions about how circuits are arranged, questions about heating and energy, and questions about safety. The teacher adds two anchor questions printed in bold at the top of the DQB: What is the relationship between voltage, current, and resistance in an electric circuit? and How is electrical energy consumed and managed safely in our homes and communities?	these questions. Some of them seem to be asking about similar things. Can someone come up and move two notes that belong together? Invite 3 to 4 students to physically move notes and explain their reasoning. Then say: I am going to add two big questions at the top that will guide our entire unit. Read the two key inquiry questions aloud slowly and say: By the end of our 12 lessons together, you will be able to answer every single question on this board — including your own. Take a photo of the DQB. We will return to it every lesson. WAIT TIME: During clustering, when students are deciding where to place a note, allow at least 8 seconds of silence before offering a prompt, respecting the cognitive work of categorisation.	board a living artefact rather than a teacher-created display.	voltage, resistance, or circuit indicate students with stronger prior knowledge who can serve as peer resources.
Model Building Phase 📹 **VIDEO:** Introduction to circuits and Ohm's law Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Electric Charge and Electric Force (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	In the final 5 minutes, students return to their notebooks and draw an Initial Particle Model: two side-by-side sketches of the inside of a wire — one labelled HIGH CURRENT (bright bulb) and one labelled LOW CURRENT (dim bulb). Students use dots to represent charge particles and arrows to represent their movement. Students write one sentence under each sketch explaining what they think is different between the two situations. These models are collected and stored in student science notebooks as Lesson 1 baseline models. Students are told that they will revise and improve this model in every lesson as they gather more evidence.	Say: Before we finish, I want you to do something brave — draw what you THINK is happening inside the wire at the particle level. You do not need to be right. This is your starting model, and starting models are always incomplete. That is normal in science. Draw two boxes in your notebook — one for high current and one for low current. Use dots for charge particles and arrows to show movement. After 3 minutes say: Write one sentence under each box explaining the difference. Take 2 volunteers to share their models under the document camera without judgment. Say: Notice how different our models are right now. By the end of this unit, we will all have a much more detailed and	Initial particle modelling externalises students implicit mental models, making them available for revision and comparison across the unit. Framing incompleteness as a feature of all starting models reduces anxiety and positions science as a process of model refinement rather than answer retrieval.	Teacher reviews collected Lesson 1 baseline models after class to identify three types of prior models: students who draw static particles with no movement, students who draw particles moving but with no difference between high and low current scenarios, and students who intuitively draw more or faster particles for high current. These three categories inform differentiated questioning strategies in Lesson 3 during the Ohm is Law investigation.

		accurate model — one that can explain everything we saw in the Huruma Estate video. Point to the DQB and say: Keep those questions in mind. They are YOUR questions and YOU will answer them. WAIT TIME: After asking for volunteers to share models, wait 7 seconds before calling on anyone, normalising the vulnerability of sharing an uncertain initial model.		

D. TEACHER REFLECTION
After the lesson, consider: Did all students commit to a written prediction before seeing the video, or did some wait to see what others wrote? Were the DQB questions student-generated or did students look to you for approval before posting? Did the water-pipe analogy for current resonate with students who have experience with Nairobi Water Supply intermittent flow, or did it need more scaffolding? Were there students who already used terms like voltage or resistance confidently — how can you leverage their knowledge without shutting down inquiry for others? Review the Observation Records and baseline particle models tonight and identify two or three student ideas that you will explicitly reference by name at the start of Lesson 2 to signal that student thinking is the driver of this unit.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	During a power restoration in Huruma Estate, Nairobi, lights glowed dimly, a kettle took longer to boil, and a bulb on a long thin extension cord was always dimmer than one plugged directly into the wall socket.
What did I learn?	Electric current is the flow of electric charge through a conductor. Current I equals charge Q divided by time t, and is measured in amperes using an ammeter connected in series.
How does this explain the phenomenon?	We cannot yet fully explain why the lights were dim or why the extension cord bulb was dimmer. We have identified that the AMOUNT of current flowing must be different in these situations, and we have posted our questions on the Driving Question Board to guide our investigation over the next 11 lessons.

LESSON 2 (40 minutes): EMF and Potential Difference: Why Does the Socket Give Less Than the Source?	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	Students will distinguish between electromotive force (EMF) and potential difference (terminal voltage) by measuring both quantities using dry cells and voltmeters, and will use this distinction to begin explaining why the voltage available at a socket is lower than expected during the Huruma Estate power restoration.
Knowledge	Define electromotive force (EMF) as the total energy transferred per unit charge by a source; define potential difference (terminal voltage) as the energy transferred per unit charge between two points in an external circuit; state that terminal voltage is always less than or equal to EMF; recall that both EMF and potential difference are measured in volts (joules per coulomb); recognise that a voltmeter is connected in parallel to measure potential difference.
Skills	Connect a voltmeter correctly in parallel across a cell and across a resistor; measure and record EMF and terminal voltage accurately; compare the two readings and identify the difference; construct a simple data table and draw a preliminary conclusion from measured values; communicate findings using correct scientific vocabulary.
Attitudes	Demonstrate curiosity when observed readings differ from predictions; show precision and care when handling electrical components; collaborate respectfully in practical groups; appreciate that real-world devices do not behave as idealised models predict.
Key Inquiry Question	What is the relationship between voltage, current, and resistance in an electric circuit?
Purpose in Storyline	In Lesson 1 students established that current is the flow of charge and that the Huruma Estate lights were dimmer than expected. This lesson introduces the concept that the energy a source can supply per coulomb (EMF) is not fully available at the terminals when the source is in use. This gap between EMF and terminal voltage is the first quantitative evidence students collect that will later be fully explained in Lesson 8 (internal resistance). It directly advances the driving question by showing that the power source itself can deliver less voltage than its rated value, helping explain why lights were dim during restoration.
Safety Notes	Ensure all connections are made with the circuit open (switch off) before the teacher checks wiring. Use only 1.5 V AA dry cells or a 3 V battery pack — never mains voltage for this activity. Voltmeter probes must be handled carefully to avoid short circuits. Cracked or leaking cells must be removed immediately and hands washed. Wires with damaged insulation must not be used.

B. LESSON OVERVIEW
Students begin by revisiting the Huruma Estate phenomenon and predicting what a voltmeter will read across a battery before and after connecting a load. They then carry out a structured investigation measuring the EMF of a dry cell (open circuit) and the terminal voltage (closed circuit) under different loads using resistors and a rheostat. The observed drop in voltage provides concrete evidence that connects to the anchoring phenomenon. Students construct an energy-per-charge mental model and update the Driving Question Board with a new piece of evidence: the source voltage reaching the circuit is lower than the source rating, especially when current demand is high.

C. LESSON IMPLEMENTATION FRAMEWORK				
Phase	**Learner Experience**	**Teacher Moves**	**Sensemaking Strategy**	**Formative Assessment Strategy**

Predict Phase 📹 **VIDEO:** Induced current in a wire Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Electromotive Force (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Students are shown the DQB from Lesson 1 and the teacher re-reads the focus question written during that lesson: 'Why did the lights in Huruma Estate glow dimly when power was first restored?' The teacher then holds up a 1.5 V AA dry cell and a voltmeter and asks: 'This cell is labelled 1.5 volts. If I connect the voltmeter across it right now, with nothing else connected, what will you expect it to read? Write your prediction in your exercise book, including the exact number you expect.' Students write individually for 90 seconds. The teacher then asks: 'Now I connect a torch bulb to the cell and measure again while the bulb is lit. Will the reading be the same, higher, or lower than 1.5 V? Write a second prediction and give a reason.' Students write for another 90 seconds. The teacher selects three volunteers to share predictions without yet revealing which is correct, and records the three different views on the board under the heading 'Our Predictions Before We Look.'	Teacher re-displays the Huruma Estate scenario image on the board and says: 'Last lesson we agreed that current is the flow of charge. Today we are asking a different question — how much energy does that charge carry?' Teacher holds up the cell and says: 'The number printed on this cell is 1.5 volts. Before you do anything, what does that number actually mean? Think silently for 30 seconds.' [WAIT 30 SECONDS — no prompting.] Teacher then takes two or three quick responses and writes key words on the board without evaluating them. Teacher then poses the two-part prediction task clearly: 'Prediction one — voltmeter reading with nothing connected. Prediction two — voltmeter reading while a bulb is lit. Write numbers, not just words.' After student sharing, teacher says: 'We are going to test both predictions right now. Keep your prediction visible so you can check yourself.'	Prediction journaling with explicit numerical commitment; public recording of conflicting predictions to create cognitive dissonance and motivate observation.	Teacher circulates during individual writing and notes whether students predict the same value for both conditions (common misconception: voltage is fixed regardless of load) or whether any student already suspects the reading will drop. This informs pacing during the Observe phase.
Observe Phase 📹 **VIDEO:** Induced current in a wire Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Electromotive Force (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Students work in groups of three or four. Each group receives: one 1.5 V AA dry cell in a holder, one voltmeter (0–3 V range), one switch, connecting wires, one 10-ohm resistor, and one 47-ohm resistor. Procedure on the task card: Step 1 — Connect the voltmeter directly across the cell terminals with the switch open (no current flowing). Record the reading in a prepared table as 'EMF (open circuit).' Step 2 — Connect the 47-ohm resistor in series with the switch across the cell. Close the switch and immediately read the voltmeter	Teacher circulates continuously. At each group, teacher asks quietly: 'Is the voltmeter in parallel or in series with the cell right now? How do you know?' [WAIT 10 SECONDS for student to answer before confirming.] If a group has the voltmeter wired in series (common error), teacher says: 'What happens to your circuit path if you remove the voltmeter? Does it break the circuit? Should it?' and waits for the student to self-correct. When a group records a terminal voltage reading, teacher asks: 'Is that reading higher or lower than your open-circuit	Structured data collection table with a built-in comparison column (Difference from EMF) that makes the pattern visible; qualitative-to-quantitative bridging through the brightness observation paired with voltage numbers.	Teacher checks each group table for correct recording. Key check: does the group show EMF greater than or equal to terminal voltage in all cases, and does terminal voltage decrease as resistance decreases (current increases)? Groups that show terminal voltage exceeding EMF have a wiring error and must be redirected before Explain phase.

	across the cell terminals. Record as 'Terminal voltage with 47-ohm load.' Step 3 — Replace the 47-ohm resistor with the 10-ohm resistor and repeat. Record as 'Terminal voltage with 10-ohm load.' Students record all three readings in a table with columns: Condition, Voltmeter Reading (V), Difference from EMF (V). Groups also note whether the bulb (if used as load) appears brighter with 10 ohms or 47 ohms and record this qualitative observation. The teacher calls time after 12 minutes and asks groups to compute the 'Difference from EMF' column before discussion begins.	reading? Was that what you predicted?' [WAIT 15 SECONDS.] Teacher does not yet explain WHY the reading drops — only prompts students to document the pattern. Teacher writes on the board midway through: 'What do you notice about the trend as the resistance decreases?' to focus student attention.		
Explain Phase 📹 **VIDEO:** Induced current in a wire Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Electromotive Force (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	The teacher leads a whole-class data share. Each group reads out their three values and a class data table is built on the board. The teacher asks the class: 'What pattern do we see across all groups?' Students identify that the open-circuit reading is highest, the 47-ohm reading is slightly lower, and the 10-ohm reading is the lowest. The teacher introduces vocabulary: 'The open-circuit reading is called the electromotive force, or EMF. We give it the symbol E. The reading when current flows is called the terminal voltage or potential difference at the terminals. We give it the symbol V.' Teacher writes on the board: EMF (E) = energy given to each coulomb of charge by the source; Terminal Voltage (V) = energy available to the external circuit per coulomb. Teacher asks: 'If the cell gives each coulomb 1.5 joules, but only 1.3 joules reach the resistor, what happened to the other 0.2 joules?' [WAIT 20 SECONDS — accept all	Teacher uses the board to build the class data table deliberately slowly, pausing after each group entry to ask: 'Is this group result consistent with the pattern we are seeing, or does it surprise you?' [WAIT 10 SECONDS after each entry.] When introducing EMF definition, teacher writes: 'E = energy per coulomb from source (unit: joule per coulomb = volt)' and says: 'Say this definition in your own words to your partner. You have 45 seconds.' [WAIT 45 SECONDS.] Teacher then cold-calls one student from a group that had NOT yet spoken. When connecting to Huruma, teacher points at the DQB and says: 'Look at the question we wrote last lesson about why the lights were dim. Does today's data give us part of an answer? Point to the specific question on the board.' Teacher does not answer for students but affirms direction: 'You are on the right track — we will build a fuller explanation by Lesson 8.'	Class data aggregation on the board to reveal a consistent pattern across groups; paired sense-making with sentence-completion anchoring; explicit vocabulary introduction tied to units (joules per coulomb equals volts) to build conceptual understanding not just label recall.	The written sentence completion ('The socket was receiving only the...') is collected or shown to the teacher as an exit check for this phase. Teacher identifies students who choose 'EMF' instead of 'terminal voltage' for targeted follow-up questioning.

	responses.] Students discuss in pairs for 60 seconds, then share. Teacher guides students toward the idea that some energy is used inside the source itself. Teacher then connects explicitly to the anchoring phenomenon: 'During the Huruma Estate power restoration, the electricity company was supplying power — like our cell — but the voltage reaching the socket was lower than expected. Using today's vocabulary, was the socket receiving the full EMF of the supply?' Students answer in writing in their books: 'The socket was receiving only the [terminal voltage / EMF] because some energy was being used [inside the source / in the external wires].' Teacher asks two students to read their sentence aloud and the class votes on which completion is correct.			
Driving Question Board (DQB) Creation 📹 **VIDEO:** Induced current in a wire Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Electromotive Force (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Students return to the class DQB created in Lesson 1. The teacher displays it and students re-read the questions posted. The teacher asks: 'Which question on our board does today evidence begin to answer, even partially?' Students identify questions related to why voltage at the socket is lower than expected or why lights are dim. The teacher invites one student to write on a sticky note: 'New evidence — the terminal voltage of a source is less than its EMF when current flows. The more current, the bigger the drop.' This is placed next to the relevant question with a sticker colour distinct from Lesson 1 entries (for example, yellow for Lesson 2 evidence). The teacher then asks: 'What NEW questions has today raised that we did not think of in	Teacher says: 'Our DQB is a living record of our growing understanding. Every lesson we add evidence AND new questions — because real science always generates more questions, not fewer.' Teacher physically stands beside the DQB and points to Lesson 1 entries, then says: 'What has changed in our thinking since yesterday? What can we now say that we could NOT say after Lesson 1?' [WAIT 20 SECONDS.] Teacher ensures the new evidence sticky note uses the words EMF and terminal voltage — if a student writes a vague phrase, teacher asks: 'Can you use the scientific vocabulary we learned today?' Teacher photographs the updated DQB for the class record.	Cumulative DQB with colour-coded evidence layers; distinguishing between answered questions and newly generated questions to model authentic scientific inquiry.	Teacher reads the new sticky notes to check that students are generating questions that connect to circuit behaviour rather than unrelated tangents. The quality of new questions reveals depth of conceptual engagement.

	Lesson 1?' Students work in pairs for 90 seconds and write new questions on sticky notes. Sample expected questions: 'Where does the lost energy go inside the cell?', 'Does the EMF of a cell change as the cell gets old?', 'Is this what happens inside the Kenya Power transformer too?' These are added to the board under a section labelled 'New questions from Lesson 2.' Teacher reads two or three aloud and says: 'These are excellent — we will return to the one about where the energy goes inside the source in Lesson 8.'			
Model Building Phase 📹 **VIDEO:** [Induced current in a wire] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Electromotive Force (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	Students individually draw a simple diagram in their exercise books showing a dry cell connected to a resistor with a switch and a voltmeter. Next to the diagram they write two labels using arrows: (1) an arrow inside the cell labelled 'EMF — energy given to each coulomb here' and (2) an arrow across the resistor labelled 'Terminal voltage (potential difference) — energy used by each coulomb here.' Below the diagram they complete a written model statement: 'The EMF of a source is [always equal to / always greater than or equal to / always less than] the terminal voltage because [some energy is used inside the source itself / the external resistance is too high / the voltmeter takes away energy].' Students choose the correct options and underline them. Students then apply the model to the anchoring phenomenon by completing a short written response: 'During the Huruma Estate power restoration, the supply EMF was approximately 240 V but the terminal voltage at	Teacher says: 'A scientific model is not just a diagram — it is a statement of what we believe is happening and why. Write yours carefully because you will update it every lesson.' Teacher circulates and checks that arrows are correctly labelled — specifically that the EMF arrow is inside the cell, not outside. If a student places it outside, teacher asks: 'Where does the cell add energy to the charge — inside itself or in the wire?' [WAIT 10 SECONDS.] For the phenomenon sentence, teacher says: 'Use only the words we defined today — EMF, terminal voltage, energy per coulomb — to complete your sentence. No vague words like power or electricity please.' At close, teacher says: 'Next lesson we are going to ask a new question: if voltage is fixed, what controls how much current flows? Bring your curiosity.'	Annotated circuit diagram with directional energy-flow arrows to make the abstract energy-per-charge concept spatial and visual; forced-choice model statement with three options to prevent vague middle-ground answers; explicit phenomenon-application writing to consolidate transfer.	Teacher collects or photographs the model diagram and written statements from at least four groups of students to assess whether the correct relationship (EMF greater than or equal to terminal voltage) is being represented and whether students can correctly apply the model to the anchoring phenomenon. Errors in the phenomenon sentences identify students needing individual support before Lesson 3.

	the socket was lower. This means that lights received [more / less] energy per coulomb than the rated supply, which is why they glowed [brighter / dimmer] than normal.' The teacher asks one or two students to share their completed model and reads the correct version aloud for the class to self-check. Teacher previews Lesson 3: 'We now know that voltage can drop. But what determines how much current flows for a given voltage? That is what we investigate next lesson using Ohm's Law.'			

D. TEACHER REFLECTION
Consider the following questions after the lesson: Did all groups correctly wire the voltmeter in parallel? If not, at what point did I notice and how quickly did I redirect? Did students articulate the definition of EMF in their own words, or did they merely copy the board definition? Were the new DQB questions generated by students substantive — pointing toward internal resistance and supply losses — or were they superficial? Were there students who still believed terminal voltage equals EMF even after the data? What follow-up question will I ask those students at the start of Lesson 3? Did I allow sufficient wait time (20 seconds) after asking why terminal voltage is lower than EMF, or did I fill the silence too quickly? Note which student groups produced the clearest annotated diagrams — these can be displayed as reference models in Lesson 3.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	When a dry cell is connected to a resistor and the switch is closed, the voltmeter reading across the cell terminals drops below the open-circuit reading. The drop is larger when a smaller resistor (higher current) is used.
What did I learn?	The energy given to each coulomb of charge by a source is called electromotive force (EMF, symbol E, unit volt). The energy available per coulomb in the external circuit is called terminal voltage or potential difference (symbol V). Terminal voltage is always less than or equal to EMF because some energy is used inside the source. Both quantities are measured in volts (joules per coulomb) using a voltmeter connected in parallel.
How does this explain the phenomenon?	During the Huruma Estate power restoration, the electricity supply had a certain EMF but the terminal voltage reaching the wall sockets was lower than expected, especially when many appliances were drawing current simultaneously. This is why lights received less energy per coulomb than normal and glowed dimly. The voltage available at the socket is not the same as the rated supply EMF — it is reduced by losses inside the source, which become larger as current demand increases.

LESSON 3 (80 minutes): Ohm's Law Investigation: Resistance as the Controller of Current	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To design and carry out a controlled experiment to verify Ohm's Law, plot V–I graphs, and calculate resistance from the gradient — connecting resistance to the dimming and brightening observed in the Huruma Estate phenomenon.
Knowledge	By the end of the lesson, the learner should be able to: (a) state Ohm's Law and the conditions under which it holds; (b) define resistance and state its SI unit (ohm, Ω); (c) explain how the gradient of a V–I graph gives resistance; (d) distinguish between ohmic and non-ohmic conductors.
Skills	By the end of the lesson, the learner should be able to: (a) design a controlled circuit using a nichrome wire, rheostat, ammeter, voltmeter, and battery; (b) collect and record voltage and current data accurately; (c) plot a V–I graph and calculate resistance from its gradient; (d) identify sources of experimental error and suggest improvements.
Attitudes	The learner appreciates the importance of systematic experimental investigation in building scientific understanding, and values accuracy, honesty in data recording, and collaborative teamwork during practical work.
Key Inquiry Question	What is the relationship between voltage, current, and resistance in an electric circuit?
Purpose in Storyline	In Lessons 1 and 2, students established that current flows in a circuit and that voltage drives that current. But they still cannot explain WHY the lights in Huruma Estate were dim — or why the bulb on the long thin extension cord is always dimmer. This lesson introduces resistance as the missing variable: students experimentally discover that for a given voltage, a higher resistance means less current, and therefore less power delivered to a device. This is the critical conceptual bridge that will later explain dim bulbs, slow kettles, internal resistance, and circuit design.
Safety Notes	Ensure all connections are made with the circuit open (switch off) before the teacher checks them. Avoid short-circuiting the battery — ammeters must always be connected in series, voltmeters in parallel. Do not leave the circuit energised for long periods as nichrome wire may become hot. Students must not touch the nichrome wire while current flows. Report any sparks or damaged insulation to the teacher immediately.

B. LESSON OVERVIEW

Students begin by predicting the relationship between voltage and current using their prior knowledge from Lessons 1 and 2. They then design and carry out a controlled experiment — connecting a nichrome wire in a circuit with a rheostat, ammeter, voltmeter, and battery — varying voltage with the rheostat and recording corresponding current values. Results are tabulated, a V–I graph is plotted, and the gradient is used to calculate resistance, formally verifying Ohm's Law. The lesson anchors every step back to the Huruma Estate phenomenon: the nichrome wire acts as a model for the resistance of wires, bulb filaments, and kettle elements in a home circuit. The Driving Question Board (DQB) is updated to record resistance as the factor that controls how much current a given voltage can drive — and therefore how much energy a device receives. The lesson advances the class's growing explanatory model of the phenomenon.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy

Predict Phase 📹 **VIDEO:** Video: Ohm's Law Source: PhET Interactive Simulations (English) 🔍 Search ARES for similar videos 📖 **HTML:** Ohm's Law: Looking at a Circuit (PLIX) Source: CK-12 🔍 Search ARES for similar readings	Students are shown a simple diagram of two circuits side by side: Circuit A has a thick copper wire connecting a bulb to a 6V battery; Circuit B has a long, thin nichrome wire connecting an identical bulb to the same 6V battery. Without any instruction, students write individual predictions in their notebooks answering three questions: (1) In which circuit will the bulb be brighter? (2) What do you think is different about the two wires that could affect the bulb? (3) How does this connect to the dimmer bulb on the long extension cord in Huruma Estate? Students then share predictions in pairs (Think-Pair-Share) before three cold-call responses are taken to the class.	Display the two circuit diagrams on the board (or ARES screen). Say: 'Before we do any experiment today, I want you to think carefully about what you already know from our last two lessons. Look at these two circuits. Write your three predictions — you have 4 minutes, working alone.' [Allow 4 minutes of silent individual writing.] Then say: 'Now turn to your partner and compare your predictions. Do you agree? Where do you disagree? You have 2 minutes.' [Allow 2 minutes pair discussion.] Cold-call three pairs — choose one pair who agreed, one who disagreed, and one who is unsure. For each, ask: 'What was your reasoning — what evidence from our previous lessons supports your prediction?' [WAIT TIME: 12 seconds after each cold call before prompting.] Record key prediction words on the board (e.g., 'thick vs thin', 'longer wire', 'more opposition', 'less current'). Do NOT correct predictions yet — say: 'We are going to test these ideas right now. Keep your predictions — we will come back to them.'	Think-Pair-Share with Anchored Prediction: By asking students to connect their predictions explicitly to the Huruma Estate phenomenon before any instruction, the teacher activates prior knowledge and creates cognitive dissonance — students must confront the gap between 'electricity works or it doesn't' and the more nuanced reality they are about to discover. Recording predictions publicly on the board creates accountability and makes the upcoming experiment feel purposeful.	Observable indicator: Students' written predictions show at least one variable (wire thickness, wire length, or material) being used as a reasoning tool. Teacher circulates during individual writing time and notes which students are connecting to the phenomenon (strong) vs. writing generic guesses (needs support). Misconception to watch for: students who write 'Circuit B bulb will not light at all' — flag these for targeted questioning during the experiment phase.
Observe Phase 📹 **VIDEO:** Video: Ohm's Law Source: PhET Interactive Simulations (English) 🔍 Search ARES for similar videos 📖 **HTML:** Ohm's Law: Looking at a Circuit (PLIX) Source: CK-12	In groups of four, students design and build a circuit comprising: a 6V battery pack, a rheostat (variable resistor), a nichrome wire of fixed length as the test resistor, an ammeter in series, a voltmeter in parallel across the nichrome wire, and a switch. Groups set the rheostat to its maximum resistance (lowest voltage across nichrome wire), close the switch, and record the voltmeter reading (V) and ammeter reading (I). They adjust the rheostat through five or six settings, recording a V–I data table each time. Students then plot a V–	Distribute equipment and a structured lab worksheet (ARES printed sheet or screen). Say: 'Your task is to find the mathematical relationship between voltage and current for this nichrome wire — the same kind of resistive wire used inside an electric kettle element. I want you to think about: what do you keep the same, what do you change, and what do you measure.' [WAIT TIME: 10 seconds.] Cold-call one group to state their variables before they begin — ask: 'What is your independent variable? Your	Structured Inquiry with Guided Graph Analysis: The experiment is structured (students follow a scaffold) but the reasoning is open — students must interpret what their graph shape means. By asking 'does a straight line through the origin fit?' the teacher is directing attention to proportionality, which is the conceptual core of Ohm's Law. The gradient calculation makes resistance a measured, concrete quantity rather than an abstract term — students literally calculate it from their own data. Connecting	Observable indicators: (1) All groups produce a data table with at least 5 V–I pairs showing increasing V with increasing I. (2) Graphs show a straight line through (or near) the origin. (3) Gradient is calculated with correct units (V/A = Ω). Misconception to watch for: groups who draw a curve or a line that does not pass through the origin — ask them to recheck their ammeter/voltmeter connections and zero readings. Groups whose gradient is very far from the expected value should be asked to check for loose

🔍 Search ARES for similar readings	I graph (V on y-axis, I on x-axis) on graph paper, draw a best-fit straight line, and calculate the gradient. They record the gradient value and label it. Finally, each group compares their experimental gradient to the accepted resistance of their nichrome wire sample (provided by teacher on a label).	dependent variable? What are you controlling?' Before groups energise circuits, circulate and check each connection — say: 'Switch must be OPEN while I check. Ammeter in series — show me. Voltmeter in parallel — show me.' Only approve circuits that are correctly wired. During data collection, prompt groups who have only 2–3 data points: 'You need at least 5 readings to draw a reliable graph — adjust your rheostat further.' When groups reach the graphing stage, say: 'Plot your points, then ask yourself: does a straight line through the origin fit your data? What does a straight line through the origin tell us about the relationship between V and I?' [WAIT TIME: 12 seconds.] For gradient calculation, prompt: 'Choose two points far apart on your best-fit line — not data points — and calculate rise over run. What are the units of your gradient?' Cold-call two groups to share their gradient values and compare them.	the nichrome wire to a kettle element ties the lab directly back to the slow-boiling kettle in Huruma Estate.	connections or whether they accidentally included the rheostat in their voltmeter measurement.
Explain Phase 📹 **VIDEO:** Video: Ohm's Law Source: PhET Interactive Simulations (English) 🔍 Search ARES for similar videos 📖 **HTML:** Ohm's Law: Looking at a Circuit (PLIX) Source: CK-12 🔍 Search ARES for similar readings	Groups share their gradients. The class computes an average resistance from all groups' results. The teacher formally introduces Ohm's Law: V = IR, deriving it from the gradient of their graphs. Students use V = IR to answer three application problems contextualised in Kenya: (1) A Kenya Power engineer measures 12 V across a stretch of copper distribution wire carrying 3 A — what is the resistance of that wire? (2) The nichrome heating element of a kettle in a Huruma Estate home has a resistance of 24 Ω. If the mains voltage is 240 V, what current flows? (3) A student	Write the class's gradient values on the board as groups report. Say: 'Look at all our results — what do you notice? Are they the same? What could explain any differences?' [WAIT TIME: 12 seconds. Cold-call 2 students.] Then say: 'The gradient of your V–I graph is a property of the wire itself. We give it a name: resistance, symbol R, unit ohm, symbol Ω. The relationship V = IR is called Ohm's Law, and it was verified by Georg Ohm in 1827 — but YOU have now verified it yourselves in this classroom, just as Kenyan engineers verify it every time they design a	Data Consolidation followed by Transfer Problem Solving: Pooling all groups' gradients normalises experimental error and gives students a class-wide result — building the idea that science is a collective enterprise. The three transfer problems are carefully sequenced: problem 1 is direct substitution, problem 2 introduces a real-world scale (240 V mains), and problem 3 requires comparative reasoning about current and brightness — directly echoing the Huruma Estate phenomenon. This scaffolded transfer ensures students are not just memorising V = IR but building	Observable indicators: (1) Students correctly rearrange V = IR to find R and I in problems 1 and 2. (2) In problem 3, students explicitly state 'the 10 Ω bulb has more current and is therefore brighter' and link this to the phenomenon. (3) During prediction review, at least 70% of students can identify that resistance was the missing variable in their original thinking. Misconception to watch for: students who say 'more resistance means more voltage' — address immediately by returning to the graph: 'If R is bigger, and V stays the same in V = IR, what must happen to I?'

	connects a bulb of resistance 40 Ω to a 6 V battery — what current flows, and how does this compare to connecting a 10 Ω bulb to the same battery? Which bulb will be brighter, and why? Students write answers individually, then discuss in groups. The class closes by returning to the original predictions on the board — teacher asks students to evaluate their own predictions.	distribution network.' Write V = IR prominently. Then say: 'Now I want you to use this law to solve three problems — these are real situations. Work individually for 6 minutes, then check with your group.' Circulate and observe written work. For problem 3, prompt: 'If resistance goes up but voltage stays the same, what happens to current? And if current goes down, what happens to the brightness — think about Huruma Estate.' [WAIT TIME: 10 seconds.] Cold-call three different students for the three answers. For the prediction review, point to the board and say: 'Look at your original predictions. Hands up — who was correct about which bulb would be brighter? What was the key idea you were missing before the experiment?'	a causal explanatory model.	
Driving Question Board (DQB) Creation 📹 **VIDEO:** [Video: Ohm's Law] Source: PhET Interactive Simulations (English) 🔍 [Search ARES for similar videos] 📖 **HTML:** [Ohm's Law: Looking at a Circuit (PLIX)] Source: CK-12 🔍 [Search ARES for similar readings]	Each student writes two sticky notes (or entries in their ARES DQB log): (1) one new piece of evidence they have gathered today that helps answer the driving question ('Why do electrical devices sometimes receive less energy than expected?'), and (2) one new question that today's lesson has raised for them. Groups post their notes on the class DQB under the column headers 'What we now know' and 'New questions we have'. The teacher reads selected entries aloud and organises them into emerging themes: resistance of wires, resistance of components, what controls current, and unanswered questions about why Huruma Estate lights recover after a few minutes.	Say: 'Take 3 minutes — write your two sticky notes. For the evidence note, start with the sentence stem: "Now I know that resistance…" For the question note, start with: "But I still wonder…"' [Allow 3 minutes silent writing.] As students post notes, quickly scan them. Read three 'evidence' notes and two 'question' notes aloud. Say: 'I am seeing a pattern — many of you now know that resistance controls how much current flows for a given voltage. And several of you are asking: where does resistance come from in a real circuit? Is it just the wire? What about the battery itself? Hold those questions — they are exactly where we are going next.' Circle any DQB notes that mention 'battery' or 'source' or 'why lights recover' — say: 'These questions are gold. We will come back to	Driving Question Board as a Living Explanation Tracker: The DQB makes the class's growing model visible and cumulative. By requiring students to write both evidence AND new questions, the teacher communicates that science advances by answering questions and generating new ones. Publicly linking today's 'new questions' to future lessons gives students a sense of narrative continuity in the inquiry — they are building toward a complete explanation of the Huruma Estate phenomenon, not just collecting isolated facts.	Observable indicators: (1) Evidence notes correctly and specifically name resistance (not just 'the wire') as the controlling variable. (2) Question notes show genuine intellectual curiosity and connection to the phenomenon — e.g., 'Why do the lights get brighter after a few minutes — does the resistance of something change?' rather than generic questions. (3) Teacher notes any student whose evidence sticky note still reflects a misconception for follow-up at the start of Lesson 4.

		them in Lesson 5.'		
Model Building Phase 📹 **VIDEO:** Video: Ohm's Law Source: PhET Interactive Simulations (English) 🔍 Search ARES for similar videos 📖 **HTML:** Ohm's Law: Looking at a Circuit (PLIX) Source: CK-12 🔍 Search ARES for similar readings	Students return to the explanatory model they began in Lesson 1 (a labelled diagram of the Huruma Estate home circuit). They now add a new layer: using the symbol for a resistor (rectangle or zigzag), they label the extension cord wire, the bulb filament, and the kettle element as resistors with resistance values. They annotate their model with the equation V = IR and draw arrows showing that for the same V, a larger R means a smaller I arrow (thinner arrow) reaching the bulb or kettle — producing less brightness or slower heating. Students write one sentence on their model: 'The long thin extension cord has a higher resistance than a short thick wire, so for the same mains voltage, less current reaches the bulb, making it dimmer.'	Say: 'Open your model from Lesson 1. You are going to upgrade it with what you learned today. Add resistor symbols to at least three components — the extension cord, the bulb filament, and the kettle element. Annotate with V = IR. Use arrow thickness to show current size — a bigger arrow means more current. You have 7 minutes.' Circulate and prompt: 'Where in your model does the resistance of the extension cord appear? How does that resistance change the current reaching the bulb?' [WAIT TIME: 10 seconds.] For students who finish early: 'Can you annotate your model with an actual number? Use the resistance you calculated today and a voltage of 240 V to calculate the current — add that number to your arrow.' Close by cold-calling two students to hold up their models and explain one annotation to the class. Say: 'By the end of this unit, this model will be complete enough to fully explain everything that happened in Huruma Estate on that evening. Today you added one of the most important pieces.'	Cumulative Annotated Model Revision: The model is not redrawn each lesson but progressively annotated — this mirrors how scientific models are built iteratively over time. The use of arrow thickness as a visual metaphor for current magnitude makes the abstract relationship between R and I perceptible. Requiring a specific sentence about the extension cord forces students to connect the symbolic model (V = IR) back to the anchoring phenomenon in plain language, bridging formal and intuitive understanding.	Observable indicators: (1) Resistor symbols appear in correct locations (in series with the component they represent). (2) Annotations correctly show that a larger R is associated with a smaller I arrow for the same V. (3) The written sentence correctly identifies the extension cord's resistance as the cause of reduced current and dimmer light — not 'less electricity' or 'weaker battery'. Teacher collects models at the end of class for a quick review before Lesson 4.

D. TEACHER REFLECTION

After the lesson, reflect on the following: (1) GRAPH QUALITY — Did most groups produce V–I graphs with a clear linear trend through the origin? If not, what connection or measurement errors occurred, and how will you address circuit-building skills before Lesson 4? (2) MISCONCEPTION TRACKING — Did any students persistently believe that 'more resistance means more voltage' rather than less current? Note their names and plan a targeted question at the start of Lesson 4 using V = IR as a self-correction tool. (3) PHENOMENON CONNECTION — Were students able to articulate, in their own words, why the bulb on the long thin extension cord is dimmer? If fewer than 70% could do so by the end of the explain phase, consider opening Lesson 4 with a 5-minute revisit using a worked example with the extension cord's resistance explicitly calculated. (4) DQB QUALITY — Did the 'new questions' on the DQB move toward internal resistance and power? If students are not yet asking about why the lights recover after a few minutes, seed this question explicitly at the start of Lesson 4. (5) TIMING — The experiment phase is the most time-pressured. If groups did not finish graphing in class, assign graph completion as homework with the gradient calculation due at the start of Lesson 4 — do not proceed to Ohm's Law application problems without all groups having their own experimental evidence in hand.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	Students observed that as the rheostat was adjusted to increase the voltage across the nichrome wire, the current through the wire also increased. When they plotted these values on a V–I graph, the points fell on a straight line passing through (or near) the origin — showing that voltage and current are directly proportional for the nichrome wire at constant temperature.
What did I learn?	Students learned that the gradient of a V–I graph gives the resistance (R) of the conductor, measured in ohms (Ω). This relationship is summarised by Ohm's Law: V = IR. Resistance is a property of the conductor that controls how much current flows for a given voltage — a higher resistance means less current for the same voltage.
How does this explain the phenomenon?	Students can now partially explain the Huruma Estate phenomenon: the long, thin extension cord has a higher resistance than a short, thick wire. Because V = IR, for the same mains voltage, a higher resistance in the cord means less current reaches the bulb — so it glows dimmer. The kettle element's resistance similarly controls how much current heats the water. Resistance is the missing variable that explains why 'electricity' can vary in its effect even from the same source.

LESSON 4 (80 minutes): Resistance and Resistivity: Why Do Thicker Cables Carry More Current?	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To investigate the factors that affect the resistance of a conductor and derive the resistivity formula, connecting findings to the Huruma Estate flickering lights phenomenon and the hazard of thin extension cords.
Knowledge	By the end of the lesson, the learner should be able to: (a) state the factors affecting resistance of a conductor (length, cross-sectional area, material/resistivity); (b) derive and apply the resistivity formula $\rho = RA/L$; (c) explain why thicker cables have lower resistance and are safer for home wiring in Kenya.
Skills	By the end of the lesson, the learner should be able to: (a) plan and carry out a fair investigation into the factors affecting resistance using wires of different gauges and lengths; (b) measure and record current and voltage to calculate resistance using Ohm's Law ($R = V/I$); (c) construct and interpret tables and graphs of resistance vs. length, and resistance vs. cross-sectional area; (d) use the resistivity formula $\rho = RA/L$ to solve quantitative problems.
Attitudes	The learner appreciates how scientific understanding of resistivity directly informs safe and efficient electrical installations in Kenyan homes and communities, and values careful, systematic investigation as a basis for sound engineering decisions.
Key Inquiry Question	What is the relationship between voltage, current, and resistance in an electric circuit?
Purpose in Storyline	In Lessons 1–3, students established that current flows, voltage drives it, and Ohm's Law links the two. Yet the neighbour's long, thin extension cord in Huruma Estate remains unexplained — why does that specific wire always produce a dim bulb? This lesson fills the gap: students discover that resistance is NOT just a property of the bulb but depends on the physical dimensions and material of every wire in the circuit. This evidence directly advances the class model by showing that the thin, long extension cord adds significant resistance, robbing the bulb of voltage and reducing current — explaining the persistent dimness even after full power is restored.
Safety Notes	Use low-voltage DC power supplies (maximum 6 V) only. Ensure all connecting wires have intact insulation. Do not allow wires to short-circuit unattended — this can cause rapid heating. Students should not touch bare wire ends when the circuit is live. Remind students that the real-world danger of thin extension cords is overheating — reinforce this as a fire-safety issue relevant to Kenyan homes.

B. LESSON OVERVIEW
Students begin by predicting which wire properties make the neighbour's extension cord dim the bulb in the Huruma Estate phenomenon. They then conduct a structured investigation — varying wire length, cross-sectional area (gauge), and material while measuring resistance via V and I readings — gathering evidence to build a quantitative model. In the Explain phase, the class derives $\rho = RA/L$ and applies it to calculate resistivities of real wiring materials. The DQB is updated to reflect that resistance is distributed throughout a circuit, not just in the bulb. Students revise their cumulative circuit model to show resistance contributions from wires, connecting all evidence to the driving question about energy delivery in Kenyan homes.

C. LESSON IMPLEMENTATION FRAMEWORK				
Phase	**Learner Experience**	**Teacher Moves**	**Sensemaking Strategy**	**Formative Assessment Strategy**
Predict Phase	Students are shown a photograph	Display the two extension cord	Prediction Wall — students' written	Teacher circulates during

📹 **VIDEO:** [Video: Resistance in a Wire] Source: PhET Interactive Simulations (English) 🔍 Search ARES for similar videos 📖 **HTML:** [Resistance in Parallel Circuits Vs. Series Circuits (PLIX)] Source: CK-12 🔍 Search ARES for similar readings	of two extension cords side by side: one thin (0.5 mm² gauge, like a cheap mkokoteni-market cord sold in Gikomba) and one thick (2.5 mm² gauge, like standard Kenya Power-approved house wiring). Both are the same length. They are asked to individually write answers to three questions in their science notebooks: (1) Which cord do you predict will make a bulb glow brighter? Why? (2) What property of the cord — its length, its thickness, or the material it is made of — do you think matters most for how much energy the bulb receives? (3) In the Huruma Estate story, the neighbour's extension cord is BOTH long AND thin. Predict whether changing only the length or only the thickness would have a bigger effect on bulb brightness. After 3 minutes of individual writing, students share predictions in pair-talk (1 minute), then the teacher cold-calls 4 pairs to voice their predictions, which are recorded on the Prediction Wall without judgment.	photograph on the projector. Say: 'Before we touch any equipment today, I want your raw thinking — there are no wrong answers at this stage, only honest predictions.' Allow 3 minutes of silent individual writing. Then say: 'Turn to your partner — 60 seconds — convince each other which factor matters most and why.' After pair-talk, cold-call exactly 4 pairs using the random name-stick method. For each response, use the revoicing move: 'So you're saying that [student's words] — class, hands up if your prediction was similar.' Tally agreements on the board. Do NOT confirm or deny any prediction. Record all four claims on the class Prediction Wall under the heading: 'We predict that resistance is affected by…' Allow 15 seconds of wait time after each cold-call before accepting a response. Close with: 'We have competing predictions. Today's investigation will help us decide which of these the evidence supports.'	and spoken predictions are publicly displayed and later revisited in the Explain phase. This strategy creates cognitive dissonance when observations contradict intuition, motivating deeper engagement with the evidence. It also makes prior conceptions visible to the teacher for formative purposes.	individual writing and reads 5–6 notebooks to identify dominant misconceptions (e.g., 'only the bulb has resistance', 'thicker wire has MORE resistance because there is more material'). These specific misconceptions are noted on a sticky note for targeted addressing in the Explain phase. Observable indicator: every student has written at least one reason for their prediction, not just a choice.
Observe Phase 📹 **VIDEO:** [Video: Resistance in a Wire] Source: PhET Interactive Simulations (English) 🔍 Search ARES for similar videos 📖 **HTML:** [Resistance in Parallel Circuits Vs. Series Circuits (PLIX)]	Students work in groups of four to conduct THREE mini-investigations using simple circuits (battery pack/DC supply at 4 V, ammeter, voltmeter, connecting leads, and sets of labelled wire samples prepared by the teacher). Each group receives a set of wire samples: (A) nichrome wires of the same gauge but different lengths (10 cm, 20 cm, 30 cm, 40 cm); (B) nichrome wires of the same length (20 cm) but different cross-sectional areas (0.2 mm², 0.5 mm², 1.0 mm²); (C) wires of the same length and gauge but different materials (nichrome, copper, and	Before students begin, demonstrate circuit setup for one wire sample (30 seconds). Say: 'Your job as scientists today is to change ONE thing at a time and measure what happens to resistance. What must you keep the same in investigation A?' (Cold-call 2 students — expect: voltage, material, thickness.) Circulate during data collection; every 4 minutes check one group's ammeter/voltmeter readings and ask: 'Does your R value make sense — is it bigger or smaller than the previous sample? Why do you expect that?' During graph	Data Table + Graph Gallery — groups collect quantitative data independently and then compare graphs across the class. This 'evidence convergence' strategy shows students that multiple groups, using different wire samples, arrive at the same pattern — building confidence that the relationship is real and not a measurement fluke. The gallery comparison also builds scientific communication skills (NGSS Practice 4: Analysing and Interpreting Data; Practice 6: Constructing Explanations).	Teacher checks each group's data table for: (a) correct calculation of R = V/I for each row; (b) correct labelling of axes and units on graphs; (c) a trend that is consistent (R increasing with L, R decreasing with A). Groups whose R values are inconsistent (e.g., due to poor contact) are prompted to re-check connections. Observable indicator: every group can state in one sentence the trend they observed for at least two of the three investigations before moving to the Explain phase.

Source: CK-12 🔍 Search ARES for similar readings	resistance wire/kanthal if available). For each sample, students connect it in a simple series circuit, record the voltmeter reading (V) across the wire and ammeter reading (I) through it, and calculate $R = V/I$. They record all data in a structured table in their notebooks. After data collection, each group plots two graphs on graph paper: R vs. length (for set A) and R vs. cross-sectional area (for set B). Groups then share graphs on the class display rail for a 'gallery comparison.' Students note: (i) R increases with length — the graph is linear; (ii) R decreases as cross-sectional area increases — the graph is a curve (inverse relationship); (iii) different materials give different R values for identical dimensions. Teacher connects each finding back to the phenomenon by asking: 'Which of these findings explains why the neighbour's thin, long extension cord in Huruma dims the bulb?'	plotting, prompt struggling groups with: 'Put length on the x-axis. What does a straight line through the origin tell you about the relationship?' After gallery walk (3 minutes), ask the whole class: 'Which of your three predictions from the Prediction Wall does the evidence support?' Allow 15 seconds wait time. Cold-call 3 students. For the cross-sectional area graph, say: 'Your graph is a curve, not a straight line — what does that tell you about the relationship between R and area?' Introduce the idea of plotting R vs. 1/A for homework or as a class extension to linearise the relationship.		
Explain Phase 📹 **VIDEO:** Video: Resistance in a Wire Source: PhET Interactive Simulations (English) 🔍 Search ARES for similar videos 📖 **HTML:** Resistance in Parallel Circuits Vs. Series Circuits (PLIX) Source: CK-12 🔍 Search ARES for similar readings	The teacher leads the class through a structured derivation of the resistivity formula. Starting from the two observed proportionalities ($R \propto L$ and $R \propto 1/A$), students combine them: $R \propto L/A$, and introduce the constant of proportionality ρ (rho), giving $R = \rho L/A$, or equivalently $\rho = RA/L$. Students learn the units: ρ is measured in ohm-metres (Ω·m). The teacher presents a data table of resistivities for real materials (copper ≈ 1.7×10^{-8} Ω·m; nichrome ≈ 1.1×10^{-6} Ω·m; rubber ≈ 10^{13} Ω·m) and asks students to rank them as conductors or insulators. Students then work in pairs on two application problems rooted in the Kenyan context: (1)	Write $R \propto L$ and $R \propto 1/A$ on the board from students' own graphs. Say: 'We have two proportionalities from YOUR data. Mathematically, how can we combine them into one equation?' Allow 15 seconds wait time. Cold-call 2 students. Guide the class to $R \propto L/A$ and introduce ρ as the 'material's own fingerprint for resistance.' Write: $\rho = RA/L$ and say: 'Rearrange this to give R — write it in your notebooks.' After pair problem-solving (5 minutes), cold-call one pair for each problem and record their working on the board. Ask the class: 'Does this answer match your prediction? If not, what did you not account for?' Explicitly return to the	Algebraic Synthesis from Student Data — students see the formula emerge from THEIR own graphs rather than being given it from a textbook. This approach (NGSS Practice 6: Constructing Explanations; Practice 5: Using Mathematics) ensures the formula is understood as a summary of evidence rather than a rule to memorise. The Prediction Wall revisit closes the cognitive loop opened in Phase 1, reinforcing metacognitive awareness of how science revises ideas.	Teacher marks pair-work calculations during the 5-minute work period. Specific indicator: students correctly convert mm² to m² ($\times 10^{-6}$) when substituting into $\rho = RA/L$ — this is a common error and the teacher notes how many groups make it. A correct answer for problem (1) is approximately $R = (1.7\times10^{-8} \times 12) / (2.5\times10^{-6}) \approx 0.082$ Ω; problem (2) gives ≈ 0.41 Ω — about 5× greater. Observable indicator: students can state in words why the Gikomba cord dims the bulb, using the terms 'resistance,' 'cross-sectional area,' and 'length' correctly.

	'A Kenya Power-approved copper house wire has cross-sectional area 2.5 mm² and length 12 m. Calculate its resistance.' (2) 'A cheap extension cord sold at Gikomba market uses wire of cross-sectional area 0.5 mm² and the same 12 m length. How much greater is its resistance?' Students compare answers, then discuss: 'How does this difference in resistance explain why one cord dims the bulb and gets hot while the other does not?' The class revisits the Prediction Wall and identifies which predictions were correct, which were partially correct, and which were refuted by evidence — using precise language from the formula.	phenomenon: 'So in Huruma Estate, the neighbour's cord has low area AND long length — what does our formula predict will happen to R?' Cold-call 3 students. Close with: 'Every wire in your house is a resistor. The formula we just derived is used by Kenya Power engineers every time they design wiring for a new estate.'		
Driving Question Board (DQB) Creation 📹 **VIDEO:** [Video: Resistance in a Wire] Source: PhET Interactive Simulations (English) 🔍 [Search ARES for similar videos] 📖 **HTML:** [Resistance in Parallel Circuits Vs. Series Circuits (PLIX)] Source: CK-12 🔍 [Search ARES for similar readings]	Each student receives two sticky notes. On the first (GREEN), they write one claim they can now make with evidence to answer the driving question: 'Why do electrical devices in our homes sometimes receive less energy than expected?' They must include the word 'resistance,' 'length,' 'area,' or 'resistivity.' On the second (ORANGE), they write one new question that has emerged — for example: 'If all wires have resistance, does the resistance of the house wiring itself affect how much energy a kettle or fan receives?' or 'Does temperature affect resistivity?' Students place their sticky notes on the class DQB. The teacher reads three green and two orange notes aloud and clusters them visually. The class notes that the DQB is growing: earlier lessons added current, voltage, and Ohm's Law; today's lesson adds that resistance is DISTRIBUTED across all wires	Distribute sticky notes and say: 'You have 3 minutes — write your green claim first, then your orange question. Green must connect to the Huruma phenomenon specifically.' After posting, read three green notes aloud using the student's own words, then ask: 'Class — does this claim match our evidence today? Thumbs up, sideways, or down.' For orange questions, say: 'These are brilliant — notice that your questions are getting more precise. In Lesson 1, we asked why the lights dimmed at all. Now you're asking HOW MUCH the wire's resistance affects the appliance — that is a much deeper question and it shows you are thinking like physicists.' Update the DQB visual cluster map on the board with labels: 'Lesson 4 Evidence: Resistance depends on L, A, and ρ.'	Claim–Evidence–Reasoning (CER) on the DQB — students must formulate a claim supported by today's evidence, not just restate a fact. The public display creates a living record of the class's growing understanding, making the storyline progression visible. Clustering orange questions around themes prepares students for upcoming lessons and gives the teacher information for planning.	Teacher reads all green sticky notes after class to assess: (a) Is the claim specific (mentions at least one factor)? (b) Does it connect to the phenomenon? (c) Is it expressed as a causal statement ('because…') rather than just a definition? Observable indicator: at least 80% of green notes include a causal link to the phenomenon (e.g., 'The thin extension cord dims the bulb because its small cross-sectional area gives it high resistance, reducing the current to the bulb').

	in the circuit, not just the appliance. The teacher previews: 'Your orange questions about temperature and about wire resistance affecting appliances will come back in Lessons 5 and 6 when we look at resistors in circuits and electrical power.'			
Model Building Phase 📹 **VIDEO:** Video: Resistance in a Wire Source: PhET Interactive Simulations (English) 🔍 Search ARES for similar videos 📖 **HTML:** Resistance in Parallel Circuits Vs. Series Circuits (PLIX) Source: CK-12 🔍 Search ARES for similar readings	Students retrieve their cumulative circuit diagram model from Lesson 3 (which showed a battery, connecting wires, and a bulb, with arrows for current and labels for voltage and resistance of the bulb). Today they revise this model to add resistance to the wires themselves. Each student draws an updated model that now shows: (a) the battery (with EMF and internal resistance — previewing Lesson 5); (b) the copper house wires represented as small resistors with a low value (labelled $R_{wire} \approx$ small); (c) the bulb as a larger resistor (R_{bulb}); (d) the neighbour's thin extension cord as a larger wire-resistor ($R_{cord} \approx$ large). They annotate: 'Total resistance = $R_{cord} + R_{bulb}$, so more R_{cord} means less current, less voltage across bulb, dimmer light.' Students compare their revised model with their Lesson 3 model and write one sentence: 'My model changed today because I now know that…' Selected students share their revised models under the document camera. The class evaluates: 'Does this model explain ALL of our observations about the Huruma Estate phenomenon so far? What is still missing?'	Say: 'Open your model from Lesson 3. Your job in the next 7 minutes is to make it more accurate using today's evidence. Every wire in your diagram must now have a resistance label.' Circulate and prompt: 'Is the cord resistance bigger or smaller than the bulb resistance in the neighbour's setup? Show that in your diagram.' After individual work, select 2 models to display under the document camera — choose one strong and one that still shows wires as having zero resistance. Ask the class: 'Compare these two models — which is more consistent with our evidence today, and why?' Cold-call 3 students. Close with: 'Your model is now more powerful — it can explain why the thin cord dims the bulb even when power is fully restored. But something is still missing: what about when the power first comes back on and everything is dim — even direct-to-wall appliances? That mystery will need Lesson 5.' Allow 15 seconds for students to write their 'My model changed because…' sentence before packing up.	Iterative Model Revision — students maintain a single evolving model across all 12 lessons, annotating changes lesson by lesson. This NGSS Practice 2 (Developing and Using Models) strategy makes scientific knowledge-building visible as a cumulative process. The 'what is still missing?' prompt ensures models are treated as work-in-progress, not finished answers — preserving the inquiry motivation for upcoming lessons.	Teacher collects (or photographs) 6–8 models at the end of the lesson. Observable indicator: the revised model correctly shows the extension cord wire as a resistor in series with the bulb, with an annotation that higher cord resistance reduces current and dims the bulb. Models that still show wires as perfect conductors (zero resistance) are flagged for individual follow-up at the start of Lesson 5.

D. TEACHER REFLECTION

After the lesson, reflect on the following questions: (1) Did all groups successfully calculate resistance from V and I readings, or did some groups have persistent measurement errors — and if so, was this due to poor connections, incorrect ammeter/voltmeter placement, or conceptual confusion? (2) How many students on the DQB green notes wrote a genuine causal claim connecting resistivity to the phenomenon, versus restating a definition? What does this tell me about the depth of sensemaking achieved? (3) Were students able to correctly handle unit conversion (mm^2 to m^2) in the $\rho = RA/L$ calculations — if not, plan a 5-minute unit conversion warm-up at the start of Lesson 5. (4) Did the Prediction Wall revisit produce genuine surprise or cognitive shift for students who had predicted 'only the bulb has resistance'? If many students still hold this misconception, build an explicit targeted discussion into the Lesson 5 opening. (5) How effectively did the model revision connect to the unresolved question about dimming at power restoration? Is the class ready to engage with internal resistance in Lesson 5, or do they need consolidation of today's content first?

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	Students observed that when wire length increases (with area and material fixed), resistance increases proportionally. When cross-sectional area increases (with length and material fixed), resistance decreases. Different materials (copper vs. nichrome) gave different resistance values for identical wire dimensions. A thin, long wire had significantly higher resistance than a short, thick wire of the same material.
What did I learn?	Students learned that resistance depends on three factors: length ($R \propto L$), cross-sectional area ($R \propto 1/A$), and material (characterised by resistivity ρ). These are combined in the formula $\rho = RA/L$ (units: $\Omega \cdot m$). Copper, used in Kenya Power-approved house wiring, has very low resistivity, making it an efficient conductor. Thin, cheap extension cords have much higher resistance due to their small cross-sectional area, which reduces current to appliances and causes energy to be wasted as heat in the cord.
How does this explain the phenomenon?	Students explained that the neighbour's long, thin extension cord in Huruma Estate has high resistance because of its small cross-sectional area and long length (both increase R according to $\rho = RA/L$). This high cord resistance means the total circuit resistance is large, reducing the current. Less current means less power delivered to the bulb ($P = I^2R$), making it glow dimly even when the mains voltage is fully restored. This is why Kenya Power standards require thick-gauge wiring in homes — to minimise wire resistance and deliver maximum energy to appliances safely.

LESSON 5 (80 minutes): Series Circuits: Same Current, Sharing Voltage — Why More Bulbs Mean Dimmer Light	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To investigate the behaviour of current and voltage in series circuits and derive the rules governing series circuit configurations.
Knowledge	Students will know that in a series circuit: (1) the current is the same at every point; (2) the total voltage is the sum of individual voltage drops; (3) the total resistance is the sum of individual resistances ($R\_total = R_1 + R_2 + \ldots + R_n$); and (4) adding more components increases total resistance and reduces current, causing each component to receive less power.
Skills	Students will be able to: (a) build a series circuit with multiple bulbs or resistors using connecting wires, a battery holder, and switches; (b) measure current at multiple points in a series circuit using an ammeter; (c) measure voltage across individual components and the source using a voltmeter; (d) record, tabulate, and analyse data to derive series circuit rules; (e) construct and annotate a circuit diagram using standard symbols.
Attitudes	Students will appreciate the importance of systematic evidence gathering in deriving physical laws, and will value careful, safe handling of electrical equipment. They will show curiosity about how the same phenomenon (dim lights in Huruma Estate) can be explained by a growing set of interconnected physical rules.
Key Inquiry Question	What is the relationship between voltage, current, and resistance in an electric circuit?
Purpose in Storyline	In Lessons 1–4, students established that voltage drives current, resistance opposes it, and Ohm's Law links the three quantities for a single resistor. But the Huruma Estate phenomenon involves multiple devices sharing the same circuit. Lesson 5 introduces the series circuit as the first multi-component configuration, giving students the evidence they need to explain why adding more bulbs on a single loop — like Christmas-tree lights — causes all bulbs to dim, and why the neighbour's long, thin extension cord (acting as an extra series resistor) always makes the connected bulb dimmer. This is a critical evidence-gathering step before students encounter parallel circuits in Lesson 6.
Safety Notes	Use 1.5 V AA batteries or a low-voltage DC power supply (maximum 6 V) only. Never connect an ammeter directly across a voltage source — it must be connected in series. Never connect a voltmeter in series — it must be connected in parallel across the component. Ensure all wires are insulated and that no bare conductors touch simultaneously. Students must switch off (open) the circuit before rewiring. Inspect all crocodile clips and connecting wires for frayed insulation before the lesson. Teacher should demonstrate safe circuit-building procedure before students begin practical work.

B. LESSON OVERVIEW

Students open the lesson by revisiting the Huruma Estate phenomenon and the Christmas-lights supporting phenomenon to make a new prediction: what will happen to bulb brightness as more bulbs are added to a single loop? They then build series circuits with 1, 2, and 3 bulbs, systematically measuring current at multiple points and voltage across each bulb and the source. Data is tabulated and analysed to derive the three cardinal rules of series circuits: (1) current is constant throughout, (2) voltages across components add up to the source voltage, (3) resistances add. Students use these rules to construct an evidence-based explanation for why Christmas-light bulbs dim when more are added and why a long extension cord dims a bulb. The lesson closes with a Driving Question Board update and a cumulative circuit model revision that now explicitly shows current, voltage, and resistance in a multi-component series loop.

C. LESSON IMPLEMENTATION FRAMEWORK				
Phase	**Learner Experience**	**Teacher Moves**	**Sensemaking Strategy**	**Formative Assessment Strategy**
Predict Phase 📹 **VIDEO:** Theoretical probability distribution example: tables Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Power Series: Introduction and Convergence (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Students receive a printed card showing two circuit diagrams side by side: Circuit A has one bulb connected to a 3 V battery; Circuit B has three identical bulbs connected in a single loop (series) to the same 3 V battery. Students individually write predictions in their science notebooks, answering three questions: (1) In Circuit B, will the current flowing through Bulb 1 be more than, less than, or the same as the current through Bulb 3? (2) Will each bulb in Circuit B be brighter than, dimmer than, or the same brightness as the bulb in Circuit A? (3) If you unscrewed one bulb from Circuit B, what would happen to the other two? Students then share predictions with a shoulder partner (Think-Pair-Share). Predictions are recorded on sticky notes and posted on the class Prediction Wall under three columns: MORE CURRENT / LESS CURRENT / SAME CURRENT.	Distribute the circuit diagram cards. Say: 'Before we touch any equipment today, I want you to think carefully about what you already know from our last four lessons. Look at these two circuits. In your notebook, answer the three questions on the board — you have 3 minutes, working silently and individually.' [Allow 3 minutes silent thinking time.] Then say: 'Now turn to your shoulder partner and compare your predictions. You have 2 minutes.' [Allow 2 minutes.] Cold-call 4 students — deliberately select students with DIFFERENT predictions to expose the range of ideas. Ask: 'Amina, what did you predict for Question 1, and what is your reasoning?' [Allow 10 seconds wait time after each student responds before probing further.] After 4 responses, say: 'We have disagreement in this room — that is exactly what science looks like before evidence. Let us collect evidence and see whose model survives.' Post all sticky notes on the Prediction Wall without correcting any responses.	Think-Pair-Share with Prediction Wall. This strategy makes prior conceptions visible and creates cognitive dissonance — students who predict 'same current' or 'brighter bulbs' will be confronted by their own data in Phase 2. The Prediction Wall is revisited at the end of the lesson to anchor the conceptual shift.	Observable indicator: Each student has written at least one prediction with a reasoning clause ('I think … because …') in their notebook before sharing. Teacher scans notebooks during the pair-share and notes the dominant misconceptions (likely: 'current gets used up' or 'bulbs closer to the battery get more current'). These are flagged for targeted questioning in Phase 3.
Observe Phase 📹 **VIDEO:** Theoretical probability distribution example: tables Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:**	Groups of 3 build series circuits on a cork tile or cardboard base using 1.5 V AA batteries (×2 in a holder = 3 V source), connecting wires with crocodile clips, three identical torch bulbs in holders, an ammeter (0–1 A range), and a voltmeter (0–5 V range). The investigation has three stages: STAGE 1 — One bulb in series with the source: Students connect the ammeter in series (measuring total current), record current and bulb brightness	Before students begin building: 'Everyone freeze and watch me for 90 seconds. I am going to show you how to connect the ammeter correctly — this is critical for your safety and for getting accurate readings.' [Demonstrate: ammeter in series, voltmeter in parallel, open switch before rewiring.] Say: 'The ammeter goes INTO the loop — it is like a gate the current must pass through. The voltmeter goes ACROSS the component — like	Structured Data Collection with Class Consolidation. By requiring students to move the ammeter to THREE positions in the same circuit before recording, students generate their own first-hand evidence that current does not change around a series loop. The class consolidation aggregates data across groups, making the pattern statistically robust and modelling real scientific practice of replication.	Observable indicator: (1) Each group's data table is complete with units and all three ammeter positions recorded before moving to Stage 3. (2) Students can state — without prompting — whether current changed when the ammeter was moved. Teacher listens for the phrase 'the current stayed the same' as a readiness indicator for Phase 3. Groups who record different currents at different positions are flagged for a

Power Series: Introduction and Convergence (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	(qualitative: bright / medium / dim). They also measure voltage across the bulb and across the source. STAGE 2 — Two bulbs in series: Students add a second bulb to the loop. They move the ammeter to three positions: before Bulb 1, between Bulb 1 and Bulb 2, and after Bulb 2 (before returning to the battery). They record current at each position and voltage across each bulb, across both bulbs combined, and across the source. They also note qualitative brightness. STAGE 3 — Three bulbs in series: Repeat the procedure with three bulbs. Students record all measurements in a structured data table (pre-printed in their workbooks). After completing the practical, each group calculates: (a) sum of individual voltage drops vs. source voltage; (b) total resistance for each stage using R = V/I. A class data consolidation occurs: each group reads out one current value and one voltage sum; the teacher records all groups' values in a class table on the board.	checking the pressure on either side of a door. Never swap these. If your ammeter needle kicks the wrong way, immediately open the switch and swap the leads.' [Circulate during practical. Crouch to eye level with each group. Ask targeted probes: 'You just moved the ammeter to a new position — did the reading change? What does that tell you?' and 'Add up your two voltage drops. Now look at your source voltage. What do you notice?' Allow 10 seconds wait time before offering hints.] During class consolidation, record all group data and ask: 'Class, are our current readings at different positions in the same circuit consistent across groups? What pattern do you see?'		circuit-check (likely a loose connection rather than conceptual error).
Explain Phase 📹 **VIDEO:** Theoretical probability distribution example: tables Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Power Series: Introduction and Convergence (Flexbook) Source: CK-12	Students return to their data and, in their groups, complete a Claim-Evidence-Reasoning (CER) frame for each of three series circuit rules: RULE 1 (Current): Claim — state what happens to current around a series loop. Evidence — cite specific ammeter readings from their data table. Reasoning — explain why, using the idea that electrons have only one path to travel. RULE 2 (Voltage): Claim — state the relationship between individual voltage drops and source voltage. Evidence — cite their calculated voltage sums. Reasoning — explain using the	Say: 'You now have data from three stages. Your job is not just to describe what you saw — your job is to BUILD A RULE that will work for ANY series circuit, not just yours. Use the CER frame.' [Allow 8 minutes for group CER writing.] After whiteboards are held up, cold-call 3 groups on Rule 1: 'Baraka's group — read your claim and one piece of evidence for Rule 1.' [10 seconds wait after each.] Probe the reasoning: 'You said electrons have only one path — can someone in another group add to that reasoning? What does one path mean for how many electrons	Claim-Evidence-Reasoning (CER) Framework with Simultaneous Mini-Whiteboard Gallery Carousel. CER forces students to distinguish between observation and interpretation, and to explicitly link data to a generalisable rule. The simultaneous whiteboard reveal allows the teacher to assess all groups at once and surfaces differences in reasoning quality without singling out individuals.	Observable indicator: Each group's mini-whiteboard contains at least ONE specific numerical value from their data table as evidence (not just a qualitative description). The Reasoning column must include a reference to either 'one path' (for current) or 'shared voltage / divides' (for voltage) or 'adds up' (for resistance). Teacher specifically looks for — and corrects — any group that writes 'current decreases as it flows through each bulb' (the 'current consumption' misconception).

🔍 Search ARES for similar readings	idea that the source voltage is 'shared' or 'divided' among the resistors in the loop. RULE 3 (Resistance): Claim — state how total resistance relates to individual resistances. Evidence — cite their R = V/I calculations for 1, 2, and 3 bulbs. Reasoning — explain why more bulbs mean more total resistance and therefore less current. Groups then apply these three rules to the two supporting phenomena: (a) Christmas lights — 'Why do all bulbs dim equally when you add more bulbs to the string?' (b) Huruma Estate extension cord — 'Why is the bulb connected via a long, thin extension cord always dimmer?' Each group writes a 2-sentence explanation for each phenomenon on a mini-whiteboard. Whiteboards are held up simultaneously (Gallery Carousel) so the teacher can scan all responses.	pass each point per second?' After Rule 3 is established, say: 'Now connect this directly back to Huruma Estate. Kesho — using only the three rules we just derived, explain in one sentence why the neighbour's bulb on the long extension cord is always dimmer.' [10 seconds wait time.] If students struggle, prompt: 'Is the extension cord adding resistance to the circuit? Is that resistance in series with the bulb?'		
Driving Question Board (DQB) Creation 📹 **VIDEO:** Theoretical probability distribution example: tables Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Power Series: Introduction and Convergence (Flexbook) Source: CK-12 🔍 Search ARES for similar	The class Driving Question Board is displayed (physical board or projected). Students individually write on sticky notes their responses to two prompts: GREEN NOTE — 'One thing I can now explain about the Huruma Estate phenomenon that I could NOT explain before today.' ORANGE NOTE — 'One new question this lesson has raised for me.' Students post their notes on the DQB. The teacher reads selected notes aloud and clusters them. The class identifies which aspect of the driving question — 'Why do electrical devices sometimes receive less energy than expected?' — is now partially answered by today's evidence. The teacher draws a visual	Say: 'Take two sticky notes — one green, one orange. Green: write one thing you can NOW explain about the Huruma Estate lights or the extension cord that you could not explain at the start of this lesson. Orange: write the new question that today's lesson has opened up for you. You have 2 minutes.' [Circulate silently.] After posting, read 3 green notes aloud: 'This group says they can now explain why Christmas lights are dim — let us hear their reasoning in one sentence.' Cold-call the author for elaboration. Then read 2 orange notes: 'This group is asking: what would happen if the bulbs were connected in different loops instead of one big loop? That is an extraordinary question	Driving Question Board Update with Progress Bar Voting. The DQB update makes the cumulative nature of the evidence-gathering sequence visible to students — they can literally see their growing explanation on the board. The progress bar voting is a metacognitive tool that prompts students to evaluate the sufficiency of their current model and motivates the next lesson.	Observable indicator: Green sticky notes must contain a specific reference to the phenomenon (not a generic 'I learned about circuits'). Orange notes reveal the quality of students' forward thinking — notes asking about parallel configurations or about independent switching show high-level anticipation of upcoming content. Teacher photographs the DQB at the end of the lesson for planning purposes.

readings	progress bar on the board labelled 'How much of the driving question can we now answer?' and invites the class to vote on what percentage they can now explain (from 0–100%). The class discusses: 'What would we need to investigate next to move this bar further?' — leading naturally toward parallel circuits in Lesson 6.	— hold that thought for Lesson 6.' Draw and label the progress bar. Ask: 'Show me with your fingers — 0 to 10 — how much of the driving question do you think we have answered?' [Scan the room.] Say: 'Most of you are saying 4 or 5 out of 10. What is missing from our explanation?' [Allow 15 seconds open wait time before cold-calling 2 students.]		
Model Building Phase 📹 **VIDEO:** Theoretical probability distribution example: tables Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Power Series: Introduction and Convergence (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Students retrieve their Cumulative Circuit Model — a running annotated diagram they have been building since Lesson 1 to explain the Huruma Estate phenomenon. Today they add a new panel labelled 'Series Circuit Rules'. In this panel they: (1) Draw a series circuit diagram with standard symbols (battery, 3 resistors, ammeter, voltmeter) using the conventions practised in the practical; (2) Annotate the diagram with arrows showing current direction and labels showing that current I is identical at every point; (3) Write the equation $V_{source} = V_1 + V_2 + V_3$ with a note explaining what this means for how voltage is 'shared'; (4) Write the equation $R_{total} = R_1 + R_2 + R_3$ and annotate it with: 'Adding more devices in series → more total resistance → less current → less brightness / slower heating'; (5) Draw a specific mini-diagram of the Huruma Estate neighbour's setup — bulb + long extension cord — labelling the cord as R_cord in series with R_bulb, and annotating why the bulb is dimmer. Students who finish early extend by calculating: if the extension cord has resistance 5 Ω and the bulb has resistance 40 Ω, and the source is 240 V, what is the	Say: 'Open your Cumulative Circuit Models. You are adding Panel 5 today — Series Circuit Rules. This panel must contain all five elements listed on the board. You have 10 minutes. Work individually first, then compare with your group.' [Write the 5 elements on the board and leave them visible throughout.] Circulate and check: 'Show me where you have written the equation for total resistance. Now show me how that equation connects to what we saw in the practical — where is your evidence?' After peer feedback: cold-call 2 students to share one peer suggestion they received and whether they agreed with it: 'Njoroge, what did your neighbour suggest you improve, and did you think it was valid? Why or why not?' Close the model-building phase by saying: 'Your model can now explain series circuits. But our driving question mentions designing circuits to deliver the RIGHT amount of energy safely and efficiently. Is a series circuit the best design for home wiring? What would happen if all your home lights were in series and one bulb blew? Hold that question — it is exactly where we are going in Lesson 6.'	Cumulative Annotated Model Revision with Peer Feedback (Star and a Step). The cumulative model is the central sensemaking artefact of the entire unit — revising it in every lesson ensures students build a coherent, connected explanation rather than isolated facts. Peer feedback using the Star and a Step protocol develops scientific communication skills and exposes students to alternative representational choices.	Observable indicator: Panel 5 of the Cumulative Circuit Model contains BOTH equations (voltage sum and resistance sum) AND a correctly labelled diagram of the extension cord scenario with R_cord in series with R_bulb. Teacher collects 6 models (2 from each ability band) at the end of the lesson for written feedback. Red flag: any model that shows different current values at different points in the series loop — this student requires a targeted follow-up conversation before Lesson 6.

	current and what is the voltage across the bulb only? Groups share their updated models with an adjacent group for peer feedback using a 'Star and a Step' protocol (one strength, one suggestion).			

D. TEACHER REFLECTION

After the lesson, reflect on the following questions and record brief notes in your planning journal: (1) CURRENT MISCONCEPTION — Did any group still record or claim that 'current gets used up' as it flows through each bulb? If yes, which students held this view, and what additional analogy or evidence might address it before Lesson 6? Consider using the 'water flow in a single pipe' analogy or revisiting the ammeter data from multiple positions. (2) VOLTAGE SHARING — Were students able to articulate WHY voltage divides (because each resistor opposes the flow and 'claims' a share of the source energy) or did they treat $V_1 + V_2 = V\_source$ as a formula to memorise without understanding? Plan a targeted warm-up question for Lesson 6 if the latter was observed. (3) PHENOMENON CONNECTION — When students applied the series rules to the Huruma Estate extension cord scenario, did they correctly identify the cord as a series resistor? If this connection was weak, begin Lesson 6 with a 2-minute revisit of this specific case before introducing parallel circuits. (4) EQUIPMENT READINESS — Note any ammeters or voltmeters that gave erratic readings today; set aside for calibration or replacement before Lesson 6. (5) DIFFERENTIATION — Were extension students stretched by the 240 V calculation problem? Consider preparing a second extension problem involving three different resistors for early finishers in Lesson 6. (6) DQB QUALITY — Review the orange sticky notes from today's DQB update. How many students spontaneously raised questions about independent switching or different loop configurations? This is a strong readiness indicator for the parallel circuits conceptual leap in Lesson 6.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	When we added more bulbs to a single loop (series circuit) and measured with the ammeter, the current reading was the same at every point in the circuit — before the first bulb, between bulbs, and after the last bulb. The bulbs became progressively dimmer as we added more bulbs to the series loop. When we summed the voltage readings across each individual bulb, the total equalled the source voltage.
What did I learn?	In a series circuit: (1) current is the same everywhere in the loop; (2) the source voltage is shared (divided) among the components so that $V\_source = V_1 + V_2 + V_3$; (3) resistances add up so that $R\_total = R_1 + R_2 + R_3$; (4) adding more components increases total resistance, which reduces current and causes each component to receive less voltage and less power — making bulbs dimmer and heaters slower.
How does this explain the phenomenon?	The neighbour in Huruma Estate whose bulb is connected via a long, thin extension cord has — without knowing it — created a series circuit. The resistance of the long thin cord adds to the resistance of the bulb in series, increasing the total resistance, reducing the current flowing through the bulb, and reducing the voltage across the bulb. Less current through a higher-resistance bulb means less power delivered to the filament, so the bulb glows dimmer. This is the same reason Christmas-light bulbs dim when more bulbs are added to the string — every new bulb is another series resistor sharing the fixed source voltage and increasing total resistance.

LESSON 6 (80 minutes): Parallel Circuits: Why Every Appliance in a Kenyan Home Gets Full Voltage	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To investigate the behaviour of voltage, current, and resistance in parallel circuits and connect these findings to the design of domestic wiring systems in Kenya.
Knowledge	By the end of the lesson, the learner should be able to: (a) state the rules governing voltage and current distribution in a parallel circuit; (b) derive the formula for equivalent resistance in a parallel circuit; (c) explain why domestic wiring in Kenyan homes uses parallel connections so that all appliances operate at 240 V independently; (d) explain why adding more appliances to a parallel circuit increases total current drawn from the supply.
Skills	By the end of the lesson, the learner should be able to: (a) build a parallel circuit using bulbs, resistors, a battery pack, ammeters, and voltmeters; (b) measure and record current at multiple branch points and at the main supply; (c) measure voltage across each parallel branch; (d) analyse data to derive the parallel-circuit rules; (e) calculate equivalent resistance for resistors in parallel.
Attitudes	The learner appreciates the importance of careful, safe circuit construction and recognises that understanding parallel circuits has direct consequences for household electrical safety and energy efficiency in Kenyan communities.
Key Inquiry Question	What is the relationship between voltage, current, and resistance in an electric circuit? How is electrical energy consumed and managed safely in our homes and communities?
Purpose in Storyline	In Lessons 1–5 students established Ohm's Law, explored series circuits, and discovered that resistance in a series path reduces current and dims bulbs — helping to partially explain the Huruma Estate phenomenon. However, a key puzzle remains: the neighbour whose bulb is always dimmer uses a long thin extension cord (added series resistance), yet ALL appliances inside the house operate normally and independently of each other. This lesson is the turning point — students discover that Kenyan home wiring is a parallel circuit, which guarantees each appliance receives the full 240 V regardless of how many other devices are switched on. This directly explains why lights and kettles inside the Huruma house behave normally once voltage is fully restored, and sets up Lesson 7's investigation of why drawing too many appliances can blow a fuse.
Safety Notes	Use battery packs (maximum 6 V) only — never mains electricity. Ensure all connecting wires have intact insulation. Students must not exceed the rated current for any component in the kit. Remind students that in real Kenyan homes, 240 V mains circuits are installed only by licensed Kenya Power electricians — students must NEVER tamper with mains wiring.

B. LESSON OVERVIEW

Students begin by predicting what will happen to voltage and current when bulbs are connected in parallel rather than in series. They then conduct a structured hands-on investigation, building a two-branch and three-branch parallel circuit, placing ammeters in the main line and each branch, and voltmeters across each branch. Data collected is analysed to derive the three rules of parallel circuits: voltage is the same across each branch; total current equals the sum of branch currents; equivalent resistance is less than the smallest individual resistance. Students use these rules to explain why Kenyan domestic wiring connects all appliances in parallel at 240 V, why adding appliances increases total current, and why the bulb on the long thin extension cord in Huruma is dimmer (a series resistance effect on one branch). The lesson closes with a model revision that updates students' circuit diagrams to reflect a simplified home wiring schematic.

C. LESSON IMPLEMENTATION FRAMEWORK				
Phase	**Learner Experience**	**Teacher Moves**	**Sensemaking Strategy**	**Formative Assessment Strategy**
Predict Phase 📹 **VIDEO:** Parsing gross domestic product Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Series and Parallel Circuits (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Students are shown two circuit diagrams on the board: Circuit A (two bulbs in series, from Lesson 5) and Circuit B (two bulbs in parallel — branches drawn but not yet labelled). They are given 3 minutes of individual silent thinking time to write predictions in their science notebooks in response to three prompt questions: (1) 'In Circuit B, if I unscrew one bulb, what happens to the other?' (2) 'Where do you think the voltage across each bulb in Circuit B will be compared to the battery voltage?' (3) 'Will the total current from the battery be more or less in Circuit B than in Circuit A — and why?' Students then share predictions with a shoulder partner for 2 minutes before a whole-class share-out. Students record predictions in their ARES notebooks using a Predict–Observe–Explain (POE) template pre-loaded in the app.	Display the two circuit diagrams and say: 'Look carefully at Circuit B. Do NOT calculate anything yet — just reason from what you already know about current and voltage.' Allow 3 minutes of silent individual writing (do not circulate — let students think independently). After 3 minutes say: 'Turn to your shoulder partner. You have 2 minutes — compare your predictions and note where you agree and disagree.' After partner talk, cold-call FOUR students (mix of high and low confidence): ask each one specifically: 'Tell us your prediction for Question 2 — voltage — and give your reasoning.' After each response, wait 12 seconds before probing with: 'Does anyone predict differently — and why?' Do NOT confirm or deny any prediction. Close with: 'We are going to find out exactly what happens. Hold your predictions — you will revisit them in Phase 3.'	Predict–Observe–Explain (POE) template: By committing predictions to writing before the investigation, students activate prior knowledge from Lessons 1–5 (Ohm's Law, series circuits) and create a cognitive hook. Comparing predictions with a partner surfaces misconceptions (e.g., 'voltage splits in parallel the same way it splits in series') that the investigation is designed to directly confront.	Circulate briefly during partner talk and scan notebooks. Look for: (a) students who predict voltage will split across parallel branches (series-circuit misconception to address in Phase 3); (b) students who correctly predict each branch gets full battery voltage; (c) students who connect total current increase to 'more paths for current to flow.' Note names of students holding the voltage-splits misconception — target them for cold-calls during Phase 3.
Observe Phase 📹 **VIDEO:** Parsing gross domestic product Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Series and Parallel Circuits (Flexbook) Source: CK-12 🔍 Search ARES	Groups of 3–4 students build parallel circuits in two stages. STAGE 1 (Two-branch parallel circuit): Using a 6 V battery pack, two identical bulbs (or 10 Ω resistors), connecting wires, one ammeter in the main line (before the junction), one ammeter in each branch, and one voltmeter across each branch — students assemble the circuit following a diagram provided on the ARES platform. They record: voltage across Branch 1 (V_1), voltage across Branch 2 (V_2), battery voltage (V_total), current in Branch 1 (I_1),	Before students begin building, say: 'Before you touch any component, your group must agree on where EACH ammeter will be placed and why. You have 90 seconds — discuss and point to the positions on your diagram.' Cold-call TWO groups to explain ammeter placement (wait 10 seconds after each response before moving on). Correct any misplacement of meters before building begins. During the activity, circulate with a checklist: verify each group has an ammeter in the MAIN LINE (not just in branches),	Structured Data Collection with Staged Complexity: Starting with two branches before three reduces cognitive load and allows students to see the pattern with fewer variables. Recording V_total alongside V_1 and V_2 in the same table forces a direct comparison. The one-bulb-unscrewed test is a critical observable that cleanly distinguishes parallel from series behaviour and links directly back to the phenomenon (in Huruma, switching off the kettle does NOT turn off the lights).	Use the circulating checklist to flag: (a) any group whose $V_1 \neq V_2$ (likely a wiring error — prompt them to check connections before assuming a real difference); (b) any group whose recorded I_total is NOT approximately equal to $I_1 + I_2$ (check for ammeter placement errors); (c) groups that correctly compute R_eq and spontaneously notice it is smaller than either individual resistance — these groups can be asked to share first in Phase 3. Take note of outlier data values to discuss as a class during analysis.

for similar readings	current in Branch 2 (I_2), and total current (I_total) in a results table. They then unscrew one bulb and observe what happens to the other. STAGE 2 (Three-branch parallel circuit): Groups add a third branch (third bulb/resistor) and repeat all measurements, adding V_3 and I_3 to their table. They also calculate equivalent resistance using R_eq = V_total / I_total and compare it to the individual resistances. Students photograph their circuit and data table using the ARES app camera feature for later reference during model building.	voltmeters are connected in PARALLEL across branches (not in series). At the midpoint, pause the class and say: 'Hold your work. Group 3 — read out your V_1 and V_2 values. Group 5 — read out yours.' Record four groups' values on the board. Ask: 'What pattern do you notice across these groups?' Wait 15 seconds. Do NOT give the answer — say: 'Keep that in mind as you complete Stage 2.' For Stage 2, prompt struggling groups: 'Where does the new branch connect — before the first junction or after?' Ensure all groups record the one-bulb-unscrewed observation clearly: 'The other bulb [stays on / goes off] — explain in one sentence why.'		
Explain Phase 📹 **VIDEO:** Parsing gross domestic product Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Series and Parallel Circuits (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Students return to their POE templates and complete the 'Observe' and 'Explain' columns based on their data. The class then participates in a Gallery Walk: each group posts their results table and a one-sentence summary of each pattern observed on the classroom wall. Students circulate (5 minutes), reading other groups' summaries and adding a sticky-note star if they agree or a sticky-note question mark if something surprises them. After the gallery walk, a whole-class discussion derives the three formal rules of parallel circuits: (1) V across each branch = V_total (voltage rule); (2) I_total = $I_1 + I_2 + I_3$ … (current rule); (3) 1/R_eq = $1/R_1 + 1/R_2 + 1/R_3$ … (resistance rule, and therefore R_eq < smallest R). Students then apply these rules to a Kenyan home context: the teacher presents a simplified schematic of a Huruma	Open the explain phase by projecting all four groups' voltage values from the board (written in Phase 2) and asking: 'In one word — what is the relationship between V_1, V_2, and V_total in every group's data?' Wait 12 seconds. Cold-call THREE students. After the third student, say: 'This is the First Rule of Parallel Circuits — write it in your own words now.' Give 60 seconds of writing time. Repeat the same process for the current data: 'Look at your I_total compared to I_1 plus I_2. What rule does this suggest?' Wait 15 seconds, cold-call FOUR students. For the resistance rule, say: 'Use your R_eq calculation. Is it bigger, smaller, or the same as R_1 alone? Does this surprise you — and can you explain why it makes sense physically?' Wait 15 seconds, take three responses. For the Kenyan home application, say: 'Mama Wanjiku in Huruma has her kettle,	Gallery Walk + Rule Derivation Sequence: The gallery walk allows students to triangulate their own data against peers' data, reducing the chance that one group's wiring error becomes the basis for an incorrect rule. Deriving the rules from DATA before stating them formally ensures students own the rules rather than receiving them as given facts. The Huruma home application immediately connects abstract rules to the anchoring phenomenon, reinforcing why this matters.	During the gallery walk, count the number of question-mark sticky notes on each group's poster — groups with many question marks may have data errors to address. During the whole-class derivation, listen for: (a) students who can state the voltage rule but cannot explain WHY (both branch ends connect to the same two nodes — address with a node diagram if needed); (b) students who can calculate I_total correctly; (c) students who correctly calculate R_eq for the Huruma home appliances and spontaneously connect the lower R_eq to higher current draw. Collect POE templates at the end of the phase to check that all students have revised their voltage prediction in writing.

	Estate house with a kettle (rated 2000 W), two LED bulbs (each 10 W), and a phone charger (5 W) all connected in parallel at 240 V. Students calculate the current drawn by each appliance, the total current, and the equivalent resistance of the household circuit. They revisit their original predictions and explicitly state what they got right and what they now revise.	two bulbs, and a charger all plugged in. Using our three rules, calculate what current Kenya Power must supply to her house.' Allow 4 minutes of group calculation, then cold-call one group to present. Ask the class: 'If Mama Wanjiku adds a second kettle — what changes, and what stays the same?' Wait 12 seconds before taking responses. Close with: 'Now go back to your prediction for Question 2. Were you right? Write one sentence correcting or confirming your prediction.'		
Driving Question Board (DQB) Creation 📹 **VIDEO:** [Parsing gross domestic product] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Series and Parallel Circuits (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	Each student receives two sticky notes. On the PINK note they write one thing they can NOW answer on the class Driving Question Board: 'Why do electrical devices in our homes sometimes receive less energy than expected?' On the YELLOW note they write one NEW question that today's lesson has raised — particularly focused on what happens when TOO many appliances are added to the parallel circuit (foreshadowing Lesson 7 on fuses and overloads). Students post their notes on the DQB under the categories 'Explained Today' and 'Still Wondering.' The teacher facilitates a 4-minute class scan of the new DQB entries, reading out selected yellow notes and asking the class: 'Should we add this to our list of questions to investigate?'	Hand out sticky notes and say: 'You have 3 minutes — individual silent thinking. On your PINK note, write the most important thing today's lesson explains about the Huruma phenomenon. On your YELLOW note, write the question you most want to investigate next.' After 3 minutes, say: 'Post your notes on the DQB now.' Once notes are posted, read THREE yellow notes aloud (choose ones that mention overloading, fuses, or safety) and ask the class: 'Does anyone else share this question?' Show of hands. Say: 'These questions are going to drive our next lesson. Hold them.' Explicitly point to the DQB entry from Lesson 5 that asked: 'Why does adding more devices to a circuit change the current?' and say: 'Has anyone answered this today? Give me a thumbs-up if you think today's lesson addresses this question.' Count thumbs. Ask one student with a thumbs-up to explain the answer in one sentence for the class.	Driving Question Board (DQB) — Cumulative Evidence Tracking: The DQB makes the progression of the inquiry visible. By physically moving questions from 'Still Wondering' to 'Explained Today,' students experience the satisfaction of growing understanding while recognising that new knowledge always generates new questions. The yellow notes provide the teacher with real-time formative data on what gaps remain and organically surface the next lesson's focus (overloading and fuses).	Read all pink notes after class. Assess whether students can correctly state: (a) voltage is the same across each branch in a parallel circuit; (b) total current increases as more appliances are added; (c) Kenyan homes use parallel wiring so each appliance gets full 240 V. Flag any pink notes that contain residual series-circuit thinking (e.g., 'voltage splits equally') for targeted follow-up at the start of Lesson 7.
Model	Students open their cumulative	Introduce the model revision by	Cumulative Model Revision with	Collect or photograph model

Building Phase 📹 **VIDEO:** Parsing gross domestic product Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Series and Parallel Circuits (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	circuit model from Lessons 1–5 in the ARES app (or in their notebooks). They are asked to make THREE specific additions or revisions: (1) Add a parallel-circuit branch to their model showing at least two appliances connected at the same two nodes, labelled with equal voltages across each branch. (2) Annotate the main supply wire with an arrow and label showing that $I_{total} = I_1 + I_2 + \ldots$ and that this total current is LARGER than any single branch current. (3) Draw a simplified schematic of the Huruma Estate house wiring, showing the mains supply (240 V), at least three parallel branches (kettle, bulb, and one other appliance), and annotate it to explain why the bulb connected via a long thin extension cord (from the original phenomenon) is dimmer — that extension cord adds a series resistance IN THAT BRANCH, reducing current and power in that branch only, while all other branches are unaffected. Students write a two-sentence 'Model Update Statement' at the bottom of their model: 'My model now explains… My model still cannot explain…'	saying: 'Your models have grown across six lessons. Today you must show that you can connect parallel circuits to the actual Huruma phenomenon — not just to abstract circuit diagrams.' Display the original Huruma phenomenon image/video still on the board. Say: 'As you draw, ask yourself: does my model explain why the dimmer-bulb neighbour's problem is a SERIES resistance in one PARALLEL branch — and why that only affects THAT branch?' Circulate and prompt with: 'Show me where the long extension cord resistance appears in your Huruma schematic.' and 'If the neighbour switches on their kettle in the same house — does your model predict the bulb gets even dimmer or stays the same? Draw that prediction into your model.' With 5 minutes remaining, cold-call TWO students to share their 'Model Update Statement.' After each, ask the class: 'Do you agree with what they say their model still cannot explain? Give a thumbs-up or thumbs-down.' Close the lesson by saying: 'In Lesson 7, we are going to find out what happens when too many appliances are drawing too much current — and why Kenya Power installs fuses and circuit breakers in every home.'	Phenomenon Re-anchoring: Requiring students to embed the extension-cord scenario into their Huruma wiring schematic forces integration of both series-resistance effects (Lesson 5) and parallel-circuit rules (Lesson 6) in a single diagram. The 'Model Update Statement' metacognitive prompt makes learning visible and honest — students must articulate both what is explained and what remains unexplained, sustaining inquiry momentum into Lesson 7.	diagrams at the end of the lesson. Evaluate against three criteria: (1) Parallel branches correctly show equal voltage labels; (2) Main supply wire correctly labelled with $I_{total} > I_{branch}$; (3) Extension cord scenario correctly drawn as a series resistor within one parallel branch, with annotation that only that branch is affected. Students who meet all three criteria are ready for Lesson 7's fuse and overload content. Students who still show voltage splitting across parallel branches need a brief one-on-one node diagram intervention before Lesson 7.

D. TEACHER REFLECTION

After the lesson, reflect on the following questions: (1) Did the staged two-branch → three-branch investigation allow all groups to collect clean, consistent data, or did wiring errors persist into Stage 2? If so, consider adding a peer-checking step before Stage 2 next time. (2) How many students held the 'voltage splits in parallel' misconception at the start of Phase 1? Were they successfully converted by the end of Phase 3, as evidenced by their corrected POE templates and pink DQB notes? (3) Did the Huruma home calculation (Mama Wanjiku's appliances) feel real and relevant to students, or did it feel like an abstract word problem? If students struggled to connect it to the phenomenon, consider opening Lesson 7 with a physical demonstration of increased current draw as appliances are added one by one. (4) Review all yellow DQB sticky notes: do students' 'Still Wondering' questions cluster around overloading, fuses, or safety? If yes, Lesson 7's focus is well primed. If questions are still

about basic parallel rules, consider a brief recap opener in Lesson 7. (5) Were all groups able to complete both Stage 1 and Stage 2 within the allocated time? If not, consider whether the data table could be pre-filled with column headers on the ARES platform to save set-up time.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	Students built two-branch and three-branch parallel circuits using battery packs, bulbs, ammeters, and voltmeters. They measured that voltage across each parallel branch equalled the battery voltage (approximately 6 V in each branch), that unscrewing one bulb left the other unaffected, and that total current in the main line equalled the sum of branch currents. They also calculated that equivalent resistance decreased as more branches were added.
What did I learn?	Students derived the three rules of parallel circuits: (1) voltage is the same across every branch and equals the supply voltage; (2) total current is the sum of all branch currents; (3) equivalent resistance is calculated using $1/R_{eq} = 1/R_1 + 1/R_2 + \ldots$ and is always less than the smallest individual branch resistance. Students connected these rules to Kenyan domestic wiring: all appliances are connected in parallel at 240 V so each operates independently at full voltage, and adding more appliances increases total current drawn from the Kenya Power supply.
How does this explain the phenomenon?	Students can now explain one key part of the Huruma Estate phenomenon: inside the house, all appliances (lights, kettle) are wired in parallel and each receives the full supply voltage independently — which is why, once voltage is fully restored, the lights and kettle all operate normally and do not affect each other. Students also explained why the neighbour's bulb on a long thin extension cord is always dimmer: the extension cord adds a series resistance within that one parallel branch, reducing current and power in that branch alone, while all other parallel branches in the house remain unaffected.

LESSON 7 (40 minutes): Series-Parallel Combination Circuits: Solving Real Kenyan Household Wiring Puzzles	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	Students will analyse series-parallel combination circuits by applying step-by-step equivalent resistance reduction methods, calculating voltages and currents at each stage, and connecting their findings to the energy delivery puzzle in the Huruma Estate phenomenon.
Knowledge	Students will be able to: (1) identify which components in a combination circuit are in series and which are in parallel; (2) apply the correct rules — R_series = R1 + R2 and 1/R_parallel = 1/R1 + 1/R2 — in the correct order to reduce a combination circuit to a single equivalent resistance; (3) use Ohm's Law at each stage to find voltage drops and branch currents; (4) explain how an internal wiring layout that mixes series and parallel elements causes some devices to receive less than full supply voltage.
Skills	Circuit analysis, step-by-step logical reduction, graph and table interpretation, collaborative problem-solving, applying mathematical formulae to real contexts, scientific communication of multi-step reasoning.
Attitudes	Persistence in multi-step problem solving, appreciation of how scientific models explain everyday experience, curiosity about energy efficiency in household wiring, responsibility in applying physics knowledge to safe and efficient design.
Key Inquiry Question	What is the relationship between voltage, current, and resistance in an electric circuit — specifically when a circuit contains both series and parallel sections?
Purpose in Storyline	Lessons 5 and 6 established the rules for pure series and pure parallel circuits. Lesson 7 bridges those two models into the realistic scenario most Kenyan homes actually present: a combination circuit in which a long extension cord (series resistance) feeds a parallel arrangement of appliances. This directly explains why the neighbour bulb on the thin extension cord is always dimmer — the cord acts as a series resistor that drops voltage before the parallel load — and updates the DQB with a unified circuit model students will carry into the internal resistance, power, and energy lessons.
Safety Notes	This lesson is paper-and-pencil / worked-example based. No live electrical equipment is used. Ensure students understand that the circuits analysed represent real household wiring and that attempting to open or rewire actual mains sockets is dangerous and illegal. Remind students that Kenya Power supply is 240 V AC and is lethal — all practical work in the unit uses low-voltage DC cells only.

B. LESSON OVERVIEW

Students begin by revisiting the Huruma Estate phenomenon and identifying a new puzzle: a circuit sketch that mixes both series and parallel sections, representing a typical Kenyan home layout where an extension cord (series resistance) connects to a multi-socket strip (parallel appliances). In the Predict phase students estimate what will happen to voltage and brightness before any calculation. In the Observe phase they work through a structured three-step reduction process on two progressively complex circuit diagrams, recording voltage and current at every node. In the Explain phase they articulate why voltage is lost across the series element and how that reduces energy delivered to the parallel load. They update the class Driving Question Board and close by building an annotated model that unifies the series and parallel rules into a single analytical framework they will use for the rest of the unit.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy

Predict Phase 📹 **VIDEO:** Theoretical probability distribution example: tables Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Resistance in Parallel Circuits Vs. Series Circuits (PLIX) Source: CK-12 🔍 Search ARES for similar readings	Students receive a hand-drawn circuit diagram on their worksheet labelled the Huruma Extension Cord Circuit. It shows a Kenya Power socket supplying 240 V, connected via a long thin extension cord represented as a single resistor R_cord = 6 ohms, which feeds a two-socket strip where Socket A has a bulb of resistance 60 ohms and Socket B has a phone charger of resistance 120 ohms connected in parallel. Students are asked three predict questions without calculating: (1) Will the bulb in Socket A receive the full 240 V? Why or why not? (2) Which socket do you think draws more current? (3) How does this connect to the neighbour with the dim bulb in the Huruma story? Students write individual predictions in their exercise books, then share with a partner using a think-pair-share structure. Pairs record their agreed prediction on a sticky note placed on the class DQB under the heading Still Unexplained.	Display the Huruma Extension Cord Circuit diagram on the board or a printed A3 poster. Say: Last lesson we proved that all appliances in a Kenyan home are wired in parallel so each gets 240 V. But look at this circuit carefully — is every part of it parallel? Give students 60 seconds of silent looking time. Then ask: Turn and tell your partner: where do you see a series element and where do you see a parallel element? Allow 90 seconds of pair talk. Cold-call three pairs and annotate the diagram on the board based on student responses. Then read the three predict questions aloud and say: Write your prediction — no numbers yet, just your reasoning based on what you already know about series and parallel circuits. Allow 3 minutes of individual writing. WAIT TIME: after each student response during cold-calling, wait a full 5 seconds before responding or redirecting — this signals that reasoning matters more than speed.	Anchoring prediction to the phenomenon: by framing the circuit as the actual extension cord in the Huruma story, students are not solving an abstract textbook problem — they are making a claim about a real event they have been investigating since Lesson 1. Annotating the diagram collaboratively surfaces prior knowledge (series: same wire path; parallel: branching paths) and creates cognitive need for a method that handles both simultaneously.	Teacher circulates and reads sticky-note predictions. Look for: Do students correctly identify the cord as a series element? Do they predict a voltage drop across the cord? Common misconception to probe: students may say the parallel branch still gets 240 V because they overgeneralise the parallel rule from Lesson 6. Note these students for targeted questioning during the Observe phase.
Observe Phase 📹 **VIDEO:** Theoretical probability distribution example: tables Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Resistance in Parallel Circuits Vs. Series Circuits (PLIX) Source: CK-12	Students work in groups of three on two structured tasks using the Step-by-Step Reduction Method introduced on the worksheet. TASK 1 — The Huruma Extension Cord Circuit (guided): The worksheet walks students through three reduction steps with blank boxes to fill. Step 1 — Combine the parallel branch: Calculate R_parallel for the 60-ohm bulb and 120-ohm charger using 1/R_p = 1/60 + 1/120; students find R_p = 40 ohms. Step 2 — Combine with series cord: Calculate R_total = R_cord + R_p = 6 + 40 = 46 ohms. Step 3 — Find total current from supply: I_total = V/R_total =	Distribute worksheets and say: You are now circuit detectives. Your job is to reduce a complicated circuit to a single resistor, like folding a map until it fits in your pocket — one fold at a time. For Task 1 you have a guide. For Task 2 you are on your own. Start with Task 1 now. Circulate during Task 1, pausing at each group for no more than 30 seconds. If a group is stuck on Step 1 ask: Which two components are sharing the same two nodes — the same two junction points? That tells you which ones are in parallel. WAIT TIME: pose this question and then	The Step-by-Step Reduction Method is a worked-example scaffold that makes the cognitive process visible: students draw a new simplified circuit at each stage, preventing the common error of losing track of what has already been combined. The two-task progression (guided then independent) follows a gradual release model. The Kenyan stockroom context in Task 2 keeps the inquiry grounded in familiar environments rather than abstract circuit diagrams.	Collect one worksheet per group at the end of the Observe phase (before the Explain phase discussion). Check: (1) Is Step 1 always combining elements that share the same two nodes? (2) Are students correctly applying the reciprocal formula for parallel and the additive formula for series? (3) Are intermediate circuits redrawn? Use these as evidence for differentiated support in the Explain phase. A student who correctly identifies which elements are in parallel but makes arithmetic errors needs different feedback than one who applies the series formula to a parallel section.

🔍 Search ARES for similar readings	240/46 = approximately 5.22 A. Step 4 — Find voltage across cord: V_cord = I_total x R_cord = 5.22 x 6 = approximately 31.3 V. Step 5 — Find voltage across parallel branch: V_parallel = 240 - 31.3 = approximately 208.7 V. Step 6 — Find individual branch currents: I_A = V_parallel / 60 = approximately 3.48 A; I_B = V_parallel / 120 = approximately 1.74 A. Students record all values in a results table and answer: How much voltage was lost in the extension cord? TASK 2 — The Nakumatt Superstore Stockroom Circuit (independent): A more complex combination circuit representing a stockroom where two parallel lamps (each 40 ohms) are connected in series with a relay switch resistor of 10 ohms, and this whole series-parallel group is in parallel with a 60-ohm motor. Students must first decide which elements to combine first, then reduce step by step, recording each intermediate R value. Supply voltage is 12 V (DC equivalent for safety modelling). Both tasks are completed in exercise books with neat circuit diagrams redrawn at each reduction stage.	step back for 10 seconds before offering any further hint. If a group finishes Task 1 early, direct them to Task 2 immediately. During Task 2 circulate and listen for groups debating which components to combine first. If a group tries to combine a series and parallel element simultaneously ask: Can you add resistors in series and parallel in the same step? What rule tells you which formula to use? After approximately 12 minutes bring the class together. Ask one group to present their Task 1 reduction at the board step by step. Pause after each step and ask the class: Does everyone agree with this step? Thumbs up, thumbs sideways, or thumbs down. Address any thumbs-down responses before moving on.		
Explain Phase 📹 **VIDEO:** Theoretical probability distribution example: tables Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Resistance in	Students return to their original predictions on the sticky notes. The teacher facilitates a whole-class sense-making discussion structured around three reveal questions displayed on the board. Question 1: You predicted whether the parallel branch would get the full 240 V. What does your calculation say — and why? Students connect V_parallel = approximately 208.7 V to the prediction that some voltage is	Return sticky-note predictions to the groups and say: Compare your prediction to your calculated answer. Were you right? What surprised you? Give 90 seconds of pair discussion. Then facilitate whole-class discussion using the three reveal questions. For Question 1 ask a student who predicted the branch would still get 240 V: What does the calculation show? How do you feel about your prediction now? Normalise	Returning sticky-note predictions creates closure on the prediction-observation cycle, making the learning feel like resolution of a genuine mystery rather than a textbook exercise. The water-pressure analogy bridges the abstract voltage-drop concept to physical intuition. The sentence-stem strategy ensures every student produces a generalisation, not just the most vocal students.	Collect the three-sentence individual explanation written in exercise books. Use the following success criteria: (1) identifies the series element as the cause of voltage drop; (2) states that the parallel load receives supply voltage minus the series drop; (3) links this explicitly to the Huruma dim-bulb observation. Students who cannot make the link to the phenomenon need re-engagement with the anchoring context before

[Parallel Circuits Vs. Series Circuits (PLIX)](#) Source: CK-12 🔍 Search ARES for similar readings	lost. Question 2: Where did the missing 31.3 V go, and what does that mean for the bulb in Socket A? Students articulate that the voltage was dropped across the series resistance of the cord, reducing the energy available to the parallel load — directly explaining the neighbour dim-bulb observation from the Huruma phenomenon. Question 3: What general rule can you now state about any circuit that has a series element feeding a parallel load? Students attempt to articulate: the series element steals voltage from the supply before the parallel load can use it, so the parallel load always receives less than the supply voltage. The teacher writes the student-generated rule on the board in their own words, then presents the formal principle alongside it. Students then individually write a three-sentence explanation in their exercise books connecting the Huruma neighbour dim bulb to the concept of series voltage drop in a combination circuit.	productive error: say In science, a wrong prediction that you can now explain is more valuable than a right guess you cannot explain. For Question 2 use the Kenyan analogy: Say imagine water flowing through a narrow pipe before reaching two taps — the narrow pipe reduces pressure at the taps. The extension cord does the same to voltage. For Question 3 use structured turn-and-talk: Say Turn to a different partner — not the one from the Predict phase — and together write one sentence that completes this stem: In a combination circuit with a series element feeding a parallel load, the voltage across the parallel load is always... Allow 2 minutes, then cold-call three pairs. WAIT TIME: after the stem is displayed, wait 60 seconds in complete silence before asking anyone to speak — this gives all students time to think before the fastest speakers dominate.		Lesson 8.
Driving Question Board (DQB) Creation 📹 **VIDEO:** [Theoretical probability distribution example: tables](#) Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Resistance in Parallel Circuits](#)	Groups retrieve their sticky notes from the DQB (under the heading Still Unexplained placed during the Predict phase) and update or replace them with new sticky notes under the heading Now We Can Explain. Each group writes one claim card: a statement of what combination circuits explain about the Huruma phenomenon, and one question card: what the combination circuit model still cannot explain. The class then reviews the DQB together. Previously unexplained questions about the dim bulb and kettle slowness are moved to the	Direct groups to the DQB and say: The DQB is our class record of what we understand and what we still need to find out. Take back your sticky note from the Predict phase. Does your prediction match what the calculation showed? If yes, upgrade it to a claim. If no, write a corrected claim. Then write a new question — something this lesson raised that we cannot yet answer. Allow 4 minutes for group writing. Then facilitate a 3-minute gallery walk where groups read each other's new question cards silently. After the gallery walk ask: Which new question appears most	The DQB functions as a public, cumulative record of the class inquiry. Moving sticky notes from Still Unexplained to Now We Can Explain gives students a concrete, visible sense of progress. The new question cards ensure the DQB is not just a record of closure but a generative tool that drives the next lesson — modelling authentic scientific practice where answering one question always raises new ones.	Examine claim cards for precision: do students say the cord drops voltage or vaguely say the bulb is dim because of the cord? Precision in causal language is an indicator of conceptual understanding. Examine new question cards: do they show curiosity about the source (internal resistance) or only about external circuit elements? This reveals how far students are projecting their model forward.

Vs. Series Circuits (PLIX) Source: CK-12 🔍 Search ARES for similar readings	Explained column. New questions that naturally arise — for example, what happens to the total current if the cord gets warm and its resistance increases? or what if the power source itself has internal resistance? — are placed in the new column labelled Next Lesson Will Help. The teacher highlights these new questions as the bridge to Lesson 8 on internal resistance.	often? Students will likely identify the question about what happens when the source itself has resistance — this is the Lesson 8 hook. Circle or highlight those question cards and say: That exact question is what we investigate next lesson. It will help us finally explain why the lights flickered when power was first restored in Huruma — not just why the extension cord dims the bulb.		
Model Building Phase 📹 **VIDEO:** Theoretical probability distribution example: tables Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Resistance in Parallel Circuits Vs. Series Circuits (PLIX) Source: CK-12 🔍 Search ARES for similar readings	In the final 5 minutes each student individually updates their Personal Circuit Model — a diagram they have been building since Lesson 1 in the back of their exercise book. They must now add a new section labelled Combination Circuits that includes: (1) a sketch of a generic series-parallel combination circuit with labels showing where voltage is dropped in the series section and where it is shared equally in the parallel section; (2) the three-step reduction method written as a personal algorithm in their own words; (3) one arrow connecting this model section to the Huruma phenomenon sketch they drew in Lesson 1, with a caption explaining which part of the Huruma scenario the combination circuit model explains. Students who finish early are prompted to shade or colour-code the series and parallel sections in their sketch to distinguish them visually.	Say: Open your exercise book to your Personal Circuit Model at the back. Every lesson you have added one layer to this model. Today you add the combination circuit layer. You have 5 minutes. Your model must show someone who was absent today exactly what a combination circuit is and how to analyse it. Circulate and give specific verbal feedback: avoid saying Good job and instead say I can see you identified the parallel nodes correctly — now label the voltage across that section, or Your algorithm says add the resistors — but which formula tells you when to add and when to use reciprocals? In the final 90 seconds ask two volunteers to hold up their models and describe one thing they drew. Close by saying: Your model is now strong enough to explain two of the three Huruma mysteries — the extension cord bulb and the parallel appliances. The third mystery — why lights flicker when power is first restored — still needs one more piece. That piece is inside the battery itself. That is where we go next lesson.	The Personal Circuit Model is a longitudinal metacognitive artefact: students see their own understanding grow across the unit. Requiring a connecting arrow to the Lesson 1 phenomenon ensures the model-building is not an isolated activity but is explicitly anchored to the phenomenon that opened the unit. The closing teacher statement creates narrative tension — the flickering lights mystery is still partly unresolved — which motivates engagement with Lesson 8.	Review Personal Circuit Models at the end of class or during the next lesson entry. Look for: Does the sketch correctly show the series section (single path) and parallel section (branching paths) as distinct? Does the personal algorithm correctly specify which formula to use for each type? Does the Huruma connection caption show causal reasoning or only descriptive labelling? These models serve as the primary formative evidence for Lesson 7 learning outcomes.

D. TEACHER REFLECTION

After the lesson ask yourself: (1) Could every student correctly identify which components were in series and which were in parallel before beginning any calculation — or did some students jump to formulas without first mapping the circuit topology? If the latter, the circuit identification step needs more scaffolding in Task 1 of the Observe phase. (2) Did the Kenyan contexts — the Huruma extension cord and the Nakumatt stockroom — feel authentic and meaningful to students, or did they feel like surface-level relabelling of a textbook problem? Authenticity is indicated by students spontaneously referring to the context in their explanations rather than using generic language. (3) Did the DQB generate genuine new questions about the power source, or did students feel the phenomenon was fully resolved? If students feel complete closure, the bridge to Lesson 8 on internal resistance needs a more provocative unresolved hook — consider adding a demonstration or video clip of a torch dimming under load to reopen curiosity. (4) Were the WAIT TIME pauses observed consistently, especially the 60-second silence before the sentence-stem sharing? If not, practise the silence deliberately — it is the single most powerful move for increasing the quality and inclusivity of student responses.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	In the Huruma Extension Cord Circuit, the voltage across the parallel appliance branch was approximately 208.7 V, not the full 240 V, because approximately 31.3 V was dropped across the 6-ohm extension cord. The bulb in Socket A and the charger in Socket B shared the reduced voltage, not the full supply voltage.
What did I learn?	When a circuit contains both series and parallel sections, the series elements reduce the voltage available to the parallel load. The correct analysis method is step-by-step equivalent resistance reduction: first combine elements sharing the same two nodes (parallel rule), then combine sequential groups (series rule), then use Ohm's Law at each stage to find voltages and currents. Total current from the supply, voltage drop across the series element, and voltage across the parallel branch can all be calculated systematically.
How does this explain the phenomenon?	The neighbour in Huruma Estate whose bulb is connected via a long thin extension cord always has a dimmer bulb because the cord acts as a series resistor in a combination circuit. It drops voltage before the current reaches the bulb, so the bulb receives less than the full 240 V supply voltage. This is not a fault in the supply — it is the predictable behaviour of a combination circuit where any series element steals voltage from the rest of the circuit.

LESSON 8 (40 minutes): Internal Resistance and EMF: Why Does the Battery Weaken Under Load?	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To investigate how internal resistance causes a drop in terminal voltage when a battery or power source supplies current, and to connect this to the flickering lights and slow kettle heating observed during the Huruma Estate power restoration.
Knowledge	Students will know that every real source of EMF has an internal resistance r; that terminal voltage V equals EMF E minus the internal voltage drop Ir; and that terminal voltage decreases as current demand increases.
Skills	Students will set up a circuit with a battery, rheostat, ammeter, and voltmeter; record terminal voltage at different currents; plot a terminal voltage versus current graph; and determine EMF and internal resistance from the graph intercept and gradient.
Attitudes	Students will appreciate that no real energy source is perfect, and will approach circuit design with awareness of limitations inside the source itself — developing honesty in recording data and precision in graphical analysis.
Key Inquiry Question	What is the relationship between voltage, current, and resistance in an electric circuit — including the resistance hidden inside the source itself?
Purpose in Storyline	This lesson provides the critical missing piece for the anchoring phenomenon: students now understand that the dim lights and slow kettle during power restoration resulted from high current demand causing a large internal voltage drop inside the supply, reducing the terminal voltage delivered to appliances. The DQB question about why voltage at the socket was lower than expected is now fully answerable.
Safety Notes	Use only low-voltage dry cells (1.5 V or 3 V battery pack). Do not short-circuit the battery — always keep a resistor or rheostat in the circuit. Ensure ammeter is connected in series and voltmeter in parallel. Warn students not to set rheostat to zero resistance. Check all connections before closing the switch.

B. LESSON OVERVIEW

Students begin by revisiting their existing prediction about why lights were dim during the Huruma Estate power restoration and why terminal voltage was lower than EMF in Lesson 2. They predict what happens to terminal voltage as more current is drawn. They then carry out a structured investigation — varying external resistance using a rheostat, recording terminal voltage and current, and plotting V versus I graphs. From the graph they extract the EMF (y-intercept) and internal resistance (negative gradient). In the Explain phase they derive V = E minus Ir algebraically and apply it to the Huruma phenomenon and to the weakening torch phenomenon. They update the DQB and refine their circuit model to include the battery as a source with both EMF and internal resistance.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase 📹 **VIDEO:** [Induced current in a wire] Source: Khan	Students are shown a photograph of a dim torch held by a child in a Mathare home, alongside a recall prompt on the board: In Lesson 2 we measured that the voltmeter	Display the dim torch photograph and the Lesson 2 voltmeter observation note. Say: In Lesson 2 we noticed something strange — the voltmeter reading across the	Prediction journaling with three structured prompts anchors prior knowledge from Lesson 2 and creates cognitive dissonance. Recording two prediction camps	Teacher scans written predictions during pair-share to identify students who already correctly associate voltage drop with internal current opposition versus

Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Resistance in Parallel Circuits Vs. Series Circuits (PLIX) Source: CK-12 🔍 Search ARES for similar readings	reading across the battery terminals dropped when the bulb was connected. Students write individual predictions in their notebooks responding to three prompts: (1) What do you think causes the voltage at the battery terminals to drop when current flows? (2) If you connect more bulbs in parallel to a battery, drawing more current, what will happen to the terminal voltage? Will it increase, decrease, or stay the same — and why? (3) How might this explain why Huruma Estate lights were dim right after power was restored, when many appliances switched on simultaneously? After writing, students share predictions with a partner, then three volunteers share with the class. The teacher records predicted patterns on the board in two columns — those predicting voltage drops with more current, and those predicting it stays constant.	battery fell as soon as we connected a bulb. We called it terminal voltage. Today we are going to find out what causes that drop and how big it can get. Pose prompt 1 and give 2 minutes of silent writing. Then say: Share your thinking with your partner for 1 minute. [WAIT 1 minute.] Then cold-call three students using name sticks. Record their ideas without evaluating. Say: We now have two camps — some of you think something inside the battery resists the current. Let us test that idea. After predictions are logged, say: Keep your prediction visible. By the end of the lesson you will tell me whether the evidence supports or contradicts what you wrote.	on the board makes student thinking public and creates a testable claim the investigation can resolve.	those who believe terminal voltage is fixed. This informs targeted questioning during the Observe phase.
Observe Phase 📹 **VIDEO:** Induced current in a wire Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Resistance in Parallel Circuits Vs. Series Circuits (PLIX) Source: CK-12 🔍 Search ARES for similar readings	Groups of four students collect: a 3 V battery pack (two D-cells in series), a rheostat (0 to 10 ohm), an ammeter (0 to 1 A), a voltmeter (0 to 3 V), a switch, and connecting wires. Students assemble the circuit: voltmeter directly across the battery terminals, ammeter and rheostat in series with the battery, switch in series. Step 1 — Open switch: record the voltmeter reading as the open-circuit voltage (this is the EMF E). Step 2 — Close switch with rheostat at maximum resistance: record V and I. Step 3 — Decrease rheostat resistance in five steps, recording V and I at each step. Students record data in a table with columns: Rheostat	Distribute equipment and a printed circuit diagram. Say: Before you close the switch, check three things: voltmeter is across the battery terminals, ammeter is in the main loop, and rheostat is at its highest resistance. I will come and verify each group before you close the switch. Circulate and approve each circuit. After data collection begins, prompt with: What do you notice about the voltmeter reading as you decrease resistance and more current flows? [WAIT 10 seconds per group.] If a group sees no change, check whether their voltmeter leads are correctly placed. During graphing, say: What does the y-intercept represent physically — what	Structured data table with a pre-built calculated column (E minus V) scaffolds students toward discovering that the missing voltage is proportional to current. Graph construction makes the linear relationship and its physical meaning (intercept = EMF, negative gradient = internal resistance) visible before the algebra is introduced.	Teacher checks each group graph for correct axis labelling, appropriate scale, and whether the best-fit line passes near the open-circuit voltage at zero current. Groups that get a flat graph are asked to check voltmeter placement — this reveals a wiring misconception in real time.

	setting, Current I (A), Terminal Voltage V (V), and a calculated column E minus V (which they will later recognise as Ir). After data collection, each group plots V on the y-axis versus I on the x-axis on graph paper, draws a best-fit straight line, identifies the y-intercept, and calculates the gradient (which will be negative). Groups label axes, include units, and write the coordinates of two points used for the gradient calculation.	current would give you that voltage? [WAIT 15 seconds.] And: What does the gradient tell us — is it positive or negative, and what does the sign mean? [WAIT 15 seconds.] Avoid giving answers; redirect with: What did your data show?		
Explain Phase 📹 **VIDEO:** Induced current in a wire Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Resistance in Parallel Circuits Vs. Series Circuits (PLIX) Source: CK-12 🔍 Search ARES for similar readings	The teacher leads a whole-class discussion anchored on three groups sharing their graphs under a document camera or on the board. The class identifies that all graphs show a downward-sloping straight line with y-intercept close to the measured EMF and negative gradient. Students are guided to read the equation of a straight line: V = c plus mI, where c is the intercept and m is the gradient. The teacher writes E for the intercept and negative r for the gradient, giving V = E minus Ir. Students copy the derivation and label each term: E is the electromotive force (energy per unit charge supplied by the source), I is the current, r is the internal resistance (the resistance of the chemical source itself), and Ir is the lost voltage (sometimes called lost volts). Students then work through three application questions in their notebooks: (1) A battery has EMF 6 V and internal resistance 0.5 ohm. Calculate the terminal voltage when it supplies a current of 2 A. (2) A battery shows an open-circuit voltage of 1.5 V but only 1.2 V when supplying 0.3 A. Calculate its internal resistance.	Project or draw a composite graph showing the class data trend. Say: Look at the shape of these graphs. What is the mathematical name for this shape? [WAIT 10 seconds.] Good — a straight line. What are the two features of a straight line equation? [WAIT 5 seconds.] Yes — gradient and intercept. Now let us name these physically. Point to y-intercept: What current is flowing at this point? [WAIT 5 seconds.] Zero current — so no load — so this is the pure EMF. Point to gradient: The line falls as I increases. What does that tell us about terminal voltage as current increases? [WAIT 10 seconds.] After deriving V = E minus Ir, say: The term Ir is called the lost volts — energy lost per unit charge inside the battery before it even reaches the circuit. Now apply this to Huruma. If 30 A is drawn suddenly, how big is the Ir drop? [WAIT 15 seconds.] Let students calculate. Say: Does this match what we predicted at the start of the lesson?	Graphical-to-algebraic bridging strategy: students derive the equation from their own data rather than receiving it as a formula. The Huruma calculation makes the abstract equation concrete and directly resolves the anchoring phenomenon. Peer comparison of answers builds collaborative verification norms.	The three application questions serve as an embedded mini-assessment. The teacher listens to the Huruma calculation discussion specifically to check whether students correctly identify the Ir term as the cause of reduced socket voltage. Students who calculate terminal voltage as equal to EMF have not applied the formula correctly and receive a targeted prompt: what does the r in the battery do to the current passing through it?

	(3) Apply V = E minus Ir to the Huruma Estate scenario: if the power supply has an effective EMF of 240 V and an internal resistance of 2 ohm, and 30 A is drawn at the moment of restoration, what is the terminal voltage at the socket? Compare to the normal 240 V and explain the dim lights. Students compare their answers with a partner, then the teacher cold-calls one student per question to present the answer and reasoning.			
Driving Question Board (DQB) Creation 📹 **VIDEO:** [Induced current in a wire] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Resistance in Parallel Circuits Vs. Series Circuits (PLIX)] Source: CK-12 🔍 Search ARES for similar readings	Students return to the class DQB displayed at the front. The teacher projects the original DQB questions from Lesson 1. Students individually read through all cards and, in their notebooks, write responses to two prompts: (1) Which questions on the DQB can we now fully answer using V = E minus Ir? Write the question and a two-sentence answer. (2) Write one new question that this lesson has raised for you — for example, about what happens to internal resistance as a battery ages, or why internal resistance matters for designing circuits with high current demand. Volunteers add new question cards to the DQB. The teacher draws attention to the original DQB card reading: Why was the voltage at the socket lower than expected when power came back? The class agrees that this question is now answered and marks it as resolved. Students also note on the DQB that the dim-torch phenomenon (The Weakening Torch) is now explained: as a torch battery discharges, its internal resistance increases, so the Ir drop grows and terminal voltage falls even with the same EMF.	Point to the Huruma-related DQB card from Lesson 1. Say: At the start of this unit you asked why the voltage at the socket was lower than expected. We have now built a model that explains it. Who can give the full explanation in one or two sentences — using E, I, and r? [WAIT 20 seconds — extended wait time to allow full sentence formulation.] Cold-call one student. Then say: Mark this card as resolved. Now, what new questions does internal resistance raise for you? Think about old batteries, about connecting many appliances, about designing circuits. [WAIT 10 seconds.] Invite three students to add new question cards. Affirm that good scientists always generate new questions as they answer old ones — this is how inquiry progresses.	DQB resolution ritual makes the progress of knowledge-building visible and rewards the iterative inquiry process. Requiring a two-sentence verbal explanation checks depth of understanding, not just recall of the formula. New question generation maintains the epistemic culture of the classroom as an inquiry community.	The teacher reads student notebook responses to prompt 1 during the DQB activity by circulating. Students who write only the formula without connecting it to the phenomenon are prompted: how does Ir explain what the Huruma residents saw? Students who correctly articulate the full causal chain (high current demand, large Ir drop, reduced terminal voltage, dim lights) are asked to share with the class.

Model Building Phase 📹 **VIDEO:** Induced current in a wire Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Resistance in Parallel Circuits Vs. Series Circuits (PLIX) Source: CK-12 🔍 Search ARES for similar readings	Students open their cumulative circuit model diagrams started in Lesson 1 and updated through Lessons 3 to 7. They now add a new component to their battery symbol: a small resistor r drawn inside the battery circle, in series with the EMF source E. They annotate the diagram with: (a) the label E for electromotive force (units: volts), (b) the label r for internal resistance (units: ohms), (c) an arrow showing where terminal voltage V is measured (across the external terminals), and (d) the equation V = E minus Ir written alongside the battery. Students then sketch a mini-scenario diagram: a battery with EMF 12 V and internal resistance 1 ohm connected to an external resistance of 5 ohm. They label all voltages (terminal voltage, lost volts) and currents on the diagram and calculate numerical values. Finally, students write a three-sentence paragraph connecting the model to the Huruma anchoring phenomenon, addressing all three original student questions: (1) why lights were dim at first, (2) why the kettle was slow, and (3) why the neighbour with the long thin extension cord always had a dimmer bulb — noting that the extension cord resistance acts like an additional series resistance in the external circuit, compounding the internal resistance effect studied today.	Say: Your circuit model has grown every lesson. In Lessons 5 and 6 you added series and parallel rules. In Lesson 7 you combined them. Today we add something new — the battery is not a perfect source. Open your model diagram. Draw the internal resistance r inside the battery. This changes everything: even the source itself resists the current it is trying to drive. Circulate and check that students draw r inside the battery symbol (not as a separate external component). Say: Now connect this updated model to the three original Huruma questions. All three should be answerable with what you now know. [WAIT 3 minutes for writing.] Ask one student to read their paragraph aloud. Prompt the class: Did they explain all three? What would you add? [WAIT 10 seconds.]	Cumulative model updating makes the knowledge-building trajectory visible. Annotated diagram with numerical values bridges graphical and algebraic understanding. The three-sentence Huruma paragraph requires students to integrate the new concept with prior learning (resistivity from Lesson 4, series rules from Lesson 5) to explain all parts of the anchoring phenomenon in a single coherent account.	Teacher reviews the mini-scenario diagram for correct numerical values: current I = 12 divided by (1 plus 5) = 2 A; terminal voltage V = 12 minus (2 times 1) = 10 V; lost volts = 2 V. Students who calculate terminal voltage as 12 V have ignored internal resistance — teacher prompts: where does the lost voltage Ir go? The three-sentence Huruma paragraph is used as an exit-ticket equivalent: collected or photographed at the end of class to assess whether students can integrate internal resistance with the anchoring phenomenon before Lesson 9.

D. TEACHER REFLECTION

After the lesson, consider: Did all groups obtain a clearly downward-sloping V versus I graph, or did some get a flat line indicating a wiring error? Were students able to identify the y-intercept as EMF without being told? Could students articulate in their own words why the Huruma socket voltage dropped — not just recite V = E minus Ir?

Review the exit-paragraph collection: students who explain only the internal resistance of the battery without mentioning the high current surge at restoration moment have a partial model. In Lesson 9, when electrical power is introduced, revisit the Ir squared power loss inside the source to deepen this model. Note any groups whose battery voltage had already dropped significantly (old cells) — this is a teaching opportunity: the apparent EMF measured may already be lower than rated, and internal resistance may be higher, foreshadowing Lesson 9 discussions on power dissipation.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	We observed that as we increased current drawn from the battery by decreasing external resistance, the voltmeter reading across the battery terminals fell steadily. The V versus I graph was a straight downward-sloping line, with the y-intercept equal to the open-circuit voltage (EMF) and a negative gradient.
What did I learn?	We learned that every real battery has an internal resistance r that causes a voltage drop of Ir inside the battery whenever current flows. The terminal voltage V equals the EMF E minus the lost volts Ir: V = E minus Ir. The EMF is the maximum voltage the source can provide (at zero current); the terminal voltage is always less when current is flowing.
How does this explain the phenomenon?	We can now explain why the Huruma Estate lights flickered dimly when power was restored: at that moment, many appliances switched on simultaneously, drawing a large surge current I. This produced a large Ir drop inside the supply, reducing the terminal voltage at the sockets below the normal 240 V, so bulbs received less voltage and glowed dimly. We also explained the weakening torch: as the battery ages, its internal resistance r increases, so the Ir drop grows and terminal voltage falls even though EMF may remain roughly constant.

LESSON 9 (80 minutes): Electrical Power: Deriving and Applying $P = IV = I^2R = V^2/R$	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	Students derive the three forms of the electrical power formula ($P = IV$, $P = I^2R$, $P = V^2/R$) and apply them to calculate the power ratings of common Kenyan household devices, explaining why wires carrying high currents heat up and why the neighbour's long, thin extension cord in Huruma Estate always causes a dimmer bulb.
Knowledge	Students will know that: (1) electrical power is the rate at which electrical energy is transferred or dissipated ($P = E/t$); (2) combining $P = IV$ with Ohm's Law yields $P = I^2R$ and $P = V^2/R$; (3) power dissipated in a resistive conductor increases with the square of current, explaining why undersized or long extension cords heat up and drop voltage; (4) the watt (W) is the SI unit of power and the kilowatt-hour (kWh) is the commercial unit of electrical energy used by Kenya Power.
Skills	Students will be able to: (a) algebraically derive $P = I^2R$ and $P = V^2/R$ from $P = IV$ and $V = IR$; (b) use all three power formulae to calculate power, current, voltage, and resistance for household devices; (c) interpret and compare power ratings on device labels (electric jiko, immersion heater, phone charger); (d) calculate heat energy dissipated in an extension cord and explain its effect on brightness and heating rate; (e) construct and annotate a quantitative circuit diagram showing power distribution in a simple domestic circuit.
Attitudes	Students will appreciate that understanding electrical power is a practical life skill that enables them to make safe and economical decisions about energy use at home, and to recognise that the physics of power dissipation is directly responsible for real hazards such as overheating extension cords and fire risks in Kenyan homes.
Key Inquiry Question	How is electrical energy consumed and managed safely in our homes and communities?
Purpose in Storyline	In Lessons 1–8 students built understanding of Ohm's Law, internal resistance, series and parallel circuits, and EMF. They can now explain why current and voltage differ across circuit elements. Lesson 9 adds the final quantitative tool: electrical power. Students discover that the dimness of the Huruma Estate lights and the long-heating kettle are not just about low voltage — they are about reduced POWER delivery. They also quantitatively explain why the neighbour's long, thin extension cord dissipates power as heat, stealing energy from the bulb. This lesson is the pivot from 'how much current/voltage?' to 'how much energy per second?' — directly answering the driving question.
Safety Notes	All circuits must use batteries (≤ 6 V) or regulated DC power supplies only. No mains (240 V) connections are to be made by students. Warn students that high-current wires become hot — this is the lesson's phenomenon, not a hazard to create. Nichrome resistance wire used in demonstrations may become warm; students must not touch it during current flow. Kenya Power safety messaging: extension cords rated for lower currents than the devices plugged into them are a leading cause of house fires — reinforce this as a real community safety message.

B. LESSON OVERVIEW

In this 80-minute lesson, students move from measuring current and voltage (Lessons 1–8) to understanding the RATE at which electrical energy is delivered or wasted — electrical power. The lesson opens by revisiting the Huruma Estate phenomenon through a new lens: students already know the long, thin extension cord has higher resistance, but now they ask, 'where does the lost energy actually go, and how fast?' Using data they collected in previous lessons, students derive $P = IV$ algebraically, then use Ohm's Law substitution to obtain $P = I^2R$ and $P = V^2/R$. They apply these formulae to realistic Kenyan household device labels (electric jiko, immersion heater, M-KOPA solar lamp, phone charger), calculate energy costs in kWh, and design a short investigation comparing power dissipated in thick versus thin resistance wire — making the 'hot extension cord' phenomenon quantitative. The lesson closes with a model-revision activity in which students annotate their cumulative circuit diagrams

with power values, completing the conceptual bridge to the driving question about safe and efficient energy delivery.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase 📹 **VIDEO:** Watch Me Source: TED-Ed Lessons 🔍 Search ARES for similar videos 📖 **HTML:** Electric Power and Electrical Energy Use (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Students are presented with three labelled household device cards (electric jiko: 1500 W / 240 V; immersion heater: 2000 W / 240 V; phone charger: 5 W / 5 V) and a fourth unlabelled card representing the neighbour's long extension cord (resistance 8 Ω, carrying 5 A). Working in pairs, students respond in their ARES journals to the prompt: 'Which device uses energy fastest? How much current does each draw? Where does the energy go in the extension cord — and how fast is it lost?' Students record numerical predictions using any formula they already know or can reason about, flagging gaps with a '?' symbol. After 4 minutes of paired prediction, each pair shares one prediction with the table group (table talk, 2 minutes). The teacher then cold-calls 4 pairs (one per table group) to share the most surprising or most uncertain prediction with the class.	Display the four device/cord cards on the board or projector. Say verbatim: 'Before we do any maths together, I want you to REASON with what you already know. You have 4 minutes — use your Ohm's Law knowledge, use your gut, use your circuit diagrams from last week. Write a number, even if you think it is wrong.' [START TIMER — 4 minutes silent paired work.] Circulate and note 2–3 interesting or misconception-revealing predictions on a sticky note for use during the Explain phase. After paired work: 'Share your most uncertain prediction with your table — 2 minutes.' [WAIT — circulate, listen.] Cold-call exactly 4 pairs: 'Table 1 — what did you predict for the extension cord? Table 3 — did your group agree on the current through the jiko?' Do NOT correct predictions. Say: 'Hold those ideas — we will come back and test every single one of them today.'	Strategy: Anchored Prediction with Flagged Uncertainty. Students commit to numerical estimates before instruction, activating prior knowledge of V = IR and E = QV. Flagging uncertainty with '?' makes knowledge gaps visible to both student and teacher, creating cognitive readiness for the derivation that follows. This prevents passive reception of formulae by ensuring students have a personal stake in resolving the discrepancy between prediction and derivation.	Observable indicator: Every student has written at least one numerical prediction and one '?' flag in their journal before sharing begins. Teacher checks during circulation that students are attempting quantitative reasoning (even if incorrect), not writing qualitative statements only. Cold-call responses reveal whether students conflate power with current, energy with power, or assume the extension cord 'loses' voltage rather than dissipates power — these misconceptions are flagged for targeted address in Phase 3.
Observe Phase 📹 **VIDEO:** Watch Me Source: TED-Ed Lessons 🔍 Search ARES for similar videos 📖 **HTML:** Electric Power and Electrical Energy Use (Flexbook)	ACTIVITY A — Reading Device Labels (10 min): Each table group receives a set of real or printed labels from common Kenyan household devices: an electric jiko (Sayona brand, 1500 W), an immersion heater (2000 W), an M-KOPA solar lamp (5 W), a phone charger (5 W / 5 V output), and a KPLC prepaid meter display showing kWh. Students read the labels and record in a structured table: Device | Power Rating (W) |	For Activity A: distribute label cards and say: 'You have 10 minutes. Read every number on these labels — they are not decoration, they are physics data. Fill your table completely. The question at the bottom — which device drains your KPLC token fastest — I want a number, not a guess.' Circulate; prompt students who are stuck: 'If power is 1500 W, and you run it for 1 hour, how many kilowatt-hours is that? Check	Strategy: Multi-Modal Evidence Triangulation. Students gather evidence across three modes — real device labels (authentic artefacts), a live teacher demonstration with measurements (direct observation), and their own prior experimental data (ARES data table). Triangulating across modes prevents the formulae from feeling abstract: the same P = IV emerges from a device label, a live wire heating up, and their own past	Observable indicator 1: Completed data table with correct calculated currents for all devices (e.g., jiko: 1500/240 = 6.25 A; immersion heater: 2000/240 = 8.33 A; phone charger: 5/5 = 1 A). Observable indicator 2: Students correctly record both ammeter and voltmeter readings for both wires and compute P = IV for each — the nichrome wire should yield significantly higher power dissipation than expected from

Source: CK-12 🔍 Search ARES for similar readings	Voltage Rating (V) | Calculated Current (A using I = P/V) | Energy used in 1 hour (kWh). They discuss: 'Which device would drain a prepaid KPLC token fastest?' ACTIVITY B — Thin Wire vs. Thick Wire Heat Demonstration (12 min): Teacher sets up two circuits side by side at the front bench: Circuit 1 uses thick copper wire (low resistance, ~0.5 Ω); Circuit 2 uses thin nichrome resistance wire (~8 Ω), both connected to the same 6 V battery and carrying measurable current (ammeter in series). Students observe — from their seats for safety — that the nichrome wire becomes warm (teacher holds a small paper strip near it; it curls). Students record ammeter readings for both circuits, note the voltage drop across each wire using a voltmeter, and calculate power dissipated (P = IV) in each wire. They connect this directly: 'This nichrome wire IS the neighbour's long extension cord — scaled down.' ACTIVITY C — ARES Platform Data Table (5 min): Students open the ARES pre-loaded data table showing results from Lesson 6 (their own current and voltage measurements across different resistors). They are asked to add a new column: 'Power dissipated (W)' by computing P = IV for each row.	the KPLC rate card — how many shillings?' For Activity B: stand at the front bench and narrate the setup: 'I am building two circuits. Both have the same 6 V supply. Watch the ammeter — tell me which circuit draws more current.' [BUILD CIRCUITS LIVE — take 2 minutes.] Ask: 'Why does Circuit 2 draw less current?' [WAIT 10 seconds — cold-call 2 students.] 'Now watch the nichrome wire carefully — do NOT touch it.' Hold paper strip near wire for 20 seconds. Ask: 'What do you observe? What does this tell us about where the energy is going?' [WAIT 12 seconds — cold-call 3 students.] Say: 'Calculate P = IV for each wire — I will give you 3 minutes.' For Activity C: 'Open ARES, go to your Lesson 6 data table. Add one column — Power. Multiply your V by your I for every row. You have 5 minutes.'	measurements. The nichrome wire demonstration is deliberately framed as a physical scale model of the extension cord, maintaining the phenomenon connection throughout.	current alone, prompting 'wait, that doesn't seem right' — a productive confusion the teacher captures for Phase 3. Observable indicator 3: ARES data table column is populated with plausible power values consistent with prior V and I measurements.
Explain Phase 📹 **VIDEO:** Watch Me Source: TED-Ed Lessons 🔍 Search ARES for similar videos 📖 **HTML:** Electric Power	PART A — Algebraic Derivation (12 min): The class derives all three power formulae together. Starting from the definition P = E/t and the relation E = QV, students show P = QV/t = IV. Teacher then guides substitution: 'If V = IR, substitute into P = IV to get P = I^2R. If I = V/R, substitute into P = IV to get P = V^2/R.' Students write	PART A: Write on board: 'P = E/t' and 'E = QV'. Say: 'These two equations you already know. Watch what happens when I combine them.' Perform derivation step by step, pausing after each line for 10 seconds to let students write. Then say: 'Now I need a volunteer — who can substitute V = IR into P = IV for me? Come to	Strategy: Worked Example → Student Practice with Prediction Reconciliation. The derivation gives students the 'why' behind the formula, not just the formula. Worked examples with Kenyan contexts (jiko, Kisumu household, Huruma extension cord) ground the algebra in familiar devices. Prediction reconciliation —	Observable indicator 1: During derivation, the board volunteer correctly writes at least the first substitution step; teacher verifies all students have the three formulae correctly written and labelled in journals. Observable indicator 2: Cold-call responses to 'what happens when current doubles?' correctly identify the

[and Electrical Energy Use (Flexbook)] Source: CK-12 🔍 [Search ARES for similar readings]	each step in their journals and annotate when to use each form: $P = IV$ (when both I and V are known); $P = I^2R$ (when I and R are known — useful for extension cord heating); $P = V^2/R$ (when V and R are known — useful for fixed-voltage devices). PART B — Worked Examples with Kenyan Context (10 min): Teacher works through three examples on the board: (1) An electric jiko rated 1500 W connected to 240 V mains — find current drawn and resistance of the heating element. (2) The neighbour's extension cord: resistance = 8 Ω, current = 5 A — find power dissipated in the cord as heat, and hence power remaining for the bulb if total supply is 120 W. (3) A phone charger: output 5 V, 1 A — find power delivered to phone battery. After each example, students check whether their Phase 1 predictions were correct. PART C — Student Practice (8 min): Students independently solve two problems from the ARES platform: Problem 1: An immersion heater in a Kisumu household draws 8.33 A from a 240 V supply. Calculate its power rating, resistance, and energy consumed in 45 minutes (in joules and kWh). Problem 2: A long, thin extension cord has a resistance of 12 Ω and carries a current of 4 A. Calculate the power dissipated in the cord. If a 60 W bulb is connected at the end, what percentage of the total power is wasted in the cord? Students are invited to correct their Phase 1 predictions in a different colour pen.	the board.' [Cold-call if no volunteer — minimum 1 student at board.] After $P = I^2R$ appears: 'Read this formula. What does it tell you — if resistance DOUBLES and current stays the same, what happens to power?' [WAIT 12 seconds — cold-call 2 students.] 'If current DOUBLES and resistance stays the same?' [WAIT 10 seconds — cold-call 2 students.] Emphasise: 'Power goes up with the SQUARE of current. This is why a thick wire carrying twice the current of a thin wire dissipates FOUR times the power as heat — not twice.' PART B: For example 2 (extension cord), say: 'This is the neighbour in Huruma Estate. We now have a number. How much power is the cord eating? Let us calculate it.' Write answer ($P = I^2R = 5^2 \times 8 = 200$ W) prominently on board. Then: 'If total available power is 120 W for the bulb, and the cord is already consuming 200 W — what does that tell us?' [WAIT 15 seconds — allow productive confusion — cold-call 3 students.] Guide to resolution: the cord has too high a resistance for this current; in reality, the extra resistance reduces current, which reduces power to the bulb. This is the quantitative explanation of dimness. PART C: Say: 'Two problems — 8 minutes — silent independent work. Use the formula that suits the information given, not always the same one.' Circulate and mark journals with a tick or a circle (circled = needs follow-up). After 8 minutes: 'Check your Phase 1 prediction. Put a star next to anything you got right. Put a question mark next to anything that still surprises you.'	correcting Phase 1 answers in a different colour — makes conceptual change visible and personally meaningful. The deliberate productive confusion in example 2 (cord dissipating 200 W when bulb only gets 120 W) surfaces the need for the fuller circuit analysis and connects back to internal resistance ideas from Lesson 8.	squared relationship (at least 3 of 5 cold-called students). Observable indicator 3: Independent practice Problem 1 answer: $P = 2000$ W, $R = 28.8$ Ω, $E = 5{,}400{,}000$ J $= 1.5$ kWh — teacher checks these specific values during circulation. Problem 2: $P\_cord = 192$ W; percentage wasted $= 192/(192+60) \times 100 = 76\%$ — students who reach this should flag it as 'alarming' — this is the target affective response.

Driving Question Board (DQB) Creation 📹 **VIDEO:** Watch Me Source: TED-Ed Lessons 🔍 Search ARES for similar videos 📖 **HTML:** Electric Power and Electrical Energy Use (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Each student writes one new sticky note (or ARES digital card) to add to the class Driving Question Board. The note must complete ONE of the following sentence starters, chosen by the student: (A) 'Now I can explain why the Huruma Estate lights dimmed because… [use power language]'; (B) 'The extension cord wastes energy because… and the formula that shows this is…'; (C) 'A new question I have about power in circuits is…'; (D) 'I used to think the kettle took longer because of [voltage/current/resistance], but now I think it is because of [power] because…'. Students then do a Gallery Walk (4 minutes) — they move around the room, reading at least 5 other cards, placing a tick on any card that 'advances the class explanation' and a '?' on any card that raises a question they share. The class reconvenes; teacher reads aloud 2–3 starred cards and 1–2 question cards, connecting them explicitly to where the unit is heading (Lesson 10: energy, kWh, Kenya Power billing; Lesson 11: circuit design; Lesson 12: final model).	Say: 'You have 3 minutes — one card, one of those four starters. Make it specific — I want numbers or formula names, not just 'I learned about power.' [TIMER — 3 minutes.] During Gallery Walk: 'Move silently. Read. Tick what advances our class understanding. Question-mark what you are not sure about.' [TIMER — 4 minutes. Circulate and identify 2–3 cards to spotlight.] After reconvening, hold up 2–3 selected cards and read verbatim: 'Listen to what [student's table group] wrote: [read card]. Class — does this explanation answer our driving question? What is still missing?' [WAIT 10 seconds — cold-call 2 students.] Then say: 'Our driving question asks about delivering the RIGHT AMOUNT of energy SAFELY and EFFICIENTLY. Today we gave it a number — power in watts. Next lesson we will ask: over a whole month, what does that cost on your KPLC prepaid meter — and what can we do to lower it?'	Strategy: Driving Question Board Gallery Walk with Peer Evaluation. The DQB is a living artefact of the class's growing explanation. Requiring students to use power language (watts, $P = I^2R$, dissipation) in their cards prevents superficial updating. Peer ticking and questioning creates a distributed peer-assessment system where the class itself identifies which ideas are load-bearing and which are still uncertain. The teacher's spotlighting move connects individual student thinking to the collective storyline.	Observable indicator 1: At least 80% of cards contain at least one formula symbol or quantitative reference (W, kW, $P = IV$, etc.) — not purely qualitative. Observable indicator 2: During Gallery Walk, teacher scans for cards that correctly connect power dissipation to the extension cord phenomenon — these indicate students have crossed the key conceptual threshold. Observable indicator 3: Student-generated question cards reveal productive next-step curiosity (e.g., 'how does Kenya Power calculate our bill from watts?' or 'if we reduce current, does the jiko still heat as fast?') — these are captured and displayed as 'Questions to Answer' in the remaining lessons.
Model Building Phase 📹 **VIDEO:** Watch Me Source: TED-Ed Lessons 🔍 Search ARES for similar videos 📖 **HTML:** Electric Power and Electrical Energy Use (Flexbook) Source: CK-12	Students retrieve their cumulative circuit model — a hand-drawn or ARES digital annotated circuit diagram that they have been revising since Lesson 1. Today's revision task: annotate the model to show POWER. Specifically: (1) Add power values (in W) for each component in their existing Huruma Estate household circuit diagram (a parallel circuit with three branches: electric jiko, immersion heater, and two bulbs, all connected to the mains supply with a long feeder cable of known	Say: 'Take out your cumulative model — the circuit diagram you have been building all unit. Today it gets a power layer.' [Display the Huruma Estate circuit schematic on the projector with component values labelled.] 'I want to see four things when I look at your model: power values on every component, a power budget bar, a red danger annotation on the cable, and one connecting sentence to our driving question. You have 12 minutes.' Circulate and narrate what you observe: 'I can see Table 2 has	Strategy: Cumulative Model Revision with Quantitative Annotation. The cumulative model is the unit's central sensemaking artefact. Each lesson adds a new layer of understanding without erasing prior work, so students can literally see their growing model. Adding power values and a proportional power budget bar makes the abstract concept of 'wasted energy' spatial and visual — students see the red danger zone eating into their power budget. The extension calculation	Observable indicator 1: Power values for at least 3 of 4 circuit components are correctly calculated and labelled (e.g., jiko: $P = V^2/R$ or $P = IV$ correctly applied). Observable indicator 2: Power budget bar is drawn and shows the feeder cable consuming a clearly visible (and correctly proportioned) slice of total power — students who draw the cable's power as negligible are flagged for follow-up. Observable indicator 3: The connecting sentence explicitly uses the word 'power' and

🔍 Search ARES for similar readings	resistance). (2) Draw a 'power budget' bar — a horizontal bar divided proportionally to show what fraction of total power goes to each device and what fraction is wasted in the feeder cable. (3) Add a red warning annotation to the feeder cable: 'DANGER ZONE — power wasted here = [calculated value] W — this heats the wire.' (4) Write one sentence connecting their power model to the driving question: 'This model shows that devices receive less energy than expected because…'. Students who finish early extend by calculating: 'If this household runs the jiko for 3 hours and the immersion heater for 1 hour per day, what is the monthly electricity bill at Kenya Power's current residential tariff (approx. Ksh 21 per kWh)?'	calculated the power in the feeder cable — now what percentage of total power is that?' [WAIT for student response.] 'Table 4 — your bar is drawn but not to scale — fix it so the bar truly shows proportion.' In the last 2 minutes, ask 2 students (cold-call, one from a higher-performing group, one from a struggling group) to hold up their models and briefly explain their power budget bar to the class. Close by saying: 'Your model now shows volts, amps, ohms, AND watts. It is becoming a full engineering tool — exactly the kind of tool that Kenya Power engineers use when they design the wiring for a new estate. Next lesson, we will use it to think about energy over time and what that means for your household bill.'	(monthly kWh bill) connects physics directly to economic reality in a Kenyan household context, reinforcing the driving question's relevance.	references either the formula or a numerical value — qualitative sentences only ('less energy goes to the device') are insufficient and teacher prompts revision. Observable indicator 4 (extension): Monthly bill calculation is in the range of Ksh 300–450 for the described usage pattern — wildly different answers prompt a quick peer check.

D. TEACHER REFLECTION

After the lesson, reflect on the following questions in your ARES teacher journal:

1. DERIVATION MOMENT: When students substituted V = IR into P = IV at the board, did they make the substitution confidently or did they hesitate? If they hesitated, which prior knowledge gap does this reveal — weak algebra, or weak recall of Ohm's Law? Plan a 5-minute recap starter for Lesson 10 if needed.
2. PHENOMENON CONNECTION: During the extension cord worked example, did students express genuine surprise or alarm when they calculated the percentage of power wasted (potentially >70%)? If students were not surprised, the phenomenon may have lost its urgency — consider assigning a home investigation: 'Feel your phone charger after it has been plugged in for 1 hour. Is it warm? Calculate why.'
3. FORMATIVE DATA: How many students' DQB cards contained formula symbols or numerical values (target: ≥ 80%)? If fewer than 60% achieved this, the class is not yet operating at the quantitative level needed for Lesson 11's circuit design task — plan an additional practice set on the ARES platform as a homework bridge.
4. EQUITY CHECK: Were students from all table groups cold-called during the derivation and model-building phases? Review your cold-call notes — if the same 4–5 students answered most questions, redesign seating or use randomised ARES name-picker for Lesson 10.
5. KENYA POWER CONNECTION: Did students engage authentically with the monthly billing extension task? If so, consider opening Lesson 10 with a real KPLC prepaid token receipt and asking students to reverse-engineer how many kWh were purchased — this makes the unit's real-world relevance undeniable.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Students observed that: (1) a thin nichrome wire connected in a circuit became warm while a thick copper wire did not, even with the same supply voltage — and voltmeter/ammeter readings confirmed higher resistance and lower current in the nichrome circuit;

	(2) real Kenyan household device labels (electric jiko, immersion heater, phone charger) carry wattage ratings that can be used to calculate the current they draw; (3) their own Lesson 6 data tables, when extended with a 'Power (W)' column, showed that components with higher resistance dissipated more power for the same current.
What did I learn?	Students learned that: (1) electrical power $P = IV = I^2R = V^2/R$ is the rate of energy transfer in a circuit, measured in watts; (2) power dissipated in a resistive conductor (such as an extension cord) increases with the SQUARE of current, meaning even a small increase in current causes a large increase in heating; (3) a long, thin extension cord with high resistance wastes a significant fraction of the total power as heat, leaving less power for the connected device — this is the quantitative explanation for the dimmer bulb in the Huruma Estate phenomenon; (4) electrical energy consumed is calculated as $E = Pt$, and is measured commercially in kilowatt-hours (kWh) by Kenya Power.
How does this explain the phenomenon?	Students can now explain: (1) why the neighbour's bulb on a long, thin extension cord is always dimmer — the cord dissipates power as heat ($P = I^2R$), reducing the power available to the bulb; (2) why the electric kettle in Huruma Estate takes longer to boil during low-voltage periods — lower voltage means lower power delivered ($P = V^2/R$), so energy is transferred to the water more slowly; (3) why undersized extension cords and overloaded circuits are genuine fire hazards — the power dissipated as heat in the wire can ignite insulation; (4) how to read a device label and calculate its operating current and resistance, and how to estimate a household's monthly electricity cost using the Kenya Power tariff.

LESSON 10 (40 minutes): Electrical Energy and Kenya Power Billing	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	Students will calculate electrical energy consumption using E = Pt, convert joules to kilowatt-hours, analyse a sample Kenya Power electricity bill, and evaluate energy conservation strategies for low-income Kenyan households.
Knowledge	Define electrical energy as E = Pt; explain the kilowatt-hour as the commercial unit of electrical energy; calculate the cost of running named household appliances from their power ratings and usage times using Kenya Power tariff rates; distinguish between power (rate of energy use) and energy (total amount used).
Skills	Calculate electrical energy in joules and kilowatt-hours; read and interpret a Kenya Power electricity bill; calculate monthly appliance running costs; design a household energy budget; present quantitative recommendations for energy saving.
Attitudes	Appreciate the economic importance of energy efficiency; develop responsibility toward household and community energy use; show empathy for the financial constraints of low-income families in Nairobi and rural Kenya.
Key Inquiry Question	How is electrical energy consumed and managed safely in our homes and communities?
Purpose in Storyline	This lesson converts the power concepts from Lesson 9 into cumulative energy over time — the quantity that Kenya Power actually charges for. Students now have a complete quantitative tool to explain the Huruma Estate phenomenon: not only was power delivery reduced during the surge, but the total energy delivered to appliances was also reduced, prolonging the time needed to boil the kettle. This sets the stage for Lesson 11 safety calculations and the Lesson 12 household wiring design task.
Safety Notes	No live electrical equipment is used in this lesson. All calculations are paper-based or use printed bill facsimiles. Ensure students do not bring actual Kenya Power bills with personal account numbers into group display.

B. LESSON OVERVIEW

Students begin by predicting how long a standard 3 kW electric jiko would take to consume one Kenya Power unit (1 kWh) and what that would cost. They then examine real appliance data cards and a printed sample Kenya Power bill to observe how energy consumption is measured and billed. Through guided calculation tasks set in Nairobi household contexts, students derive and apply E = Pt, convert between joules and kilowatt-hours, and calculate monthly costs for common appliances. The lesson closes with students updating the Driving Question Board: they can now explain why the Huruma Estate kettle took longer to boil in terms of both reduced power AND reduced energy delivery per minute, and they set up the question of how fuses and safe current limits protect against dangerous energy dissipation.

C. LESSON IMPLEMENTATION FRAMEWORK				
Phase	**Learner Experience**	**Teacher Moves**	**Sensemaking Strategy**	**Formative Assessment Strategy**
Predict Phase 📹 **VIDEO:** WATCH: Humans and Energy Source: Khan	Students are presented with an Elbow-Partner Predict card (one per pair) showing: a 3000 W electric jiko (Kenya ceramic charcoal-electric hybrid), a 60 W	Teacher displays the three appliance images on the board using a printed poster or projector. Teacher says: 'Before we do any calculations today, I want to know	Prediction cards activate prior knowledge of power ratings from Lesson 9 and create cognitive dissonance by requiring students to integrate time — a variable not	Teacher scans prediction cards during circulation. Key diagnostic: do students recognise that energy depends on BOTH power AND time, or do they rank appliances by

Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Electric Power and Electrical Energy Use (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	bulb, and a 750 W immersion heater, each running for 2 hours. The card asks: (1) Which appliance uses the most electrical energy in 2 hours? (2) If electricity costs Ksh 20 per unit, predict the cost for each appliance for the 2-hour period. (3) Return to the Huruma Estate phenomenon: the kettle rated at 2200 W took 8 extra minutes to begin boiling on the night of power restoration. Predict whether this wasted energy or saved energy for the household, and why. Students write individual predictions first (2 minutes), then share with their partner (1 minute). A show-of-hands temperature check is taken for question 1 before any instruction begins.	what your gut tells you. Look at these three appliances — the electric jiko your family might use at home, a standard bulb, and an immersion heater. Which one do you think uses the most electricity in 2 hours? Write your answer and a one-sentence reason.' WAIT 2 MINUTES in silence — circulate and note which misconceptions appear (common error: students confuse high power with high energy without considering time; some may say the bulb because it runs longer in practice). After partner sharing, teacher asks: 'Hands up if you chose the electric jiko.' Then: 'Hands up for the immersion heater.' Then: 'Hands up for the bulb.' Teacher does NOT confirm or deny. Teacher says: 'Hold your prediction — we are going to find out exactly who is right and why. And at the end of the lesson, I want you to come back to your Huruma Estate prediction and see if the kettle situation was about power, energy, or both.' WAIT 30 SECONDS for students to re-read their prediction before moving on.	yet formally combined with power in their mental model. The Huruma Estate callback maintains narrative coherence.	power rating alone? This informs pacing in the Explain phase.
Observe Phase 📹 **VIDEO:** WATCH: Humans and Energy Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Electric Power and Electrical Energy Use (Flexbook) Source: CK-12	Students are given two printed artefacts: (A) an Appliance Data Table listing six common Kenyan household devices — 2200 W electric kettle, 60 W CFL bulb, 3000 W electric jiko, 750 W immersion heater, 5 W phone charger, 1000 W flat iron — with typical daily usage times as reported in a Kenya National Bureau of Statistics household survey excerpt (adapted for classroom use); and (B) a Sample Kenya Power Electricity Bill facsimile for a fictional household 'Wanjiru wa Kamau, Huruma	Teacher distributes printed artefacts and says: 'You are now going to do what a Kenya Power billing engineer does every month. Everything you need is on these two sheets. Your job is to read the bill, understand what each line means, and then calculate the energy cost for each appliance in Wanjiru wa Kamau home.' Teacher pauses and says: 'Before you calculate anything — spend 90 seconds just reading the bill silently. Notice what units are used. Notice what the numbers mean.' WAIT 90 SECONDS.	The authentic Kenya Power bill artefact grounds abstract energy calculations in a real-world financial document students may have seen at home but never analysed. Appliance data from a Kenyan context makes the numbers meaningful. The observation task requires students to read quantitative information from a document rather than a contrived textbook table, developing scientific literacy.	Observation Recording Sheets are collected at the end of the phase. Teacher checks: (1) correct subtraction of meter readings to get kWh consumed; (2) correct identification of the tariff rate; (3) correct use of E = Pt with consistent units (convert watts to kilowatts before multiplying by hours). Groups making unit errors (e.g., using watts instead of kilowatts) are flagged for targeted support in the Explain phase.

🔍 Search ARES for similar readings	Estate, Nairobi' showing: meter readings (previous and current), units consumed (kWh), Energy Charge at Ksh 12.00 per kWh (Domestic Lifeline Tariff block), Fuel Cost Charge, REP Levy, Fixed Charge, VAT, and Total Amount Due of Ksh 1 240. Students work in groups of four to complete an Observation Recording Sheet with three tasks: (1) Calculate the total kWh shown on Wanjiru wa Kamau bill from the meter readings. (2) Identify which line on the bill represents the cost per unit and record it. (3) Using the Appliance Data Table, calculate the daily energy consumption in kWh for each appliance, then estimate its monthly cost using the Ksh 12.00 per kWh tariff. Groups record all working in their exercise books.	Teacher then asks: 'What unit does Kenya Power use to charge you? Shout it out.' (Expected: kilowatt-hour or units.) Teacher writes kWh on the board and says: 'Good. This is the unit of electrical ENERGY. Not power — energy. Who can tell me the difference between power and energy from last lesson?' WAIT 20 SECONDS for hands. Teacher cold-calls one student, then redirects to a second student for elaboration. Teacher says: 'Exactly. Power is the RATE — how fast energy is used, measured in watts. Energy is the TOTAL amount used — how much altogether, measured in joules or kilowatt-hours. Now go ahead and complete your observation sheet. I will come round.' Teacher circulates, asking probing questions: 'What does the meter reading difference tell you?', 'Why does the phone charger cost so little even though you use it every day?', 'Why is the electric jiko the biggest energy user even though it is not the highest power device on this list?'		
Explain Phase 📹 **VIDEO:** WATCH: Humans and Energy Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Electric Power and Electrical Energy Use (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Students are guided through a structured whole-class derivation and worked-example sequence. Part 1 — Derivation (5 minutes): Starting from P = E/t (from Lesson 9), students rearrange to get E = Pt. Teacher demonstrates on the board that if P is in watts and t is in seconds, E is in joules; if P is in kilowatts and t is in hours, E is in kilowatt-hours. Students record both forms in their notebooks. Part 2 — Worked Example Set (10 minutes): Teacher works through three graded examples on the board, each connected to the Huruma Estate story: Example 1	During derivation, teacher says: 'From last lesson, power is energy divided by time. I am going to rearrange that formula. Watch carefully — tell me when I make a step.' Teacher writes slowly, pausing after each algebraic step. After writing E = Pt, teacher says: 'This is one of the most useful equations in physics AND in household budgeting. Write it down in a box in your notebook. Underline it.' During Example 1, teacher narrates unit conversion explicitly: 'I have 2200 watts. Kenya Power charges in kilowatts. What do I do? Divide by 1000.	Worked examples are sequenced from straightforward calculation (Example 1) to phenomenon-connected extension (Example 2 revisiting Huruma Estate with quantitative precision) to proportional reasoning (Example 3). The energy conservation discussion develops higher-order thinking and connects physics to household economics — a core CBC value.	Peer-marked individual practice task gives immediate formative data. Teacher notes how many students correctly convert watts to kilowatts (most common error), whether students multiply daily consumption by 30 correctly, and whether students can apply the tariff rate. Three targeted questions on the board guide peer marking: (1) Did your partner convert W to kW? (2) Did your partner multiply by the correct number of hours? (3) Did your partner multiply kWh by Ksh 12.00?

	— 'Wanjiru wa Kamau 2200 W kettle runs for 5 minutes to boil water. Calculate the energy in joules and in kWh.' Example 2 — 'On the night of power restoration, due to low terminal voltage the kettle received only 1800 W effective power. It ran for 13 minutes to achieve the same boil. Calculate the extra energy consumed in kWh and the extra cost at Ksh 12.00/kWh.' Example 3 — 'The 3 kW electric jiko runs 3 hours per day for 30 days. Calculate the monthly cost and express it as a fraction of Wanjiru wa Kamau total bill of Ksh 1 240.' Part 3 — Student Practice (5 minutes): Students individually solve a parallel task: 'Otieno in Kisumu uses a 750 W immersion heater for 45 minutes every morning and a 60 W bulb for 6 hours every evening. Calculate his monthly energy consumption in kWh and monthly cost at Ksh 12.00 per kWh.' Students swap books with a neighbour for peer marking using the answer displayed on the board. Part 4 — Energy Conservation Discussion (3 minutes): Students briefly discuss in pairs: 'Suggest TWO changes Wanjiru wa Kamau could make to reduce her bill by at least 20% without removing any appliance entirely.' Pairs share suggestions aloud; teacher scribes on the board.	2200 divided by 1000 is 2.2 kilowatts. Write that step every time.' WAIT 15 SECONDS for students to write. During Example 2, teacher says: 'Here is the connection back to Huruma Estate. Because the terminal voltage was lower — remember Lesson 8 — the effective power delivered to the kettle was lower. Lower power means LESS energy per minute. So to get the same job done — boiling the same water — the kettle had to run LONGER. And running longer means MORE energy used, MORE cost. Does that surprise you?' WAIT 20 SECONDS for student reactions. Teacher cold-calls two students for responses. During the conservation discussion, teacher says: 'You are not engineers yet, but you are about to think like one. If you were advising Wanjiru wa Kamau on how to cut her bill, what would you say? Think about which appliances use the most energy and whether the usage time can be reduced.' WAIT 2 MINUTES for pair discussion before taking responses.		
Driving Question Board (DQB) Creation 📹 **VIDEO:** [WATCH: Humans and Energy] Source: Khan	Each student receives a sticky note of a designated colour (green for new evidence, orange for updated questions). Students are asked to respond to two prompts: Green sticky: 'Write one piece of new evidence from today that helps explain why the Huruma	Teacher says: 'Today you added a new tool to your model of electricity. You can now calculate not just how fast energy is used, but how much energy is used over time — and what it costs. Let us add that to our Driving Question Board.' Teacher hands out sticky	The colour-coded DQB protocol makes visible both the accumulation of evidence and the generation of new questions — the two engines of scientific inquiry. The gallery read builds community knowledge and validates student contributions. The teacher re-	The most-ticked green stickies reveal which concepts have been successfully consolidated across the class. The most-ticked orange stickies reveal shared misconceptions or genuine curiosity gaps that should inform Lesson 11 planning. Teacher

Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Electric Power and Electrical Energy Use (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Estate kettle took longer to boil — use numbers if you can.' Orange sticky: 'Write one question this lesson has raised for you about electricity safety or billing that you still want answered.' Students post their notes on the class DQB under two columns: 'New Evidence' and 'New Questions'. The teacher leads a 3-minute gallery read where students read five sticky notes from others (not their own) and place a small tick on any they agree with or find surprising. The teacher then reads aloud the three most-ticked green stickies and the three most-ticked orange stickies. The teacher circles orange questions related to safety and fuses, saying: 'You will notice several of you are asking: what stops a circuit from drawing too much current and using dangerous amounts of energy? That is exactly what Lesson 11 is about — the heating effect of current and electrical safety.'	notes and reads the two prompts aloud, then says: 'You have 90 seconds. Write something specific — not just a general comment. Use a number, a formula, or a name from today.' WAIT 90 SECONDS in silence. After posting, teacher says: 'Walk to the board and read five notes that are not yours. Put a tick on ones you agree with or find interesting. You have 2 minutes.' WAIT 2 MINUTES. Teacher then reads the top-ticked notes aloud, validating student language before restating in precise physics vocabulary. For example, if a student wrote 'less power means more time to boil so more money wasted,' teacher says: 'Excellent — let me make that even more precise: lower terminal voltage led to lower effective power delivered to the kettle, so to transfer the same thermal energy to the water, the kettle had to operate for a longer time, consuming more electrical energy and incurring greater cost. That is a complete physics explanation.' Teacher explicitly links orange safety questions to Lesson 11 to maintain storyline coherence.	voicing move models scientific language without dismissing student language.	photographs the board before the end of lesson.
Model Building Phase 📹 **VIDEO:** WATCH: Humans and Energy Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Electric Power and Electrical	Students update their Personal Circuit Model — a cumulative diagram started in Lesson 1 that now includes labels for EMF, terminal voltage, internal resistance, Ohm's Law, series and parallel rules, and power dissipation. Today students add a new annotated layer: an Energy Accounting Box. In this box students draw and label: (1) the formula $E = Pt$ with units in joules and kilowatt-hours; (2) a simple two-column table showing Power	Teacher says: 'Open your Circuit Model to the latest page. Today we are adding an Energy Accounting Box. This is the layer that connects physics to economics — it is how Kenya Power actually thinks about your household circuit.' Teacher draws the Energy Accounting Box template on the board and walks through each element: 'Formula first — E equals P times t. Write it large. Now fill in your three-appliance table using numbers	The cumulative Personal Circuit Model makes knowledge growth visible over the unit. The Energy Accounting Box metaphor links to financial literacy, a cross-cutting CBC competency. The Model Claim writing task develops scientific argumentation — students must distil a lesson into a defensible, falsifiable statement. The class voting protocol on claim completeness mirrors the social process of scientific consensus-building.	Teacher collects three to four Personal Circuit Models at random to check: (1) correct placement of $E = Pt$ with both unit systems; (2) numerically correct appliance table; (3) accurate and specific Huruma Estate speech bubble; (4) a Model Claim that includes power, time, kilowatt-hours, and cost. Feedback is written on a sticky note attached to the model and returned at the start of Lesson 11.

Energy Use (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	(W) vs Energy per hour (kWh) for three appliances from today; (3) a speech bubble from the Huruma Estate house saying: 'Low terminal voltage meant lower power to the kettle AND more energy needed per boiling cycle AND higher cost per successful boil.' Students also write a one-sentence Model Claim at the bottom of their model page: 'Electrical energy is the product of power and time, measured in kilowatt-hours commercially, and its cost depends on both the power rating of a device and how long it is used.' A selected student reads their Model Claim aloud for class comparison, and the class votes on whether it is complete, partially complete, or needs revision.	from today. Then write the speech bubble from the Huruma house. Make it sound like someone who now understands the physics — use the words terminal voltage, power, energy, and cost.' WAIT 4 MINUTES for students to complete their models. Teacher circulates, checking that students are using correct units and connecting the Huruma Estate phenomenon explicitly. Teacher says: 'Now write your one-sentence Model Claim. It should summarise what electrical energy IS and how we calculate its COST. You have 60 seconds.' WAIT 60 SECONDS. Teacher selects one student to read aloud. Teacher says to the class: 'Does this claim tell us everything we need? Thumbs up if it is complete, sideways if partially complete, down if something important is missing.' WAIT 15 SECONDS for responses. Teacher facilitates a brief revision of the claim if needed, saying: 'What word or idea could we add to make this claim even stronger?' WAIT 20 SECONDS for suggestions.		

D. TEACHER REFLECTION

After the lesson, consider: Did students successfully distinguish power (rate) from energy (total amount) — the most common confusion in this topic? Did the Kenya Power bill artefact feel authentic and accessible, or did students struggle with the document format? Were the worked examples paced appropriately, or did unit conversion (W to kW) consume too much time? Did the DQB sticky note protocol generate genuine new questions pointing toward Lesson 11 safety content? Which groups struggled most with the energy conservation discussion — was it a mathematics issue or a conceptual issue about what variables can be changed? For Lesson 11, prepare to reference the orange DQB sticky notes explicitly by reading two or three aloud at the lesson opening to show students their questions were heard and will be answered.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Students examined a sample Kenya Power electricity bill for a fictional Huruma Estate household and an appliance data table showing power ratings and daily usage times for six common Kenyan household devices including an electric jiko, kettle,

	immersion heater, and phone charger.
What did I learn?	Electrical energy E = Pt, where energy in kilowatt-hours equals power in kilowatts multiplied by time in hours; one kilowatt-hour (1 kWh) is one unit on a Kenya Power bill charged at approximately Ksh 12.00 on the domestic lifeline tariff; the total cost of running an appliance depends on BOTH its power rating AND the duration of use; the Huruma Estate kettle consumed more total energy during the power restoration event because lower terminal voltage reduced the power delivered, requiring a longer running time to boil the same water.
How does this explain the phenomenon?	The Huruma Estate kettle took longer to boil because the lower terminal voltage during the power surge reduced the effective power delivered to the heating element — less power per second meant less energy transferred to the water per minute, so more minutes were needed to transfer enough thermal energy to reach boiling point, and this longer running time also meant greater total electrical energy consumed and greater cost on the Kenya Power bill.

LESSON 11 (80 minutes): Heating Effect of Electric Current & Electrical Safety	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To investigate the heating effect of electric current, evaluate its practical applications in Kenyan households, and analyse electrical safety devices and practices that prevent fires and accidents in homes.
Knowledge	By the end of the lesson, the learner should be able to: (a) explain how electrical energy is converted to heat energy when current flows through a resistor (Joule heating: $Q = I^2Rt$); (b) describe the operating principles of electrical heating appliances (electric cooker, immersion heater, toaster); (c) explain how fuses, circuit breakers, and earthing protect household circuits; (d) analyse common causes of electrical fires in Kenyan homes and propose safety recommendations.
Skills	By the end of the lesson, the learner should be able to: (a) conduct and record a fair investigation into the heating effect of current for different resistors; (b) calculate heat produced using $Q = I^2Rt$ and cross-check with $P = I^2R$; (c) design a simple electrical safety guide for a Kenyan household; (d) interpret fuse ratings and select the correct fuse for a given appliance.
Attitudes	The learner appreciates the dual nature of the heating effect of current — its useful applications in everyday Kenyan life and the serious hazards it poses when circuits are poorly designed or maintained — and demonstrates responsibility by committing to safe electrical practices at home and in the community.
Key Inquiry Question	How is electrical energy consumed and managed safely in our homes and communities?
Purpose in Storyline	In previous lessons, students built understanding of Ohm's Law, internal resistance, series and parallel circuits, and electrical power. They know that resistance causes energy losses and that current through a resistor dissipates power as $P = I^2R$. In this lesson, students now investigate the thermal consequence of that power dissipation — Joule heating — and connect it directly to the anchoring phenomenon: the neighbour's long, thin extension cord (high resistance) heats up dangerously while delivering less energy to the bulb. This lesson also explains why the underpowered kettle during the brownout took longer to boil — less voltage meant less current, so $Q = I^2Rt$ yielded less heat per unit time. Students cap the evidence-gathering sequence by studying the safety systems (fuses, circuit breakers, earthing) that prevent the heating effect from causing fires, directly addressing the 'safely and efficiently' clause of the driving question.
Safety Notes	SAFETY: All experiments use low-voltage DC supplies ($\leq$ 6 V) or teacher-demonstrated setups only. No mains voltage is to be handled by students. Nichrome wire becomes hot — students must not touch it during or immediately after current flows. Warn students never to use damaged extension cords or overload sockets at home. If using a candle or match to demonstrate an analogy, follow school fire-safety protocols. Dispose of any melted fuse wire safely.

B. LESSON OVERVIEW

Students investigate the heating effect of electric current through a guided experiment using nichrome wire, low-voltage DC sources, and thermometers (or temperature sensors). They use $Q = I^2Rt$ to calculate heat produced and connect this to how household appliances (electric cookers, immersion heaters, toasters) work. They then pivot to the hazard side: analysing why electrical fires start in Kenyan homes (overloaded sockets, thin extension cords, worn insulation) and how safety devices — fuses, circuit breakers, earthing — interrupt dangerous currents before fires ignite. The lesson anchors throughout to the Huruma Estate phenomenon: the long, thin extension cord is now fully explained as a high-resistance element that both wastes energy as heat AND poses a fire risk. Students conclude by designing a household electrical safety guide for a Huruma Estate family.

C. LESSON IMPLEMENTATION FRAMEWORK				
Phase	**Learner Experience**	**Teacher Moves**	**Sensemaking Strategy**	**Formative Assessment Strategy**
Predict Phase 📹 **VIDEO:** Introduction to work and energy Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **PDF:** topical-test-form-3-physics-heating-effect-of-electric-current-answerse Source: Kenya Curriculum Tools 2025 🔍 Search ARES for similar readings	Students are shown two images side by side on the board: (1) a glowing coil inside an electric toaster in a Nairobi kitchen, and (2) a charred, melted extension cord photographed after an electrical fire in Huruma Estate. Teacher poses the prediction prompt: 'Both of these involve current flowing through a wire. In your science notebook, write answers to these two questions — (A) Why does the toaster coil glow orange-hot while the copper supply wires stay cool? (B) The neighbour in Huruma Estate used a long, thin extension cord. We know it caused the bulb to be dimmer. Could that same cord be dangerous? Why or why not?' Students write individual predictions silently for 4 minutes, then share with a shoulder partner for 2 minutes. Three pairs are cold-called to share. Teacher records key prediction words on the DQB column labelled 'What we THINK we know — Lesson 11'.	Display both images clearly. Read the two prediction questions aloud slowly. Say: 'You have four minutes to write your own prediction — no talking yet. Use what you already know about resistance and power from our last lessons.' Set a visible 4-minute timer. After timer: 'Now share your prediction with your shoulder partner — you have 2 minutes.' After partner talk: Cold-call exactly 3 pairs using a name-stick system. For each pair, ask: 'What was your prediction, and what earlier evidence made you think that?' WAIT 12 seconds after each question before accepting the answer. Probe with: 'You said the thin cord might get hot — which formula from Lesson 10 supports that idea?' Record student language verbatim on the DQB. Do NOT correct misconceptions yet — annotate them with a question mark for later revisiting.	STRATEGY — Predict-Discuss-Share (PDS): Students first commit to individual written predictions, activating prior knowledge of $P = I^2R$ from Lesson 10. Partner discussion forces articulation and immediate peer challenge before whole-class sharing. Recording predictions verbatim on the DQB creates a public accountability record that students will revise at the end of the lesson, making conceptual change visible.	Observable indicator: Every student has two written predictions in their notebook before partner talk begins. Teacher circulates during silent writing and scans notebooks — look for whether students spontaneously reference $P = I^2R$ or resistance. Flag students who write 'thin wires carry less electricity' (common misconception about current vs. energy) for targeted questioning during the Observe phase. Cold-call responses reveal whether students are connecting resistance to heat generation.
Observe Phase 📹 **VIDEO:** Introduction to work and energy Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **PDF:** topical-test-form-3-physics-heating-effect-of-electric-current-	PART A — Heating Effect Experiment (25 min): In groups of four, students set up a simple circuit: low-voltage DC power supply (3 V and 6 V settings), a 30 cm length of nichrome wire (SWG 28) coiled around a small glass test tube containing 20 mL of water, an ammeter in series, and a voltmeter in parallel across the nichrome coil. A thermometer is placed in the water. Groups record: initial water temperature, current (A), voltage (V), and temperature after 3 minutes at 3 V,	Before distributing equipment, demonstrate the setup at the front bench for 90 seconds. Say: 'Your nichrome wire is the heating element — it is like the coil inside a toaster. The copper wires are like the supply cables. Watch what happens to temperature as we increase voltage.' Circulate during the experiment. At the 3-minute mark for the 3 V trial, prompt groups: 'Read your ammeter and thermometer NOW — record the values before you change voltage.' After both trials, ask groups:	STRATEGY — Data-to-Pattern Bridge: Students generate real temperature-change data from two voltage conditions, then immediately compare patterns (higher V → higher I → greater ΔT). By also calculating appliance resistances from real-world Kenyan product ratings, students bridge lab data to everyday objects they recognise from their homes. The contrast between the nichrome wire (hot) and copper connecting wire (cool, same current) provides a direct, tactile	Observable indicators: (1) Results tables are complete with units — flag groups with missing resistance column or incorrect ammeter readings. (2) During appliance card task, listen for groups correctly applying $R = V^2/P$; if groups incorrectly use $R = V/P$, intervene immediately with: 'Which formula relates power, voltage, and resistance? Let us check your formula sheet.' (3) The cold-called groups should articulate that higher resistance in the nichrome coil means more energy is

answerse Source: Kenya Curriculum Tools 2025 🔍 Search ARES for similar readings	then repeat at 6 V. They record all data in a results table: | Trial | Voltage (V) | Current (A) | Resistance (Ω) | Time (s) | ΔT (°C) |. Students also briefly touch the nichrome wire housing (not the wire itself) to notice warmth, then touch the copper connecting wires — comparing the two. PART B — Appliance Case Studies (10 min): Each group receives a card describing a Kenyan household appliance: Group 1 — Ramtons electric kettle (2200 W, 240 V); Group 2 — Sayona electric cooker hotplate (1500 W, 240 V); Group 3 — Nunix immersion water heater (1500 W, 240 V); Group 4 — a locally made toaster (800 W, 240 V). Groups calculate the resistance of their appliance's heating element using $R = V^2/P$, then calculate the current drawn using $I = P/V$. They record these values and discuss: 'Why does the heating element need HIGH resistance while the supply wires need LOW resistance?'	'Compare the temperature rise at 3 V versus 6 V — what happened to the current when you doubled the voltage?' WAIT 10 seconds. For the appliance cards, say: 'Use $R = V^2/P$ — the formula you derived in Lesson 10. Then find the current. Think about why a heating element should have high resistance.' Cold-call 2 groups to share their calculated resistance values. Ask the class: 'Why would a kettle element with high resistance get very hot, but the copper flex cord carrying the same current stays cool?' WAIT 15 seconds.	observation that resistance — not current alone — determines heating. This embeds NGSS SEP 3: Planning and Carrying Out Investigations and SEP 4: Analysing and Interpreting Data.	converted to heat per unit charge — if they say 'it attracts more electricity,' redirect to the $P = I^2R$ relationship.
Explain Phase 📹 **VIDEO:** Introduction to work and energy Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **PDF:** topical-test-form-3-physics-heating-effect-of-electric-current-answerse Source: Kenya Curriculum Tools 2025 🔍 Search ARES for similar	PART A — Joule Heating (10 min): Teacher leads a whole-class explanation of the Joule heating equation: $Q = I^2Rt$, connecting it explicitly to $P = I^2R$ (which students derived in Lesson 10): 'Power is the rate of energy transfer — multiply by time to get total heat energy.' Students work in pairs to solve two contextual problems: (1) 'A Ramtons kettle draws 9.2 A from a 240 V supply. Its element has resistance 26 Ω. How much heat does it produce in 3 minutes (180 s)? Is this enough to boil 1.5 L of water starting at 20°C?' [Use $Q = mc\Delta T = 1.5 \times 4200 \times 80 = 504{,}000$ J; compare with $Q = I^2Rt = 9.2^2 \times 26 \times 180 \approx 394{,}000$ J —	Write $Q = I^2Rt$ on the board and say: 'We already know $P = I^2R$. Power times time equals energy — so this is simply that equation extended. Let us use it to explain the Huruma brownout.' Assign Problem 1 to pairs — set a 4-minute timer. Circulate and prompt: 'Your answer for Q should be in joules — does your unit work out?' After 4 minutes, cold-call one pair to present their whiteboard. Ask the class: 'Is the heat produced enough to boil the water? What does that tell us about efficiency?' WAIT 12 seconds. Transition: 'Now let us go back to Huruma — the voltage was lower. Use $I = V/R$ first, then	STRATEGY — Worked Example → Student Application → Phenomenon Link (WE-SA-PL): Teacher models one Joule heating calculation with full unit tracking, then immediately transfers responsibility to student pairs for a phenomenon-tied problem. The deliberate choice of the Huruma brownout as the calculation context ensures the sensemaking is not abstract — students are calculating exactly why the kettle was slow on the night of the power restoration. This embeds NGSS SEP 5: Using Mathematics and Computational Thinking and SEP 6: Constructing Explanations.	Observable indicators: (1) Mini-whiteboards show correct substitution into $Q = I^2Rt$ with consistent units (J, A, Ω, s). Flag pairs who mix up P and Q, or who forget to square the current. (2) In Problem 2, students should calculate $I = 180/26 \approx 6.9$ A (versus 9.2 A normal) and $Q \approx 222{,}000$ J versus ~394,000 J at normal voltage — if pairs do not notice the dramatic reduction, ask: 'How does this number compare to what you got in Problem 1? What would a resident notice?' (3) During the safety section, listen for students who can connect fuse rating to maximum safe current — a student who says 'fuse melts so

readings	prompt discussion about efficiency losses.] (2) 'During the Huruma Estate brownout, voltage dropped to 180 V instead of 240 V. The same kettle element has R = 26 Ω. Calculate the new current and the heat produced in 3 minutes. Why did the kettle take longer to boil?' Students present solutions on mini-whiteboards. PART B — Electrical Safety Devices (10 min): Teacher uses a labelled diagram on the board (or projected slide) to explain: fuse (melts when I exceeds rating, breaks circuit), circuit breaker (trips electromagnetically), earthing (provides safe path for fault current through earth wire rather than through a person). Students annotate the diagram in their notebooks. Teacher connects to phenomenon: 'The long, thin extension cord in Huruma — if it carried too much current, what would happen to the cord itself? How would a fuse in that circuit protect the family?'	$Q = I^2Rt$.' After Problem 2 is solved, ask: 'How much LESS heat was produced during the brownout compared to normal? Roughly what percentage?' For the safety section, say: 'A fuse is a deliberate weak point — it sacrifices itself to save the circuit. A circuit breaker is a reusable fuse.' Draw a simple household ring-main diagram with a fuse box. Cold-call students: 'If a 13 A fuse is in this circuit and the appliance draws 20 A — what happens?' WAIT 10 seconds.		it doesn't waste electricity' still holds a misconception and needs redirection.
Driving Question Board (DQB) Creation 📹 **VIDEO:** [Introduction to work and energy] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **PDF:** topical-test-form-3-physics-heating-effect-of-electric-current-answerse Source: Kenya	The class returns to the class Driving Question Board. Teacher reveals the 'What we THINK we know' predictions recorded at the start of the lesson and asks students to evaluate each one: 'Which predictions were correct? Which were partially right? Which do we now need to revise?' Students use sticky notes (green = confirmed, orange = revised, red = was a misconception) to annotate the DQB. Students then add new question sticky notes in a dedicated column: 'What NEW questions does today's evidence raise?' Examples expected: 'What is the right fuse rating for each appliance at home?', 'Can solar	Point to the DQB and say: 'Let us audit our predictions from the start of the lesson.' Read each recorded prediction aloud and ask: 'Green, orange, or red — and WHY?' For any prediction marked red (misconception), say: 'This was a brilliant prediction to make — now let us write a corrected version together.' Invite 3 students to share their one-sentence driving question connection. Listen for: mentions of $Q = I^2Rt$, fuse/circuit breaker function, resistance of extension cord, brownout current reduction. Cold-call one student who has been quieter in previous phases. Say: 'I want to hear from someone who hasn't shared yet	STRATEGY — Colour-Coded DQB Audit (Confirm-Revise-Question): The traffic-light annotation system makes conceptual change tangible and public. Students must justify their colour choice with evidence gathered in today's lesson, practising NGSS SEP 7: Engaging in Argument from Evidence. The new-questions column sustains curiosity and intellectual humility — students see that good science always generates new questions. Connecting back to the phenomenon in a one-sentence synthesis consolidates the lesson's contribution to the full evidence sequence.	Observable indicators: (1) Every student applies at least one sticky note with a written justification — scan for vague notes like 'it was wrong' without a reason. (2) New question sticky notes should be phenomenon-relevant, not random; off-topic questions indicate the student has not yet connected today's content to the unit storyline — note these students for Lesson 12 support. (3) One-sentence driving question answers in notebooks should reference either $Q = I^2Rt$, fuse/circuit breaker operation, or the extension cord hazard — generic answers ('electricity can be dangerous') indicate insufficient

Curriculum Tools 2025 🔍 Search ARES for similar readings	panels cause electrical fires too?', 'Why does Kenya Power use very thin wires in some areas?'. Teacher reads aloud one student-generated question that connects strongly to the driving question and asks the class: 'How does what we learned today move us closer to answering WHY the Huruma Estate lights were dim AND why the extension cord was dangerous?' Students write a one-sentence answer in their notebooks as an exit thought before the final phase.	today.' WAIT 15 seconds for their response. After the DQB update, explicitly preview: 'In Lesson 12 — our final lesson — we will bring ALL our evidence together into a complete explanation and a final model. Today's safety knowledge will be essential for the design task.'		conceptual integration.
Model Building Phase 📹 **VIDEO:** Introduction to work and energy Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **PDF:** topical-test-form-3-physics-heating-effect-of-electric-current-answerse Source: Kenya Curriculum Tools 2025 🔍 Search ARES for similar readings	Students retrieve their cumulative circuit model from Lesson 10 (a labelled diagram showing a source with internal resistance, external resistance, current flow, and power dissipation). They now ADD two new layers to this model: (1) A HEAT ANNOTATION layer — students annotate every resistive element in their model with an arrow labelled '$Q = I^2Rt$' pointing away from the component, with the arrow size proportional to the resistance of that element (the long thin extension cord gets a thick arrow; the copper supply wires get a tiny arrow). (2) A SAFETY DEVICE layer — students add a fuse symbol in series near the source, an earth wire connected to the metal casing of any appliance, and a circuit breaker symbol at the consumer unit (fuse box). Students also sketch a small inset diagram of a Huruma Estate household ring-main circuit showing: consumer unit with circuit breakers, earth wire, two parallel branch circuits (lights and sockets), and the problematic extension cord with its fuse recommendation labelled.	Distribute students' Lesson 10 models (stored in portfolios). Say: 'Your model is like a map of the circuit — today we add two new features: where heat escapes, and where safety devices sit.' Demonstrate on a large class model at the board: draw a fuse symbol, say 'A fuse is drawn as a small rectangle with a line through it — add this in series, close to the positive terminal of your source.' Walk around and check that heat arrows are correctly sized (bigger arrow on high-R components). After 8 minutes of individual model work, say: 'Find your model-partner group and do a 3-minute peer review — one glow, one grow, written on a sticky note on their model.' After peer review: 'Update your model now — you have 3 minutes.' Close by asking the whole class: 'Has anyone drawn the extension cord of the Huruma neighbour? Show me where the heat is escaping and where the fuse is.' Cold-call two students to hold up models and describe their extension cord annotation. WAIT 12 seconds after asking for volunteers before cold-	STRATEGY — Cumulative Model Revision with Peer Critique: The multi-lesson cumulative model serves as a running scientific explanation — each lesson adds one layer of mechanistic detail, mirroring how scientists iteratively revise models with new evidence. Adding proportional heat arrows operationalises $Q = I^2Rt$ visually, while the safety layer connects abstract circuit-protection concepts to a concrete household layout. Peer critique using glow/grow language develops NGSS SEP 8: Obtaining, Evaluating, and Communicating Information and builds collaborative scientific discourse norms. The extension cord annotation directly closes the loop on the anchoring phenomenon.	Observable indicators: (1) Fuse symbol is placed IN SERIES, not in parallel — a fuse drawn in parallel is a critical misconception; correct immediately with: 'A fuse must be in series so all current flows through it. If it is in parallel, current bypasses the fuse entirely — the protection is lost.' (2) Heat arrows should be larger on the nichrome/extension cord components than on copper wires — if all arrows are the same size, the student has not operationalised $Q \propto R$ (for constant I and t); prompt: 'Which component has the highest resistance in your circuit? Should its heat arrow be bigger or smaller than the copper wire?' (3) The Huruma ring-main inset should show parallel branches — if a student draws all appliances in series, revisit the Lesson 9 parallel circuit evidence and ask: 'What happens to all the lights in a series circuit if one bulb blows?'

	Groups share their updated models with an adjacent group and give one piece of 'glow' feedback (what is strong) and one piece of 'grow' feedback (what could be clearer). Students revise their models based on peer feedback.	calling.		

D. TEACHER REFLECTION

After the lesson, reflect on the following questions: (1) Did students successfully connect $Q = I^2Rt$ back to the brownout scenario, or did the calculation feel disconnected from the phenomenon? If the latter, consider opening Lesson 12 with a quick re-run of the Problem 2 numbers displayed on the board as a warm-up anchor. (2) Were there students who could calculate Joule heating correctly but could not explain WHY the thin extension cord is dangerous (i.e., they treated $Q = I^2Rt$ as a formula game rather than a physical explanation)? These students may need a targeted analogy: 'A thin pipe under high water pressure is more likely to burst — a thin wire under high current is more likely to overheat.' (3) How many students' cumulative models now correctly show: (a) heat escaping from high-resistance components, (b) fuse in series, (c) parallel branches in the household circuit? Use this as a readiness diagnostic for Lesson 12's final explanation task. (4) Were the safety recommendations students generated relevant to real Kenyan home hazards (overloaded sockets from multi-adapter plugs, use of worn extension cords, DIY wiring without a licensed electrician)? If the recommendations were generic, the Lesson 12 design task should provide a more structured scaffold. (5) Note which students generated the strongest new DQB questions — invite them to open Lesson 12 by reading their questions aloud to orient the final evidence synthesis discussion.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Students observed that when current flowed through nichrome wire coiled around a water-filled test tube, the water temperature increased — and the increase was greater at higher voltage (higher current). They observed that nichrome wire became warm while copper connecting wires carrying the same current stayed cool. They examined photographs of a charred extension cord from Huruma Estate and appliance rating plates for Kenyan household devices (Ramtons kettle, Sayona cooker, Nunix immersion heater).
What did I learn?	Students learned that electrical energy is converted to heat energy in any resistor carrying current, quantified by $Q = I^2Rt$ (Joule's Law). High-resistance elements (nichrome, toaster coil, kettle element) convert more electrical energy to heat than low-resistance conductors (copper wires), which is why heating appliances use high-resistance alloys deliberately. During the Huruma brownout, lower voltage reduced current ($I = V/R$), which dramatically reduced heat output ($Q \propto I^2$), explaining why the kettle was slow. The long, thin extension cord of the neighbour had high resistance, so it dissipated significant heat — making it both a dim-bulb cause AND a fire hazard. Fuses (melt and break the circuit), circuit breakers (trip electromagnetically), and earthing (divert fault current safely to ground) are safety devices that prevent dangerous overcurrents from causing fires or electrocution.
How does this explain the phenomenon?	The Huruma Estate phenomenon is now more fully explained: when power was first restored after the Kenya Power blackout, the supply voltage was below the standard 240 V (brownout condition). Lower voltage → lower current through each appliance → less power delivered ($P = V^2/R$) → less heat generated ($Q = I^2Rt$) → dimmer lights and a slower kettle. The neighbour's long, thin extension cord has high resistance, so even at full voltage it drops a significant portion of the supply voltage across itself ($V = IR$), reducing voltage across the bulb and wasting energy as heat in the cord. This same cord, if overloaded, could reach ignition temperatures — the fuse in the circuit is designed to melt and break the circuit before this happens, protecting the household from fire.

LESSON 12 (80 minutes): Model Building, DQB Resolution & Final Explanation: Designing Safe and Efficient Circuits for Kenyan Homes

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	To consolidate all sub-strand concepts — Ohm's Law, resistivity, series and parallel circuit rules, internal resistance, electrical power and energy, and safety — into a unified explanatory model that fully resolves the anchoring phenomenon of the flickering lights and slow kettle of Huruma Estate.
Knowledge	Students will know: (a) the relationship between voltage, current, and resistance as described by Ohm's Law; (b) how internal resistance causes terminal voltage to drop under load; (c) why domestic wiring is connected in parallel to ensure independent operation, low effective resistance, and consistent voltage across appliances; (d) how electrical power ($P = IV = I^2R = V^2/R$) and energy ($E = Pt$) determine appliance performance and cost; (e) safety principles including fuse ratings, earthing, and safe current limits for household circuits.
Skills	Students will be able to: (a) construct and justify a final annotated circuit model integrating all unit concepts; (b) design a safe and efficient wiring plan for a model Kenyan household; (c) calculate total current draw, appropriate fuse ratings, and monthly energy costs using Kenya Power tariff data; (d) compare their Lesson 1 initial model with their final model to articulate conceptual growth; (e) present a coherent final explanation for all anchoring and supporting phenomena.
Attitudes	Students will appreciate: (a) the relevance of physics to everyday Kenyan life and home safety; (b) the importance of evidence-based reasoning over intuition; (c) responsible use of electrical energy in the context of Kenya's energy resources and load-shedding challenges; (d) the value of collaborative, iterative model-building as a scientific practice.
Key Inquiry Question	Why do electrical devices in our homes sometimes receive less energy than expected, and how can we design circuits to deliver the right amount of electrical energy safely and efficiently?
Purpose in Storyline	This is the capstone lesson of the storyline. Students have spent eleven lessons gathering evidence about current, voltage, resistance, internal resistance, power, energy, and safety. Today they synthesise all of that evidence into a final explanatory model and a summative performance task. The Huruma Estate phenomenon — dim lights, slow kettle, dim bulb on long extension cord — is fully explained and resolved. The Driving Question Board is closed out. Students leave with a complete, coherent mental model of current electricity and the engineering skills to apply it.
Safety Notes	All circuit diagrams and wiring plans are paper-based or simulation-based — no live mains wiring is constructed by students. Emphasise to students that real household wiring must be done by a licensed electrician. Remind students that the safety principles they apply in their designs (correct fuse ratings, earthing, not overloading sockets) are Kenya Power and Kenya Bureau of Standards (KEBS) requirements. No electrical equipment beyond low-voltage demonstration kits (≤6 V) should be energised during this lesson.

B. LESSON OVERVIEW

In this final lesson of Sub-Strand 3.2, students complete a four-part capstone sequence. First, they revisit their original Lesson 1 circuit models and compare them with their current understanding — articulating every conceptual shift across the unit. Second, they deliver a structured Final Explanation for the Huruma Estate anchoring phenomenon, drawing on Ohm's Law, internal resistance, parallel wiring logic, power dissipation, and safety rules. Third, they close out every Driving Question Board card with evidence-referenced answers. Finally, they complete a summative performance task: designing a safe, efficient wiring plan for a model Kenyan household (the 'Wanjiku Family Home') with a given appliance list, a Kenya Power monthly budget of KES 2,000, and a 15 A mains fuse — calculating current draws, selecting fuse ratings, justifying parallel configuration, and estimating monthly energy costs. The lesson ends with a whole-class gallery walk of completed plans and a brief

metacognitive reflection on how their thinking changed from Lesson 1 to Lesson 12.

C. LESSON IMPLEMENTATION FRAMEWORK				
Phase	**Learner Experience**	**Teacher Moves**	**Sensemaking Strategy**	**Formative Assessment Strategy**
Predict Phase 📹 **VIDEO:** Video: Ohm's Law Source: PhET Interactive Simulations (English) 🔍 Search ARES for similar videos 📖 **HTML:** River Ferry and Current Vectors (PLIX) Source: CK-12 🔍 Search ARES for similar readings	Students retrieve their Lesson 1 initial circuit drawings and written predictions about why the Huruma Estate lights flickered. Working individually for 5 minutes, they annotate their original model with a red pen, circling every idea they now know to be incomplete or incorrect and starring every idea that turned out to be on the right track. They then turn to a partner and share: 'The biggest thing I was wrong about in Lesson 1 was… because…' Partners listen without interrupting, then swap. The class is cold-called (5 pairs) to share one red-circle item each. Teacher records key misconceptions on the board in a 'Then vs. Now' T-chart. Students write a one-sentence bridge statement in their notebooks: 'In Lesson 1 I thought _______, but now I know _______ because the evidence showed _______.'	Distribute Lesson 1 model sheets (pre-collected or from student notebooks). Say: 'Before we write our final explanation, I want you to be honest scientists. Look at what you believed on Day 1 of this unit. Use your red pen — no shame in being wrong; that is what learning looks like.' Allow 5 minutes of silent individual annotation. [WAIT TIME: 10 seconds after posing the partner-share prompt before releasing students to talk.] Circulate and identify 3–4 compelling 'then vs. now' contrasts to highlight. Cold-call 5 pairs using name cards: 'Amina, what did you circle in red, and why?' Record responses verbatim in the T-chart under 'Lesson 1 Misconception.' After 5 pairs, ask the class: 'Do you see a pattern in what we all misunderstood?' [WAIT TIME: 12 seconds.] Guide students to notice that most Lesson 1 models treated electricity as an on/off switch with no concept of resistance, internal resistance, or voltage drop — exactly what the Huruma phenomenon exposed.	Strategy: 'Then vs. Now Model Comparison.' By placing their original and current models side by side, students externalise conceptual change — making implicit learning explicit and building metacognitive awareness. This also surfaces any remaining gaps before the final explanation is written.	Observable indicator: Each student has at least 3 red-circle annotations and 1 star on their Lesson 1 model. The bridge sentence is complete and references at least one specific piece of evidence (e.g., 'the ammeter readings in our series circuit experiment'). Teacher listens during cold-calls for students who still conflate voltage and current — these students receive a targeted prompt card during the performance task.
Observe Phase 📹 **VIDEO:** Video: Ohm's Law Source: PhET Interactive Simulations (English) 🔍 Search ARES for similar videos	A rapid 'Evidence Parade' is projected on the board — a slide deck of 8 key data sets and diagrams collected across Lessons 1–11: (1) the Huruma Estate video still; (2) the V-I graph from the Ohm's Law experiment (Lesson 2); (3) the resistivity comparison table of copper vs.	Project each slide for 40 seconds, then advance. Say: 'You collected all of this evidence over twelve lessons. Today, all of it comes together.' After the parade, ask: 'Which single piece of evidence do you think is most important for explaining why the Huruma kettle was slow? Write your answer —	Strategy: 'Evidence-to-Phenomenon Mapping.' Students explicitly connect each data artefact to the anchoring phenomenon, practising the scientific skill of using multiple lines of evidence to build a single coherent explanation. This mirrors how real engineers at Kenya	Observable indicator: The Evidence Reference Sheet is at least 80% complete with correct phenomenon-to-evidence links. Teacher scans 6 sheets during circulation. Any student who cannot link internal resistance to the kettle's slow heating receives a targeted question: 'What happens

📖 **HTML:** River Ferry and Current Vectors (PLIX) Source: CK-12 🔍 Search ARES for similar readings	nichrome (Lesson 4); (4) ammeter/voltmeter readings from the series vs. parallel circuit practical (Lessons 5–6); (5) the internal resistance graph showing terminal voltage drop vs. current (Lesson 7); (6) the power dissipation data from the kettle and bulb comparison (Lesson 9); (7) the energy cost calculation for a Nairobi household (Lesson 10); (8) the completed safety checklist from the fault-diagnosis task (Lesson 11). Students have 6 minutes to review the parade in pairs, filling in a one-page 'Evidence Reference Sheet' — matching each data set to the Driving Question Board card it answers and to the part of the Huruma phenomenon it explains. The reference sheet will be used as a resource during the performance task.	do not shout it out.' [WAIT TIME: 15 seconds.] Cold-call 4 students with different answers to create productive disagreement: 'Korir says the internal resistance graph. Fatuma says the series-circuit ammeter data. Let's hold that tension — your final explanation will tell us who is right, or whether both matter.' This primes students for the synthesis they will do in Phase 3.	Power diagnose grid problems using multiple sensor data streams.	to terminal voltage when a large current is drawn from a battery or transformer with internal resistance — and what does lower voltage do to the power delivered to the kettle?'
Explain Phase 📹 **VIDEO:** Video: Ohm's Law Source: PhET Interactive Simulations (English) 🔍 Search ARES for similar videos 📖 **HTML:** River Ferry and Current Vectors (PLIX) Source: CK-12 🔍 Search ARES for similar readings	Students individually write a structured Final Explanation (15 minutes) using the four-part scaffold posted on the board: (A) CLAIM — state what is happening in the Huruma phenomenon in one sentence; (B) EVIDENCE — cite at least four specific data points or experimental results from the unit; (C) REASONING — use physics principles (Ohm's Law, internal resistance, parallel circuits, power formula) to explain why the evidence supports the claim; (D) CONNECTION TO DESIGN — explain how knowing this physics allows engineers and home-owners to design circuits that prevent the Huruma problem. Students must address all three Huruma observations: dim lights at power restoration, slow kettle, and permanently dim bulb on long thin	Post the four-part scaffold clearly. Say: 'This is your moment. You are the scientist who has done the experiments. Write your explanation as if you are advising Kenya Power on why this keeps happening in estates like Huruma.' [WAIT TIME: 10 seconds before students begin writing — allow re-reading of the scaffold.] Circulate during the 15-minute writing period. Target students who are vague: 'You wrote 'resistance is high' — high compared to what? Give me a number or a comparison from our data.' During round-robin, listen for whether students correctly explain that: (1) at power restoration, the supply voltage is initially low (or internal resistance is high under surge current), reducing power delivered to bulbs and kettle; (2) a long thin	Strategy: 'Claim-Evidence-Reasoning-Design (CERD) Framework.' The structured scaffold ensures students do not conflate description with explanation. The 'Design' column elevates explanation to engineering application — connecting physics knowledge to real household circuit planning, which is the core competency of the unit.	Observable indicator: Final Explanation addresses all three Huruma observations using at least four distinct physics concepts. Teacher uses a 4-point rubric during cold-call presentations: 1 = claim only, 2 = claim + evidence, 3 = claim + evidence + reasoning, 4 = claim + evidence + reasoning + design connection. Target score for end of unit: 3–4. Students scoring 1–2 are flagged for a brief reteach conversation during the performance task.

	extension cord. After writing, groups of 4 share explanations in a 'round-robin read-aloud' — each member reads one part (A, B, C, or D) of their own explanation aloud while others listen for gaps or contradictions. Groups flag one unresolved disagreement to bring to the whole class. Two groups are cold-called to present their full explanation. Teacher facilitates a class consensus-building discussion to produce one agreed 'Class Final Explanation' posted on the wall beside the original Huruma Estate photograph.	extension cord has significant resistance in series with the bulb, causing a voltage drop across the cord and less voltage — and therefore less power — across the bulb; (3) parallel domestic wiring ensures each appliance receives the full supply voltage regardless of other appliances. Cold-call Group 1 and Group 2. After each presentation, ask the class: 'What did this group say that your explanation did not include? Add it in a different colour.' Scribe the consensus explanation with student input: 'Who can give me the exact equation that links voltage drop to cord resistance? Amara?' Ensure the final class explanation explicitly mentions: $V = IR$, $P = V^2/R$ or $P = IV$, internal resistance r, terminal voltage $V_t = EMF - Ir$, parallel circuit rules (same voltage, independent branches), and at least one safety principle.		
Driving Question Board (DQB) Creation 📹 **VIDEO:** Video: Ohm's Law Source: PhET Interactive Simulations (English) 🔍 Search ARES for similar videos 📖 **HTML:** River Ferry and Current Vectors (PLIX) Source: CK-12 🔍 Search ARES for similar readings	The full Driving Question Board — maintained since Lesson 1 — is brought to the front of the room. All remaining open cards (questions generated by students across the unit that have not yet been formally answered) are read aloud by the teacher. Students vote silently (thumbs up / sideways / down) on whether each question has now been answered by the evidence. For questions with consensus 'thumbs up,' a volunteer writes a one-sentence answer on a green sticky note and places it over the question card. For 'sideways' questions (partially answered), students write what is still unknown on an orange sticky note — these become 'beyond-this-unit' questions that feed into	Read each DQB card with full energy: 'Lesson 3, Otieno asked: why does a thicker wire carry more current safely? Has the unit answered this?' After the vote, if thumbs are mixed, probe: 'Wanjiku, you gave a sideways thumb — what is still missing from our explanation?' [WAIT TIME: 10 seconds.] For the three anchor questions, say: 'These are the questions that started everything. We are going to answer them the way scientists do — with evidence. I want three volunteers who feel they can give our best possible answer.' Accept class nominations. Each selected student stands, reads the anchor question, and delivers their answer using the class Final Explanation as a	Strategy: 'Ceremonial DQB Closure.' Publicly resolving the questions students themselves generated throughout the unit gives ownership of the learning to the class. The green/orange sticky-note distinction honours the reality that science always generates new questions — modelling intellectual humility alongside intellectual achievement.	Observable indicator: At least 90% of DQB cards receive a green or orange sticky note by end of phase. The three anchor question answers are coherent, evidence-referenced, and use correct physics vocabulary (voltage, resistance, current, power, internal resistance, parallel circuit). Teacher notes any anchor question answer that is still vague or incorrect and addresses it explicitly before moving to the performance task.

	future strands (e.g., electronics, energy systems). The three original Driving Question Board anchor questions — (1) Why dim lights at restoration? (2) Why slow kettle? (3) Why dim bulb on long cord? — are answered last, ceremonially, by three different students selected by class vote as 'most improved explainer' across the unit. The completed DQB is photographed and displayed in the classroom as a record of the class's inquiry journey.	reference. After each, ask: 'Class, does this answer fully satisfy the question? What, if anything, would you add?' This creates a moment of collective intellectual closure. Close by saying: 'In Lesson 1, nobody in this room could explain why the lights in Huruma were dim. Today, all of you can. That is what physics education is for.'		
Model Building Phase 📹 **VIDEO:** Video: Ohm's Law Source: PhET Interactive Simulations (English) 🔍 Search ARES for similar videos 📖 **HTML:** River Ferry and Current Vectors (PLIX) Source: CK-12 🔍 Search ARES for similar readings	Students receive a one-page task brief: the Wanjiku family in Githurai, Nairobi, is moving into a new three-room house. They have the following appliances: a 100 W bulb (living room), a 60 W bulb (bedroom), a 1,200 W electric kettle (kitchen), a 300 W phone charging unit (bedroom), and a 750 W iron box (kitchen). Kenya Power supplies 240 V AC. The household fuse is rated 15 A. The family has a monthly electricity budget of KES 2,000 (using Kenya Power's domestic tariff of approximately KES 22 per kWh for the first 50 kWh). Students must: (1) draw a complete, labelled circuit diagram for the house showing all appliances connected in parallel to the 240 V mains, with individual switches and a main fuse; (2) calculate the current drawn by each appliance (I = P/V); (3) calculate the total current when all appliances run simultaneously and determine whether the 15 A fuse is safe; (4) calculate the monthly energy cost if the kettle runs 1 hour/day, the iron 0.5 hours/day, the bulbs 5 hours/day, and the charger 3 hours/day — and advise the family whether this	Distribute the task brief and a blank circuit diagram template (pre-drawn house outline with three rooms and mains entry point). Say: 'You are now the electrical engineer. The Wanjiku family is counting on you to keep their home safe and their electricity bill within budget. Every decision you make must be backed by a calculation or a physics principle we have studied.' [WAIT TIME: 12 seconds for students to read the brief silently before any questions.] Clarify only procedural questions — do not re-teach content at this stage. During circulation (every 5 minutes), use targeted prompts only: 'Show me the calculation for the kettle current.' / 'You have all appliances in parallel — now calculate the total current. Is 15 A enough?' / 'Your diagram is missing the earth wire on the kettle — why does a kettle specifically need earthing?' At the 20-minute mark, give a 10-minute warning. For the gallery walk, say: 'Walk slowly. Read the plan carefully. Your glow must reference a specific correct calculation or design feature. Your grow must suggest something specific — not	Strategy: 'Engineering Design Performance Task with Gallery Walk Peer Review.' The performance task requires students to apply — not just recall — every major concept from the unit in an authentic Kenyan household context. The gallery walk introduces peer feedback as a scientific community practice, mirroring how engineering designs are reviewed before implementation. Annotating the diagram with phenomenon-referenced reasoning ensures conceptual understanding drives the design, not just procedural calculation.	Observable indicator (summative): (1) Circuit diagram correctly shows all five appliances in parallel with individual switches and a main fuse — labelled with correct symbols. (2) All five current calculations are correct (I = P/V): bulb 100W → 0.42 A; bulb 60W → 0.25 A; kettle → 5.0 A; charger → 1.25 A; iron → 3.125 A; total ≈ 10.04 A — correctly concluded to be within the 15 A fuse rating. (3) Monthly energy cost is correctly calculated at approximately KES 792, correctly advised as within budget. (4) Two safety features are identified and justified with physics reasoning. (5) Parallel justification explicitly references voltage consistency, independent operation, and the Huruma Estate phenomenon. Teacher uses a 5-criterion checklist rubric. Students meeting 4–5 criteria demonstrate mastery of sub-strand SLOs.

	fits their budget; (5) identify two safety features in their design (e.g., correct fuse, earthing of kettle) and justify each using physics; (6) annotate their diagram with a brief explanation of why parallel — not series — wiring is essential for this home, referencing the Huruma Estate phenomenon. Students work individually (30 minutes) then participate in a 10-minute gallery walk, leaving one 'glow' (strength) and one 'grow' (suggested improvement) sticky note on two peers' plans.	just 'add more detail." After the gallery walk, cold-call 3 students to share the most interesting 'grow' comment they received and how they would respond to it. Close by connecting back to the phenomenon: 'If you had been the electrical engineer for Huruma Estate from the beginning, using everything in this plan, would those lights have flickered? Why or why not?'		

D. TEACHER REFLECTION

After delivering this lesson, reflect on the following questions: (1) Did students' Final Explanations genuinely integrate all unit concepts (Ohm's Law, internal resistance, parallel circuits, power, energy, safety), or did explanations remain partial — focusing on one or two concepts only? Which concepts remain weakest? (2) Were the DQB anchor question answers delivered by students accurate, confident, and evidence-referenced? If not, which part of the storyline — which specific lesson or experiment — failed to build sufficient understanding, and how would you strengthen it? (3) In the Wanjiku Family Wiring Plan, did students correctly reason that parallel wiring provides voltage consistency across all appliances, or did some still design series arrangements? What additional analogy or activity in earlier lessons might have solidified this concept? (4) Did the gallery walk produce substantive peer feedback, or were 'glow and grow' comments superficial? How might you scaffold peer review more explicitly in future iterations? (5) Consider equity: did all students — including those who struggled earlier in the unit — find an entry point in the performance task? Were the targeted prompt cards sufficient support, or do some students require a follow-up reteach session on core calculations? (6) Most importantly: when you asked the closing question — 'If you had been the electrical engineer for Huruma Estate from the beginning, would those lights have flickered?' — how did students respond? Did their answers demonstrate genuine transfer of learning from physics principles to real-world engineering thinking? Record three student responses verbatim for use in curriculum review documentation.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Students reviewed their Lesson 1 circuit models and annotated them with red-pen corrections, participated in an Evidence Parade revisiting eight key data sets from Lessons 1–11 (including the Ohm's Law V-I graph, the internal resistance terminal voltage graph, the series vs. parallel ammeter readings, and the Kenya Power energy cost data), and completed a gallery walk of peer-designed household wiring plans for the Wanjiku Family Home in Githurai, Nairobi.
What did I learn?	Students synthesised all sub-strand concepts into a unified Final Explanation: that the Huruma Estate dim lights and slow kettle at power restoration were caused by high effective internal resistance in the supply network reducing terminal voltage and therefore power delivered to appliances ($P = V^2/R$); that the permanently dim bulb on the long thin extension cord resulted from resistive voltage drop across the cord in series with the bulb, leaving less voltage — and therefore less power — across the bulb; and that domestic wiring must be in parallel so that every appliance receives the full 240 V supply voltage, operates independently, and benefits from the low effective resistance of parallel branches. Students also learned to calculate total household current draw,

	assess fuse safety, and estimate monthly energy costs using Kenya Power tariff rates.
How does this explain the phenomenon?	Students explained the complete Huruma Estate anchoring phenomenon using the Claim-Evidence-Reasoning-Design (CERD) framework, referencing Ohm's Law ($V = IR$), terminal voltage ($V_t = EMF - Ir$), power equations ($P = IV = V^2/R = I^2R$), parallel circuit rules (same voltage across branches, total current as sum of branch currents), and safety principles (correct fuse rating, earthing of metal-cased appliances). They closed every Driving Question Board card with evidence-referenced answers and justified a complete parallel wiring design for a model Kenyan household — demonstrating mastery of the sub-strand's core competency: designing safe, efficient electrical circuits for everyday Kenyan life.

DIFFERENTIATION AND INCLUSION		
Learner Need	**Support Strategies**	**Extension Strategies**
English Language Learners	Provide bilingual glossaries; allow use of first language for initial model drawing.	Write extended explanations of observations; lead group discussions in English.
Students with learning difficulties	Provide structured note frames; pair with a supportive partner during investigations.	Mentor peers; create additional visual models to explain concepts.
Gifted and advanced learners	Access to additional reading materials; challenge questions during DQB.	Lead model-building discussions; research and present extension topics to class.